PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. Under the Panjab University Act VII of 1947—enacted by the Govt. of India)

SYLLABI

FOR

B.Sc./B.Sc. (Hons)/B.Sc. (Hons. with research)
Major & Minor Courses

SECOND YEAR
(SEMESTER SYSTEM)
EXAMINATIONS, 2025-26

(SEMESTER : THIRD AND FORTH)

i.e

Third Semester: November/December, 2025
Forth Semester: April/May, 2026

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Table of Contents

Sr. No.	Particulars	Page No.
1.	Agriculture	03
2.	Bio-Chemistry	10
3.	Bioinformatics	19
4.	Biotechnology	26
5.	Botany	33
6.	Chemistry	46
7.	Computer Science and Applications	76
8.	Environment Conservation	89
9.	Industrial Chemistry	93
10.	Mathematics	110
11.	Microbiology	126
12.	Physics	133
13.	Statistics	154
14.	Zoology	170

Agriculture

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY

B.A./B.Sc., B.A./B.Sc. HONOURS and B.A./B.Sc. HONOURS WITH RESEARCH IN AGRICULTURE

(Table 2*)

Sem	Level	Course Code	Course Name	Credits (Th+Pr+Tut)	Total Credits
III	L-200	AGR-DSC-3-Maj/Min-301	Agronomic Practices for Cultivation of Kharif Crops	3+1=4 (x3)	24
		AGR-DSC-4-Min-302	Soil Science	3+1=4	
		AGR-MDC-3-301	Statistical Techniques in Agriculture	2+1=3	
		AGR-SEC-3-301	Agricultural Extension	2+1=3	
		AEC-3-301	Eng-3/MIL-1	2	
IV	L-200	AGR-DSC-5-Maj/Min-401	Agronomic Practices for Cultivation of Rabi Crops	3+1=4 (x3)	24
		AGR-DSC-4-Maj-402	Soil Science	3+1=4 (x2)	
		AGR-VAC-3-401	-	2	
		AEC-4-401	Eng-4/MIL-2	2	
The student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secures 4 credits from summer internship					

Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.

9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

B.A./B.Sc.(General) Agriculture (Major/Minor)

Course Title : AGRONOMIC PRACTICES FOR CULTIVATION OF *KHARIF* CROPS **Course Code: (AGR-DSC-3 Maj/Min-301)**
Semester: Third

Learning Outcomes: To give student basic agronomic principles and practices for cultivation of *Kharif* crops.

Max. Marks: 100 **Course Credits** = 3+1=4

1. Theory: 75 Written Exam: 67 Internal Assessment: 8
2. Practical: 25 Written Exam: 22 Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 10short answer type questions of one and a half mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

UNIT – I **12 Hours**

Distribution, Climatic requirement, Soil requirements, Rotations, Improved varieties, Agronomic practices (land preparation, seed rate & seed treatment, weed control, fertilizer application, irrigation) and harvesting of :- Cereals – Rice, Basmati Rice, Maize. Bajra

UNIT – II **11 Hours**

Distribution, Climatic requirement, Soil requirements, Rotations, Improved varieties, Agronomic practices (land preparation, seed rate & seed treatment, weed control, fertilizer application, irrigation) and harvesting of :- Oilseeds- Groundnut, Sesame. Cotton (American and Desi), Sugarcane

UNIT – III **11 Hours**

Distribution, Climatic requirement, Soil requirements, Rotations, Improved varieties, Agronomic practices (land preparation, seed rate & seed treatment, weed control, fertilizer application, irrigation) and harvesting of :-Pulses –Moong, Mash, Arhar, Soyabean.

UNIT – IV **11 Hours**

Distribution, Climatic requirement, Soil requirements, Rotations, Improved varieties, Agronomic practices (land preparation, seed rate & seed treatment, weed control, fertilizer application, irrigation) and harvesting of :-Green manuring Crop (Dhaincha, cowpea and sunhemp). Fodders : Maize, Sorghum, Bajra, Guara

GUIDELINES FOR PRACTICAL EXAMINATION

Time- 3 Hrs **Max Marks: 25**
Practical: 22 **Internal Assessment: 03**

1. Crop identification based on seed and morphological characteristics.
2. Different inorganic fertilizers, fertilizers recommendation for *Kharif* crops and calculating the fertilizer requirement of various crops as per the recommendation using different combinations.
3. Practice in different methods of fertilizer application.
4. Sowing of important *kharif* crops.
5. Identification of *kharif* weeds and chemical method of weed control.
6. Visit to different farms.
7. Study of morphological characterstics of rice, maize, bajra, groundnut and cotton.

Suggested Readings:-

Reddy S.R., *Principles of Crop Husbandry,*Kalyani Publishers, Ludhiana, 2009.
Handbook of Agriculture, I.C.A.R. Publications, New Delhi, 2008.
Weeds of North India I.C.A.R. Publications, New Delhi, 2008.
Package of Practices for kharif crops, P.A.U. Publications Ludhiana, Corresponding year.

B.A./B.Sc.(General) Agriculture (Major/Minor)

Course Title: Agronomic practices for Cultivation of *Rabi* Crops

Course Code: (AGR-DSC-5Maj/Min-401)

Semester: Fourth

Learning Outcome: To give student basic knowledge of principles and practices for raising vegetable crops.

Max. Marks: 100 **Course Credits** = 3+1=4

1. Theory: 75 Written Exam: 67 Internal Assessment: 8
2. Practical: 25 Written Exam: 22 Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 10short answer type questions of one and a half mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

UNIT – I **12 Hours**

Climatic requirement, Soil requirements, Rotations, Improved varieties, Agronomic practices (land preparation, seed rate & seed treatment, weed control, fertilizer application, irrigation) and harvesting of :- Cereals – Wheat, Barley, Winter Maize.

UNIT – II **11 Hours**

Climatic requirement, Soil requirements, Rotations, Improved varieties, Agronomic practices (land preparation, seed rate & seed treatment, weed control, fertilizer application, irrigation) and harvesting of Oilseeds — Rapeseed and Mustard (Toria, GobhiSarson, Raya, Taramira, Linseed, and Sunflower.

UNIT – III **11 Hours**

Climatic requirement, Soil requirements, Rotations, Improved varieties, Agronomic practices (land preparation, seed rate & seed treatment, weed control, fertilizer application, irrigation) and harvesting of :- Pulses — Gram, Field Pea and Lentil, Summer moong.

UNIT – IV **11 Hours**

Climatic requirement, Soil requirements, Rotations, Improved varieties, Agronomic practices (land preparation, seed rate & seed treatment, weed control, fertilizer application, irrigation) and harvesting of sugarcane and Fodders : Barseem, Shaftal, Lucerne, Oat and Senji.

GUIDELINES FOR PRACTICAL EXAMINATION

Time- 3 Hrs **Max Marks: 25**
Practical: 22 **Internal Assessment: 03**

1. Crop identification based on seed and morphological characteristics
2. Different inorganic fertilizers, fertilizers recommendation for *Rabi* crops and calculating the fertilizer requirement of various crops as per the recommendation using different combinations.
3. Different methods of fertilizer application used in *Rabi* crop.
4. Sowing of important *Rabi* crops
5. Identification of *Rabi* weeds and chemical method of weed control.
6. Study of morphological characteristics of important *Rabi* crops.
7. Visit to different farms.

Suggested Readings:

Reddy S.R., *Principles of Crop Husbandry,*Kalyani Publishers, Ludhiana, 2009.
Handbook of Agriculture, I.C.A.R. Publications, New Delhi, 2008.
Weeds of North India I.C.A.R. Publications, New Delhi, 2008.
Package of Practices for Rabi crops, P.A.U. Publications Ludhiana, Corresponding year.

B.A./B.Sc.(General) Agriculture (Major)

Course Title : SOIL SCIENCE **Course Code: (AGR-DSC-4 Maj-402)**
Semester: Fourth

Learning Outcome: To give student elementary information on basic principles of soil sciences.

Max. Marks: 100 **Course Credits** = 3+1=4

1. Theory: 75	Written Exam: 67	Internal Assessment: 8
2. Practical: 25	Written Exam: 22	Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 10short answer type questions of one and a half mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

UNIT – I 12 Hours

Introductory knowledge of soil: definition of soil, component of soil
Definition, occurrence and classification of soil forming rocks & minerals.
Weathering of rocks and minerals.
Soil Profile, Horizons. Classification of soil based on soil taxonomy.

UNIT – II 11 Hours

Important physical properties of soil: Soil separates/particles, their mechanical analysis and characteristics.
Soil texture, soil densities and porosity of soil.
Classification of Soil structure, factors effecting soil structure and its importance.
Soil Organic Matter origin, its nature and composition.

UNIT – III 11 Hours

Decomposition of various organic compounds: Factors affecting decomposition, decompositions of water soluble and water insoluble organic matters.
Advantages and disadvantages of organic matter.
Concept of humus formation.
Soil reactions:- Methods of expressing and factors affecting Soil reactions, its influence on nutrient availability & plant growth.
Buffer action & buffering capacity of soils.Factors affecting buffering capacity of soils.

UNIT – IV 11 Hours

Formation & problems of Salt effected soils: Different types of Salt effected soils, nature & sources of soluble salt in saline soil, appraisal of saline &sodic soil. Calcareous soil.
Reclamations of saline and alkali or sodic soils: Chemical reactions involving reclamations of salt affected soil. Gypsum requirement.
Plant growth concept and factors affecting plant growth.
Plant nutrition: Essential elements, classification & general function of plant nutrients, forms of nutrient elements absorbed by plants. Process of nutrient uptake.

GUIDELINES FOR PRACTICAL EXAMINATION

Time- 3 Hrs **Max Marks: 25**
Practical: 22 **Internal Assessment: 03**

1. Identification of Important rocks and minerals.
2. Examination of soil profile in the field.
3. Soil sampling and use of augers.
4. Determination of water and soil texture.
5. Study of soil colour using Munsell soil colour charts.
6. Study of soil texture and soil structure of a given sample.
7. Calculations of soil densities and porosity of soil.
8. Calculation of gypsum requirement for given sample
9. Soil survey of a farm.

Suggested Readings:

Brady, N.C. and Weil, R.R., *The Nature and Properties of Soil: 13thedn.,*Pearson education Pte.Ltd.New Delhi, 2002.
Oswal, M.C., *Soil Physics,* Oxford & IBH publishing Co.Pvt.Ltd.New Delhi, 1994.
Biswas, T.D., and Mukherjee, S.K., *Text book of soil science,* Tata McGraw Hill Publishing Co. Ltd., New Delhi, 1997.

Biochemistry

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR(Table 1*)
and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY(Table 2*)

B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH IN BIOCHEMISTRY

Table 1*						Table 2*					
Sem	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**
III	L-200	(BCH-DSC-3 Maj/Min-301)	Intermediatory Metabolism	3+1=4	24	III	L-200	(BCH-DSC-3 Maj/Min-301)	Intermediatory Metabolism	3+1=4 (x3)	24
		(BCH-DSC-4 Min-302)	Enzymology & Bioenergetics	3+1=4				(BCH-DSC-4 Min-302)	Enzymology & Bioenergetics	3+1=4	
		(BCH-MDC-3-301)	Basic Cell Biology	2+1=3				(BCH-MDC-3-301)	Basic Cell Biology	3	
		(BCH-SEC-3-301)	Analytical Techniques in Biochemistry	0+3=3				(BCH-SEC-3-301)	Analytical Techniques in Biochemistry	2+1=3	
		(AEC-3-301)	Eng-3/MIL-1	2				(AEC-3-301)	Eng-3/MIL-1	2	
IV	L-200	(BCH-DSC-5 Maj/Min-401)	Membrane of Biochemistry & Neurobiology	3+1=4 (x3)	24	IV	L-200	(BCH-DSC-5 Maj/Min-401)	Membrane of Biochemistry & Neurobiology	3+1=4 (x3)	24
								(BCH-DSC-4 Maj-402)	Enzymology & Bioenergetics	3+1=4 (x2)	
		(VAC-3-401)	-	2				(VAC-3-401)	-	2	
		(AEC-4-401)	Eng-4/MIL-2	2				(AEC-4-401)	Eng-4/MIL-2	2	
The student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secures 4 credits from summer internship											

Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

Semester III

Major/Minor

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:

1. Examiner will set total of NINE questions comprising TWO questions from each unit and ONE compulsory question.
2. All questions carry equal marks.
3. The compulsory question will consists of 12 short questions of one mark each covering the whole syllabi.
4. Besides Question No. 1, all other questions may contains 2-3 sub-parts.
5. Questions should be uniformly spread over the entire syllabus.
6. The students are required to attempt FIVE questions in all, ONE question from each unit and the Compulsory question.

(BCH-DSC-3 Maj/Min-301): Intermediatory Metabolism (3+1)

Marks: 75 (60+15 Int.)

Objective: To understand Metabolism of amino acids, their derivatives and nucleic acids.

Learning Outcomes:
Upon successful completion of this course, students will be able to:

- Comprehend the basic types of catabolic and anabolic pathways of carbohydrates,aminoacids,fatty acids and nucleic acids.
- Learn to perform various clinical testsrelated to metabolism.

SECTION-I

(Lectures: 10)

Introduction to Metabolic pathways. Digestion and absorption carbohydrates, Metabolism of carbohydrates, glycolysis and alcoholic fermentation, the pentose phosphate pathway, metabolism of fructose, galactose, TCA cycle and pyruvate dehydrogenase complex Glucuronate and glyoxylate pathway, gluconeogenesis, Cori cycle, glycogenolysis& glycogenesis, biosynthesis of starch.

SECTION-II

(Lectures: 10)

Digestion and absorption of lipids, beta-Oxidation of fatty acids including odd chain fatty acids.Alpha and omega oxidation of fatty acids.Formation and utilization of ketone bodies.Biosynthesis of saturated and unsaturated fatty acids Biosynthesis and degradation of phospholipids and triglycerides.Cholesterol biosynthetic pathway Biosynthesis of prostaglandins, thromboxanes, leukotrienes.

SECTION-III

(Lectures: 10)

Digestion & absorption of proteins, nitrogen balance, general reactions of amino acids: deamination, transamination and decarboxylation. Urea cycle. Catabolism of glucogenic and ketogenic amino acid Biosynthesis of nutritionally essential amino acids: glutamate and

glutamine, aspartate and asparagine, proline, alanine, cysteine &selenocysteine, tyrosine, serine, glycine. Conversion of amino acids to specialized products: catecholamine, glutathione, creatinine, serotonin.

SECTION-IV

(Lectures: 10)

Nucleic Acids: Degradation and biosynthesis of purines and pyrimidines nucleotides *Denovo* and salvage pathway, formation of deoxyribonucleotides, biosynthesis and degradation of heme. Disorders of heme, purine & pyrimidine metabolism, anticancer drugs.

PRACTICALS:

(BCH-DSC-3 Maj/Min-301): Intermediatory Metabolism

Marks: 25 (20 + 5 Int.)

1. Estimation of blood glucose
2. Demonstration of Amylase activity
3. Estimation of total cholesterol
4. Estimation of pyruvic acid.
5. Assay of Serum Transaminases (SGPT and SGOT)
6. Estimation of Serum Urea
7. Estimation of Serum Uric Acid
8. Estimation of Serum Creatinine
9. Estimation of DNA

Suggested Books:

1. Lippincott's Illustrated Reviews: Biochemistry by P.C. Champe, R.A. Harvey & amp; D.R. Ferrier, Indian Edition, 2017.
2. Harper's Illustrated Biochemistry by R.K. Murray, P.A. Mayes, D.K. Granner and V.W. Rodwell,31th Ed. 2018, Prentice Hall International Ltd. (U.K.).
3. Voet D and Voet JG. Biochemistry 5 th Edition, John Wiley and Sons Inc., 2021.
4. Lehninger A.L., Nelson D.L. and Cox M.M. (2021)8th edition Principles of biochemistry (W.H. Freeman, USA).
5. Stryer.L. J.M. Berg, J.L. Tymoczko (2023)10th edition, Biochemistry (W.H. Freeman Company, New York).
6. Conn E.E.and Stumpf P.K. (2006)5th edition Outlines of Biochemistry, (John Wiley and Sons, USA).
7. R.H. Garett and C.M. Grisham (2023)7th Edition Biochemistry, (Cengage Learning, US).
8. Zubay G.L., Parson W.W, Dennis E. Vance (William C. Brown, U.S.)1998 Principles of Biochemistry volume-1.

Minor

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:

1. Examiner will set total of NINE questions comprising TWO questions from each unit and ONE compulsory question.
2. All questions carry equal marks.
3. The compulsory question will consists of 12 short questions of one mark each covering the whole syllabi.
4. Besides Question No. 1, all other questions may contains 2-3 sub-parts.
5. Questions should be uniformly spread over the entire syllabus.
6. The students are required to attempt FIVE questions in all, ONE question from each unit and the Compulsory question.

(BCH-DSC-4 Min-302): Enzymology and Bioenergetics (3+1)

Marks: 75 (60+15 Int.)

Objective: To learn principles of thermodynamics, nature of enzymes, their mode of action and enzyme kinetics.

Learning Outcomes:

Upon successful completion of the course, students will be able to:

- Understand the concepts of enzymes, their mechanisms and kinetics of enzymatic reactions.
- Understand the mechanism of oxidative phosphorylation and ATP synthesis
- Learn to perform enzymology experiments and learn to calculate enzyme activityand kinetics.

SECTION-I

(Lectures: 10)

Introduction to enzymes.General characteristics of enzymes' system of nomenclature and classification of enzymes.Prosthetic groups.Holoenzyme, apoenzyme, cofactors, isoenzymes and their clinical significance and zymogens. Coenzymes (NAD/NADP, FMN/FAD, coenzyme A, TPP, pyridoxal phosphate, tetrahydrofolate) and their biochemical functions, assay of enzyme activity, units of enzyme activity. Active site of enzymes.

SECTION-II

(Lectures: 10)

Factors affecting velocity of enzyme catalyzed reactions: enzyme concentration, pH and temperature. Michaelis-Menten equation.Determination of Km and its significance.

Enzyme inhibition: Various types of enzyme inhibitions. Determination of Ki value.Enzyme inhibitors and their importance.Introduction to multisubstrate enzyme, Allosteric enzymes properties and dregulation.

SECTION-III

(Lectures: 10)

Enzymes as catalysts. Theories of enzymes catalysis: proximity and orientation effects, acid base catalysis, covalent catalysis. Role of metals in enzyme catalysis Monomeric enzymes, oligomericenzymes &multienzyme complexes.Mechanism of action of chymotrypsin, abzymes and lysozyme Introduction to multisubstrate enzymes.Allosteric enzymes and enzyme regulation.

SECTION-IV

(Lectures: 10)

Concepts of bioenergetics, principles of thermodynamics & their application in Biochemistry, concept of free energy.Biological oxidations.Redox potential. ATP and High energy phosphate compounds: introduction, phosphate group transfer potential, enzymes and co-enzymes involved in oxidations and reductions. Mitochondrial electron transport chain and oxidative phosphorylation

Suggested Books:

1. T. Palmer Enzymes: Biochemistry, Biotechnology and Clinical Chemistry, 2nd Ed. (2018) Affiliated East-West press Pvt. Ltd. New Delhi.
2. L.Stevens and Nicholas C.Price Fundamentals of Enzymology: The cell and molecular Biology of catalytic Proteins (2000) 3rd edition (Oxford University Press, USA).
3. Robert A. Copeland: A practical Introduction to Structure, Mechanism and Data Analysis 2nd edition (John Wiley and Sons Inc., Publications)
4. Harper's Illustrated Biochemistry by R.K. Murray, P.A. Mayes, D.K. Granner and V.W. Rodwell, 31th Ed. 2018, Prentice Hall International Ltd. (U.K.).
5. Voet D and Voet JG. Biochemistry 5th Edition, John Wiley and Sons Inc., 2021.
6. LehningerA.L.,NelsonD.L.andCox M.M.(2021)8th edition Principles of biochemistry (W.H. Freeman, USA)
7. StryerL., J.M. Berg, J.L. Tymoczko (2023)10th edition, Biochemistry (W.H. Freeman company, New York)
8. Conn E.E. and Stumpf P.K. (2006)5th edition Outlines of Biochemistry, (John Wiley and Sons, USA)
9. R.H. Garett and C.M. Grisham (2023)7th Edition Biochemistry, (Cengage Learning, US)

PRACTICALS:

(BCH-DSC-4 Min-302): Enzymology and Bioenergetics

Marks: 25 (20 +5 Int.)

1. Assay of Enzyme Activity and Specific Activity e.g. Acid Phosphatase
2. Effect of pH on enzyme activity.
3. Effect of temperature on enzyme activity and determination of energy of activation.
4. Effect of substrate concentration on enzyme activity and determination of Km.
5. Partial Purification of Acid Phosphatase from Germinating Mung Bean
6. Effect of inhibitors on enzyme activity.
7. Demonstration of potato polyphenol oxidase activity.

Semester IV

Major/Minor

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:

1. Examiner will set total of NINE questions comprising TWO questions from each unit and ONE compulsory question.
2. All questions carry equal marks.
3. The compulsory question will consists of 12 short questions of one mark each covering the whole syllabi.
4. Besides Question No. 1, all other questions may contains 2-3 sub-parts.
5. Questions should be uniformly spread over the entire syllabus.
6. The students are required to attempt FIVE questions in all, ONE question from each unit and the Compulsory question.

(BCH-DSC-5-Maj/Min-401): Membrane Biochemistry & Neurobiology (3+1)

Marks: 75 (60+15 Int.)

Objective: to make students aware of structural and functional aspects of membranes. Highlighting the significance of their compositional heterogeneity and to various cell functions.

Learning Outcomes:
Upon successful completion of this course, students will be able to:

- Understand the composition, structure and various properties of membrane
- Electrochemical and ionic changes that power the various membrane like RBC and neurons functions.
- They will be able to independently perform various experiments related to isolation and purification of membrane.

SECTION-I

(Lectures -10)

Introduction: membrane structure and organization: models of membranes structure. Protein- lipid interactions in membranes, fluid mosaic model of membrane structure, its merits and demerits. Constituents of biological membranes in prokaryotes and eukaryotes.Asymmetry of biomembranes.Structure of RBC membrane and ethrocytedeformities.

SECTION –II

(Lectures-10)

Membrane dynamics: Physical properties of membrane components, membrane fluidity, Phase transitions, cytological and biochemical methods to study membrane fluids. Membrane transport: Typical properties of membrane transport, diffusion and osmosis: uniport and synport, passive diffusion, facilitated diffusion, Carrier mediated active transport. Role of Na, K- ATPase.Ca-ATPase & H^+-ATPase.

SECTION –III

(Lectures-10)

Membrane analysis: isolation and characterization of cell membrane, criteria of membrane purity. Solubilization of membranes and their reconstitution.Labeling of membrane probes and determination of transbilayer distribution of membrane components. Liposomes: unilamelar and multilamelar vesicles. Methods of their preparation, characterization and their applications in targeting drugs and gene therapy.Ionophores and siderophores, structure and functions.

SECTION –IV

(Lectures-10)

Neurobiology: Central nervous system, peripheral nervous system, action potential, neurotransmitters, synapses & nerve gases. Mechanism of nerve impulse transmission: structure, classification and functions of neurons and neuroglias. The resting membrane potential & action potential. Properties of nerve fibers: excitability, conductivity, all or none law, accommodation, adaptation, summation, refractory period, indefatigability. Synapses: types, structure, synaptic transmission of the impulse, synaptic potentials, neurotransmitters, cotransmitters, neuromodulators. The neuromuscular junction: structure, transmission, end-plate potential.

Suggested Books:

1. Jain, MK. Introduction to Biological membranes, John Wiley and sons New York, 1988
2. Lodish H. Molecular Cell biology, WH Freeman and Company, 2021.
3. Bruce alberts, Molecular Biology of the cell 7th edition 2022
4. Geoffrey M. Cooper 9th edition 2023The cell: A molecular approach. Oxford University Press
5. Gerald Karp: Cell and molecular Biology 9th Edition 2019 John Wiley and sons
6. Harper's Illustrated Biochemistry by R.K. Murray, P.A. Mayes, D.K. Granner and V.W. Rodwell, 31th Ed. 2018, Prentice Hall International Ltd. (U.K.).
7. Voet D and Voet JG. Biochemistry 5th Edition, John Wiley and Sons Inc., 2021.
8. Lehninger A.L., Nelson D.L. and Cox M.M. (2021)8th edition Principles of biochemistry (W.H. Freeman, USA)
9. StryerL., J.M. Berg, J.L. Tymoczko (2023)10th edition, Biochemistry (W.H. Freeman company, New York)

PRACTICALS:

(BCH-DSC-5-Maj/Min-401): Membrane Biochemistry & Neurobiology

Marks: 25 (20+5 Int.)

1. Demonstration of hypotonic, hypertonic and osmotic media.
2. Preparation of red cell ghost membrane.
3. Cell lysis by sonication.
4. Preparation of RBC Ghost Cell.
5. Isolation of Chloroplast from Spinach Leaves and Estimation of total Chlorophyll.
6. Extraction and Separation of Photosynthetic Pigments by Partition Chromatography (TLC)
7. Isolation of Mitochondria by cell fractionation method.

Major

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:

2. Examiner will set total of NINE questions comprising TWO questions from each unit and ONE compulsory question.
3. All questions carry equal marks.
4. The compulsory question will consists of 12 short questions of one mark each covering the whole syllabi.
5. Besides Question No. 1, all other questions may contains 2-3 sub-parts.
6. Questions should be uniformly spread over the entire syllabus.
7. The students are required to attempt FIVE questions in all, ONE question from each unit and the Compulsory question.

(BCH-DSC-4-Maj-402): Enzymologyand Bioenergetics (3+1)

Marks: 75 (60+15 Int.)

Objective: To learn principles of thermodynamics, nature of enzymes, their mode of action and enzyme kinetics.

Learning Outcomes:
Upon successful completion of the course, students will be able to

- Understand the concepts of enzymes, their mechanisms and kinetics of enzymatic reactions.
- Understand the mechanism of oxidative phosphorylation and ATP synthesis.
- They will learn to perform enzymology experiments and learn to calculate enzyme activityand kinetics.

SECTION-I

(Lectures: 10)

Introduction to enzymes.General characteristics of enzymes' system of nomenclature and classification of enzymes.Prosthetic groups.Holoenzyme, apoenzyme, cofactors, isoenzymes and their clinical significance and zymogens.Coenzymes (NAD/NADP, FMN/FAD, coenzyme A, TPP, pyridoxal phosphate, tetrahydrofolate) and their biochemical functions, assay of enzyme activity, units of enzyme activity. Active site of enzymes.

SECTION-II

(Lectures: 10)

Factors affecting velocity of enzyme catalyzed reactions: enzyme concentration, pH andtemperature.Michaelis-Menten equation.Determination of Km and its significance.Enzyme inhibition: Various types of enzyme inhibitions. Determination of Ki value.Enzyme inhibitorsand their importance.Introduction to multisubstrate enzyme, Allosteric enzymes properties andregulation.

SECTION-III

(Lectures: 10)

Enzymes as catalysts. Theories of enzymes catalysis: proximity and orientation effects, acid base catalysis, covalent catalysis. Role of metals in enzyme catalysis Monomeric enzymes, oligomeric enzymes &multienzyme complexes.Mechanism of action of chymotrypsin, Abzymes and Lysozyme.

SECTION-IV

(Lectures: 10)

Concepts of bioenergetics: principles of thermodynamics & their application in Biochemistry, concept of free energy. Biological oxidations.Redox potential. ATP and High energy phosphate compounds: introduction, phosphate group transfer potential, enzymes and co-enzymes involved in oxidations and reductions. Mitochondrial electron transport chain and oxidative phosphorylation.Mechanism of oxidative phosphorylation. Inhibitors and Uncouplers of Oxidative Phosphorylation

Suggested Books:

1. T. Palmer Enzymes: Biochemistry, Biotechnology and ClinicalChemistry, 2nd Ed. (2018) AffiliatedEast-West pressPvt. Ltd. New Delhi.
2. L.Stevens and Nicholas C.Price Fundamentals of Enzymology:The cell and molecular Biology of catalytic Proteins(2000) 3rd edition (Oxford University Press,USA)
3. Robert A.Copeland: A practical Introduction to Structure ,Mechanism and Data Analysis 2ndedition(John Wiley and Sons Inc.,Publications)
4. Harper's Illustrated Biochemistry by R.K. Murray, P.A. Mayes, D.K. Grannerand V.W. Rodwell, 31th Ed. 2018, Prentice Hall International Ltd. (U.K.).
5. Voet D and Voet JG. Biochemistry 5th Edition, John Wiley and Sons Inc., 2021.
6. LehningerA.L.,NelsonD.L.andCox M.M.(2021)8th edition Principles of biochemistry(W.H.Freeman,USA)
7. StryerL.,J.M.Berg,J.L.Tymoczko (2023)10th edition,Biochemistry(W.H.Freemancompany,New York)
8. Conn E.E.and Stumpf P.K.(2006)5th edition Outlines of Biochemistry,(John Wiley and Sons,USA)
9. R.H.Garett and C.M.Grisham (2023)7th Edition Biochemistry,(CengageLearning,US)

PRACTICALS:

(BCH-DSC-4-Maj-402): Enzymologyand Bioenergetics

Marks: 25 (20 +5 Int.)

1. Preparation of casein from milk
2. Determination of achromatic point of saliva
3. Assay of alkaline phosphatase activity.
4. Effect of pH on enzyme activity.
5. Effect of temperature on enzyme activity and determination of energy of activation.
6. Effect of substrate concentration on enzyme activity and determination of Km.
7. Effect of inhibitor on enzyme activity
8. Demonstration of potato polyphenol oxidase activity.

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BIOINFORMATICS

COURSE OUTLINE & SYLLABUS FOR BIOINFORMATICS

Students opting for Single major and Single minor(Table 1*)and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY(Table 2*)
B.Sc., B.Sc. Honours and B.Sc. Honours withResearch(1st Year Programme)

Sem	Level	Paper Code	Paper Name	Credits (Th+Pr)
I	L-100	BINF-DSC-1 (Maj/Min-101)	Bioinformatics-I	3+1=4
		BINF-MDC-1-101	Introduction to Bioinformatics-I	2+1=3
		BINF-SEC-1-101	Tools for Biological Data Analysis	0+3=3
II	L-100	BINF-DSC-2 (Maj/Min-201)	Bioinformatics-II	3+1=4
		BINF-MDC-2-201	Introduction to Bioinformatics-II	2+1=3
		BINF-SEC-2-201	Computational Tools for Small and Macromolecules	0+3=3
III	L-200	BINF-DSC-3-Min-301	Computational Methods for Protein Sequence and Structure Analysis	3+1=4
		BINF-DSC-4-Min-302	Programming Language in C	3+1=4
		BINF-MDC-3-301	Bioinformatics Methods for Biomolecule Analysis	2+1=3
		BINF-SEC-3-301	Artificial Intelligence Tools for Biological Data Analysis	0+3=3
IV	L-200	BINF-DSC-5-Min-401	Computational Methods for Gene and RNA Structure Analysis	3+1=4

Notes: -

10. MDC/IDC subject is offered by The Department of Bioinformatics for the students of allied/other streams. The students opting for Bioinformatics as DSC subject will select MDC/IDC from the list of courses offered by the allied/other departments.
11. Language paper chosen as MIL-1, MIL-2, MIL-3 and MIL-4 should be in the same subject.

SEMESTER - III

Paper: BINF-DSC-3-Min-301: COMPUTATIONAL METHODS FOR PROTEIN SEQUENCE AND STRUCTURE ANALYSIS

External Theory Marks: 60
Internal Theory Marks: 15
Total Theory Marks: 75
Exam Duration:3 Hours

Course Duration:

1 Semester = 15 weeks which will have 45 hours of Theory Lectures (3 hours/week)and 30 hours of practical classes (2 hours/week)
1 Credit = 15 hours of theory lectures or 30 hours of practical classes
Total Credits: 4 (3 Credits for Theory Lectures and 1 Credit for Practical)
Each credit is equal to 25 marks

Learning Outcomes:

Computational Methods for Protein Sequence and Structure Analysisis to equip students with computational tools and methodologies for analyzing protein sequences and predicting their structures and functions.

General Instruction:

- The question paper will have seven questions, and each question have 12 marks. The first question would be compulsory having sub-parts covering the entire syllabus in the form of short and objective type questions.
- The remaining six questions will be set out of the Units I and II consisting of three questions from each unit.
- A candidate is required to attempt five questions in all by selecting two questions, from each unit and the first compulsory question.

UNIT - I

Motif and Domains: Basic concept of Protein family, types of motifs and domains.

Sequence patterns and profiles: Basic concept and definition of sequence patterns and profiles, various types of pattern representations viz. consensus, regular expression (Prosite-type), PHI-BLAST.
Sequence profiles: PSSM, profile-based database searches using PSI-BLAST. Applications of Pattern and Profile.

Domaindatabases: CDD, SMART, InterPro, Prosite
Tools for searching pattern and profile: InterPro Scan and Scan Prosite and MEME.

UNIT – II

Secondary Structure Prediction methods:
Ab-initio based: Chou Fasman and GOR, Homology based and Neural Network (PSIPRED).

Tertiary Structure Prediction Methods: Homology Modeling, Threading/Fold Recognition, Ab-initio Methods, AI-Based Tertiary Structure Prediction (AlphaFold)

Structural Validation and Quality Assessment Tools: PROCHECK and Verify3D.

Recommended Books:

1. David W. Mount (2004). Bioinformatics: Sequence and Genome Analysis, 2nd Edition, Cold Spring Harbor Laboratory Press.
2. Richard Durbin, Sean R. Eddy, Anders Krogh, and Graeme Mitchison (1998). Biological Sequence Analysis: Probabilistic Models of Proteins and Nucleic Acids, Cambridge University Press.
3. Carl Branden and John Tooze (1999). Introduction to Protein Structure, 2nd Edition, Garland Science.
4. Jin Xiong (2006). Essential Bioinformatics, Cambridge University Press.
5. BharathRamsundar, Peter Eastman, Patrick Walters, and Vijay Pande (2019). Deep Learning for the Life Sciences: Applying Deep Learning to Genomics, Microscopy, Drug Discovery, and More, O'Reilly Media.
6. Tiago Antao (2018). Bioinformatics with Python Cookbook, 2nd Edition, PacktPublishing.

BINF-DSC-3-Min-301: COMPUTATIONAL METHODS FOR PROTEIN SEQUENCE AND STRUCTURE ANALYSIS – (PRACTICAL)

External Practical Marks: 20
Internal Practical Marks: 05
Total Practical Marks: 25

1. MEME for motif identification and Prosite for motif analysis.
2. Sequence patterns and profiles by PSI-BLAST and PHI-BLAST
3. Use InterPro and SMART to annotate protein domains.
4. Compare CHOU-FASMAN, GOR, and PSIPRED for a given protein sequence.
5. Perform homology modeling using SWISS-MODEL.
6. Structure prediction by Phyre2, I-TASSER and AlphaFold
7. Validate modeled structures using PROCHECK and Verify3D

BINF-DSC-4-Min-302: PROGRAMMING LANGUAGE IN C

External Theory Marks: 60
Internal Theory Marks: 15
Total Theory Marks: 75
Exam Duration:3 Hours

Course Duration:

1 Semester = 15 weeks which will have 45 hours of Theory Lectures (3 hours/week)and 30 hours of practical classes (2 hours/week)

1 Credit = 15 hours of theory lectures or 30 hours of practical classes

Total Credits: 4 (3 Credits for Theory Lectures and 1 Credit for Practical)
Each credit is equal to 25 marks

Learning Outcomes:

To introduce basic concepts of "C" Language, which is required to do programming and solve problems related to Bioinformatics.

General Instruction:

- The question paper will have seven questions, and each question have 12 marks. The first question would be compulsory having sub-parts covering the entire syllabus in the form of short and objective type questions.
- The remaining six questions will be set out of the Units I and II consisting of three questions from each unit.
- A candidate is required to attempt five questions in all by selecting two questions, from each unit and the first compulsory question.

UNIT - I

Introduction to 'C' Language: Character set, variable and Identifiers, Built-in data types, Variable Definition, Arithmetic operations and Expressions, Constants and Literals, Simple assignment statement. Basic input/output statement.

Conditional statement and Loops: Decision making, Conditions, Relational Operators. Logical connectives If Statement, If-else statement Loops, while loop, do while loop, for loop, Nested loops. Switch statement, structural programming Programs related to decision making statements, loops.

Array: One dimensional arrays, Array manipulation; Searching, Insertion, Deletion of an element from an array. Finding the largest /smallest element in an array; Two dimensional arrays, multi-dimensional array, Addition / Multiplication of two matrices, Transpose of a square matrix, Strings, operation on strings.

UNIT - II

Functions: Top-down approach and Bottom-up approach of problem solving, Modular programming and functions, Standard library of C functions, Prototype of a function: Formal parameter list, Return Type function Call, Block Structure, Passing arguments to a function: Call by reference, call by value, Recursive functions, Arrays as function arguments.

Structure and Unions: Structure variables, initialization, structure assignment, nested structure, structures and functions, structures and arrays, arrays of structures, structures containing arrays, unions.

Pointers: Address Operators, Pointer type declaration, pointer assignment, pointerinitialization, pointer arithmetic, functions and pointers, Arrays and Pointers, Pointer arrays, Dynamic memory allocation malloc() and calloc() and free().

File Processing: Concept of Files, Files operating in various modes and closing of a file, reading from a file, writing to a file.

Recommended books:

1. Byron Gottfried "Programming wit 'C' Second addition. Tata McGrawhill, (2000).
2. RG Dromey "How to solve it by computer", Prentice Hall of India, 7th Edition, (2001).
3. E. Balaguruswami "Programming with Ansi-C", Pearson Education, 1st Edition, (2002).
4. Kamthane "Programming with ANSI and Turbo C", Pearson Education, First Edition, (2002).
5. Venugopal and Prasad " Programming with C" Tata McGrawhill, 1st Edition, (1997).
6. BW Kemighan and DM Ritchie "The C Programming Language", Prentice Hall of India, Second edition, (2001).

BINF-DSC-4-Min-302: PROGRAMMING LANGUAGE IN C(PRACTICAL)

External Practical Marks:	20
Internal Practical Marks:	05
Total Practical Marks:	25

1. Writeaprogramtocalculateaprimenumber is evenandoddnumbers
2. Writeafunctiontoreadamatrixofsizemxnfromthekeyboard and rotate the array clock-wise.
3. Writeaprogramtoaddandsubtractatwo-threematrixsimultaneously
4. Writeaprogramtofind factorial of number using recursion.
5. ProgrammingproblemsonArray,PointerandFiles
6. Write a program to translate protein sequence using DNA.
7. Write a program to convert protein or DNA sequence into FASTA FORMAT
8. Write a program to cut a DNA using restriction enzyme like EcoRII.

SEMESTER - IV

BINF-DSC-5-Min-401: COMPUTATIONAL METHODS FOR GENE AND RNA STRUCTURE ANALYSIS

External Theory Marks: 60
Internal Theory Marks: 15
Total Theory Marks: 75
Exam Duration:3 Hours

Course Duration:

1 Semester = 15 weeks which will have 45 hours of Theory Lectures (3 hours/week)and 30 hours of practical classes (2 hours/week)

1 Credit = 15 hours of theory lectures or 30 hours of practical classes

Total Credits: 4 (3 Credits for Theory Lectures and 1 Credit for Practical)
Each credit is equal to 25 marks

Learning Outcomes:

Computational Methods for Gene and RNA Structure Analysis is to equip students with computational tools and methodologies for analyzing gene structure and predicting their structures and functions.

General Instruction:

- The question paper will have seven questions, and each question have 12 marks. The first question would be compulsory having sub-parts covering the entire syllabus in the form of short and objective type questions.
- The remaining six questions will be set out of the Units I and II consisting of three questions from each unit.
- A candidate is required to attempt five questions in all by selecting two questions, from each unit and the first compulsory question.

UNIT - I

Basic Concepts of Gene Structure: Exons, introns, promoters, enhancers, and untranslated regions (UTRs). Differences between prokaryotic and eukaryotic gene structures.

Nucleic acid sequence analysis: Reading frames, Concept of ORF, Translational andtranscriptional signals.

Computational methods for gene prediction: Ab initio methods: Markov Model, HMM, Artificial Neural Networks and Discriminant Analysis. Homology-based approaches, Consensus Based Programs. Performance Evaluation: Real vs Predicted gene, Sensitivity, Specificity.

Prediction tools: Genscan, GenMark and Grail.

UNIT – II

Promoter analysis: Promoter and Regulatory Elements in Prokaryotes and Eukaryotes. Predictions of promoters in prokaryotic and eukaryotic Genomes: Ab initio based and Phylogenetic Footprinting.

Basics of RNA structure: Primary, secondary, and tertiary structures of RNA.

RNA secondary structure prediction: RNA secondary structure thermodynamics. Ab-initio and Comparative methods.

Tools of RNA secondary structure prediction: Mfold, Sfold and Vienna RNA package.

Post transcriptional Gene regulation: miRNA and siRNA.

Recommended Books:

1. **Jin Xiong (2006).** Essential Bioinformatics, Cambridge University Press.
2. Bioinformatics: Sequences, structure and databanks by Des Higgins and WillieTaylor, Oxford University Press, (2000).
3. Bioinformatics by David Mount, Cold Spring Harbor Laboratory, (1989).
4. Bioinformatics: A practical guide to the analysis of genes and proteins. Ed. By Baxevanis, 3
5. Pevsner, Jonathan. ***Bioinformatics and Functional Genomics***. 3rd ed. Hoboken, NJ: Wiley-Blackwell, 2015

BINF-DSC-5-Min-401: COMPUTATIONAL METHODS FOR GENE AND RNA STRUCTURE ANALYSIS (PRACTICAL)

External Practical Marks: 20
Internal Practical Marks: 05
Total Practical Marks: 25

1. Gene Prediction Tools: GenMark, GenScan and GENEID.
2. Promoter 2.0 prediction server.
3. Tools of RNA secondary structure: Mfold
4. Transcription factor databases like Transfac.
5. miRDB.

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Biotechnology

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR(Table 1*)
and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY(Table 2*)
B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH

Table 1*						Table 2*					
Sem	Level	Course Code	Course Name	Credits (Th+Pr+Tut)	Total Credits	Sem	Level	Course Code	Course Name	Credits (Th+Pr+Tut)	Total Credits
III	L-200	BIOT-DSC-(MAJ/MIN)-3-301	Molecules of Life	3 +1	24	III	L-200	BIOT-DSC-3 (MAJ/MIN)-301	Molecules of Life	3 + 1	24
		BIOT-DSC-(MAJ/MIN)-4-302	Cell & Molecular Biology	3 +1				BIOT-DSC-4 (MAJ/MIN)-302	Cell & Molecular Biology	3 + 1	
		*MIN-B-3/ MIN-C-3		3 +1				Major		3+1	
		*MIN-B-4/ MIN-C-4		3 +1				Major		3+1	
		BIOT-MDC-3- 301	Bioanalytical Techniques	2 +1				MDC-3-301	MDC	2 + 1	
		**BIOT-SEC-3 - 301	Techniques in Biochemistry & Immunology	2+1				SEC-3-301	SEC	2+1	
		AEC-3-301	Eng-3/MIL-1	2				AEC-3-301	Eng-3/MIL -1	2	
IV	L-200	BIOT-DSC-(MAJ/MIN)-5-401	Basic Immunology	3 +1	24	IV	L-200	BIOT-DSC-5 (MAJ/MIN)-401	Basic Immunology	3 + 1	24
		BIOT-DSC-(MAJ)-6-402	Biochemical Metabolism	3 +1				Major		3+1	
		BIOT-DSC-(MAJ)-7-403	Fundamentals of Genetics	3 +1				Major		3+1	
		BIOT-DSC-(MAJ)-8-404	Biosafety & Bioethics	3 +1							
		* MIN-B-5/ MIN-C-5		3 +1				Major		3+1	
		BIOT-VAC-3-401	Value Added Course	2				Major		3+1	
		AEC-4-401	Eng-4/MIL-2	2				VAC-3-401	Value Added Course	2	
								AEC-4-401	Eng-4/MIL -2	2	

The student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secures 4 credits from summer internship

*Minor subjects to be selected from the three streams (Biochemistry, Microbiology & Bioinformatics as per annexure-I) However PU Colleges having other subjects from science faculty can also offer them as Minor.

**Candidate has to select all the SECs offered by the Department of Biotechnology in colleges as proposed and approved by BOS Biotechnology PU colleges.

Semester III

BIOT-DSC-(MAJ/MIN)-3-301: MOLECULES OF LIFE

Learning Outcomes: To make student conversant with the biochemical aspect of cell, chemical structure & functions of various biomolecules. The course encompasses the overall perspective on the major biomolecules namely carbohydrates, lipids, proteins and nucleic acids, their characteristic properties and organization in carrying out all cellular functions. Students will develop the practical skills related to estimation of proteins, carbohydrates, sugars and lipids.

Course Duration
45 hours of lectures for Theory + 30 hours of practical
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)

Credits : 4 credits (Each credit is equal to 25 marks)

External Theory	Internal Theory	External practical	Internal practical
68	07	22	03

Theory Paper: The exam of this course will be of 3 hours duration.
Instructions for paper setters and candidates:
The syllabus of this paper has been divided into FOUR units.
Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. The students are required to attempt one question from each unit.
Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each

UNIT – I

Water: Physicochemical properties of water, structure of liquid water. pH and buffers
Carbohydrates: Structure of important mono, di-, oligo- and polysaccharides, glycoproteins, peptidoglycan, glycolipids and lipopolysaccharides. Stereoisomerism of monosaccharides, Mutarotation and the anomeric forms of D-glucose. Storage polysaccharides and cell walls.Enantiomers and anomers.Acid mucopolysaccharides.Reaction of monosaccharides.

UNIT II

Lipids: Classification of lipids and fatty acids, general functions and structure of major lipid subclasses, acylglycerols, phosphoglycerols, phosphoglycerides, sphingolipids, glycosphingolipids and terpenes, sterols, steroids: Prostaglandins. Characteristics of fatty acids and fats (saponification, iodine, acid, acetyl and peroxide values).Rancidity of fats, refractive index, melting point and boiling point of fats and their physical properties.

UNIT – III

Proteins: Structure of amino acid
Structural organization of proteins (primary, secondary, tertiary and quaternary structure), Protein function.Forces stabilizing primary, secondary and tertiary structure.

UNIT IV

Nucleic Acids: Structure of purine and pyrimidine bases. Nucleosides and nucleotides. Different forms of DNA and RNA. Physical and chemical properties of nucleic acids.

SUGGESTED READING

1. Nelson, D. L., Lehninger, A. L., &Lehninger, C. M. M. (2017). Principles of biochemistry (7th ed.). New York: W.H Freeman & Co.
2. Stryer L, J. M. Berg, J.L. Tymoczko (2001). Biochemistry (W.H. Freeman and Company, New York).
3. Botham, K., Mayes, P., Murray, R. K., &Granner, D. K. (2006). Harper's illustrated biochemistry (27th ed.). New York: McGraw-Hill Companies.
4. Voet, D., &Voet, J. G. (2011). Biochemistry (4th ed.). Singapore: John Wiley & Sons Inc.
5. Berg, J. M., Tymoczko, J. L., &Stryer, L. (2012) Biochemistry (7th ed.). New York: W. H Freeman & Co.

BIOT-DSC-(MAJ/MIN)-3-301: MOLECULES OF LIFE(Practicals)

1. Preparation of physiological buffers.
2. Estimation of Glucose in given solution by DNSA method.
3. Estimation of carbohydrates in given solution by Anthrone method.
4. Quantification of nucleic acids by colorimetry (orcinol)
5. Estimation of protein using Lowry's method

BIOT-DSC-(MAJ/MIN)-4-302:CELL & MOLECULAR BIOLOGY

Learning Outcomes:

This paper will help understand the detailed structure of eukaryotic and prokaryotic cell and its inner components. It will impart an in depth knowledge andhelp understand the processes of cell transport and cell locomotion. This paper will communicate the knowledge of chromosomal structure and composition. Practical skills will be imparted to the students through critically designed practical related to the subjects.

Course Duration

45 hours of lectures for Theory + 30 hours of practical
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)

Credits : 4 credits (Each credit is equal to 25 marks

External Theory	Internal Theory	External practical	Internal practical
68	07	22	03

Theory Paper: The exam of this course will be of 3 hours duration.

Instructions for paper setters and candidates:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. The students are required to attempt one question from each unit.

Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each

Unit I

Cell as a basic unit of living systems: the cell theory, pre-cellular evolution, broadclassification and ultrastructure of cell types (PPLOs, Bacteria, eukaryotic microbes, plant and animal cells)
Structure and function of cell membrane,Cellulartransport:Passive&activetransport (sodium pump), Co-transport symport, antiport, endocytosisand exocytosis,entryof virusesandtoxinsintocells.

UnitII

Structure and function of cytosol, Golgibodies, vacuoles, endoplasmicreticulum(roughandsmooth), ribosomes, cytoskeletal structures (actin, microtubules), mitochondria, chloroplast, lysosomes, peroxisomes, nucleus(nuclear membrane, nucleoplasm, nucleolus).
Celllocomotion:Amoeboid,FlagellarandCilia.

Unit III

DNA and RNA: Chemical composition of prokaryotic and eukaryotic DNA and RNA, DNA structure-single stranded DNA, detailed account of double stranded DNA,A-DNA, B-DNA, Z-DNA and their importance.
Prokaryotic DNA replication: Initiation, Elongation and termination. DNA polymerases, composition and features, replication factors. Replication of single stranded DNA, M13 viral DNA.
Eukaryotic DNA replication: Initiation replication (initiation complexes and their assembly), Elongation and termination: DNA polymerases and their composition. Telomerase and mode of action.

UNIT-IV

Structural features of rRNAs- prokaryotic and eukaryotic.tRNAs: structural features, their anticodon feature. mRNAs: prokaryotic and eukaryotic mRNAs, modifications of eukaryotic mRNA, structural features.
Transcription: Initiation, Elongation and termination in prokaryotes and eukaryotes.
Gene Regulation in prokaryotes: Operon concept: Lactose and Tryptophan operon.
Characteristics of the genetic code, structure and charging of tRNA.
Translation: Overview and mechanism of translation process (Initiation, Elongation and termination) in prokaryotes and Eukaryotes.

Reference Books:

1. The World of the Cell (2008) Becker, Klein Smith & Hardin Pearson education Inc
2. Cell and Molecular Biology (2010) E.D.P De Robertis and E.M.F. De Robertis, Jr.
3. Molecular Cell Biology (2007) Lodishetal. Freeman & Co
4. Molecular Biology of the Cell (2007) Bruce Alberts, Alexander Johnson, Julian Lewis, Martin Raff, Keith Roberts, PeterWalter.
5. The Cell: A Molecular Approach (2013) Geoffrey M.Cooper and Robert E.Hausman
6. Cells and Cell Function: Advanced Level (2007) Sally Morgan
7. Lewin, B., 2004, Genes VIII, Oxford Univ. Press, Noida, India.
8. Ingraham, John.L.2004, Introduction to Microbiology, 3 Ed., Thomson brocks/Cole Inc.
9. Malacinski, George M., 2005, Freifelder's. Essentials of Molecular Biology, Narosa Publishing House, ND. Karp, Gerald, 2005, Cell and Molecular Biology, Wiley International, USA.
10. Snusted and Simmons, 2006, Principles of genetics, John Wiley and Sons, Inc Brown, T.A, 2007, Genomes 3, Garland Science.
11. Weaver R.F., Molecular biology (2005), McGraw
12. Molecular cloning: A laboratory manual Vol. I-III Sambrook J. and Russel, David N. 2001 Cold Spring Harbour.

BIOT-DSC-(MAJ/MIN)-4-302:CELL BIOLOGY& MOLECULAR BIOLOGY(Practicals)

1. To study the parts and function of a light microscope
2. To prepare a wet mount of onion peel for microscopy
3. Counting of cells using haemo-cytometer
4. To study cell locomotion of amoeba
5. Preparation of Reagents for DNA isolation.
6. Isolation of genomic DNA from Bacteria
7. Agarose gel electrophoresis of DNA.
8. Plasmid DNA isolation.

Semester IV

BIOT-DSC-(MAJ/MIN)-5-401: BASIC IMMUNOLOGY

Learning Outcomes: To understand general aspects of immune system like different components of the immune system, generation and functions of these components. It will impart the knowledge of basic immunological techniques. It will introduce the students to the defence mechanism of the body which protects the human beings from infections.

Course Duration

45 hours of lectures for Theory + 30 hours of practical
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)

Credits : 4 credits (Each credit is equal to 25 marks

External Theory	Internal Theory	External practical	Internal practical
68	07	22	03

Theory Paper: The exam of this course will be of 3 hours duration.

Instructions for paper setters and candidates:

The syllabus of this paper has been divided into FOUR units.
Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. The students are required to attempt one question from each unit.

Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each

Unit-I

Overviews of immune system – Historical perspective
Innate and acquired immune Response, Clonal nature of immune response.
Cells of the immune system: Hematopoiesis and differentiation, B-Lymphocytes, T-lymphocytes, macrophages, Eosinophils, Basophils, Neutrophils, dendritic cells, Natural killer cells, &Mast cells.
Lymphocyte Trafficking: Cell Adhesion molecules (Integrin, Selectin, Mucin family and Ig Superfamily).

Unit-II

Organs of the immune system: Primary and secondary lymphoid organs, Structural features and Functions.
Antigen – Immunogenicity Vs. antigenicity, factors effecting immunogeniticity, nature of immunogen, epitopes, Heptans and antigenicity, pattern recognition receptors.

Unit-III

B Cell Activation, Differentiation: B-Cell activation and proliferation, *In vivo* sites for induction of humoral responses, T-dependent and T-independent antigens.
Immunoglobulins: Structure of antibody, antibody effector function, antibody classes and biological activities, antigenic determinants on Immunoglobulins, Production of Monoclonal Antibodies, applications of polyclonal and monoclonal antibodies.

Unit-IV

Major histocompatibility complex: General organization and inheritance, MHC molecules and genes, genetic map, cellular distribution, regulation of MHC expression and disease susceptibility.
T-Cell Maturation, Activation, and Differentiation: T-Cell Receptors: Structure and Roles, T- Cell Receptor Complex, Thymic Selection of the T-Cell Repertoire, TH-Cell Activation, T-Cell Differentiation, Cell Death and T-Cell populations, Peripheral-T-Cells.

Reference Books:

1. Kuby Immunology (2006) by Thomas J. Kindt, Richard A. Goldsby, Barbara A. Osborne, Janis Kuby (W.H. Freeman).
2. Immunology- A short course (2009) by Richard Coico, Geoffrey Sunshine (Wiley Blackwell).
3. Fundamentals of immunology (1999) by William Paul (Lippincott Williams & Wilkins).
4. Immunology (2001) by Ivan Maurice Roitt, Jonathan Brostoff, David K. Male (Mosby).
5. Kindt TJ, Goldsby RA, Osborne BA, Kuby J. Immunology. (6th Ed.) WH Freeman & Co. New York; 2006.
6. Delves PJ, Martin SJ, Burton DR, Roitt IM. Roitt's essential immunology.WileyBlackwell; 2011

BIOT-DSC-(MAJ/MIN)-5-401: BASIC IMMUNOLOGY (Practicals)

1. Lymphoid organs and their microscopic organization.
2. Differential leucocytes count.
3. Separation of serum from blood.
4. Separation of plasma from blood
5. Agglutination (Blood group testing).
6. To isolate mononuclear cells from peripheral blood using ficollpaque.

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Botany

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR(Table 1*)

and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY(Table 2*)

B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH IN BOTANY

Table 1*						Table 2*					
Sem	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**
III	L-200	(BOT-DSC-3 Maj/Min-301)	Morphology of Angiosperms	3+1=4 (x2)	24	III	L-200	(BOT-DSC-3 Maj/Min-301)	Morphology of Angiosperms	3+1=4 (x3)	24
		(BOT-DSC-4 Maj/Min-302)	Cytogenetics	3+1=4 (x2)				(BOT-DSC-4 Min-302)	Cytogenetics	3+1=4	
		(BOT-MDC-3-301)	Natural Resource Management	3				(BOT-MDC-3-301)	Natural Resource Management	3	
		(BOT-SEC-3-301)	Organic Farming	2+1=3				(BOT-SEC-3-301)	Organic Farming	2+1=3	
		(AEC-3-301)	Eng-3/MIL-1	2				(AEC-3-301)	Eng-3/MIL-1	2	
IV	L-200	(BOT-DSC-5 Maj/Min-401)	Reproductive Biology of Angiosperms	3+1=4 (x2)	24	IV	L-200	(BOT-DSC-5 Maj/Min-401)	Reproductive Biology of Angiosperms	3+1=4 (x3)	24
		(BOT-DSC-6 Maj-402)	Phycology	3+1=4				(BOT-DSC-4 Maj-402)	Cytogenetics	3+1=4 (x2)	
		(BOT-DSC-7 Maj-403)	Mycology	3+1=4							
		(BOT-DSC-8 Maj-404)	Cell Biology	3+1=4							
		(BOT-VAC-3-401)	Plants and Human Welfare	2				(BOT-VAC-3-401)	Plants and Human Welfare	2	
		(AEC-4-401)	Eng-4/MIL-2	2				(AEC-4-401)	Eng-4/MIL-2	2	
The student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secures 4 credits from summer internship											

***Note: The candidate will opt for either Template 1 or 2 from Semester III onwards.**

Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

SEMESTER – III

Course Title: MORPHOLOGY OF ANGIOSPERMS **Course Code: BOT-DSC-3 Maj/Min-301**

Learning Outcome: This paper deals with the morphology and reproductive structures in flowering plant forms. The course work of this paper covers vegetative and reproductive morphology of these plants and will familiarize the students with plants bearing the enclosed seeds.

Max. Marks: 100 **Course Credits** = 3+1=4

1.	Theory: 75	Written Exam: 67	Internal Assessment: 8
2.	Practical: 25	Written Exam: 22	Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration (Two hours)

Instructions for Paper Setters:

Question Paper will have four sections. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 15 questions having 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teaching methodology includes series of lectures making use of charts, transparencies, LCD, Models, slides, practical demonstrations, extension lectures from experts, field visits, discussions, quiz competitions etc.

UNIT – I **12 Hours**

General characters of angiosperms, Diversity in plant forms: annuals, biennials and perennials. Diversity on the basis of size, habit and habitat. Monocots and dicots.

UNIT-II **11 Hours**

Root: Characteristics and functions. Tap root and adventitious root system and their various types; structural modifications for storage, respiration and reproduction

UNIT – I **11 Hours**

Stem: Characteristics and functions, branching pattern. Modifications of aerial, subaerial and underground stem.

UNIT-IV **11 Hours**

Leaf: Parts of Leaf, Venation, phyllotaxy, simple and compound leaves, functions and modifications of leaves.

Suggested Readings:

1. Bhojwani, S.S. and Bhatnagar, S.P. The Embryology of Angiosperms, 5th edition. Vikas Publishing House Pvt. Ltd., Delhi, 2009
2. Goyal J.P. and Saini, A. Angiosperms: Structure Development and Reproduction, Trueman Book Co. Jalandhar & New Delhi, 2006.
3. E.J.Eames. Morphology of Vascular Plants, Standard University Press.
4. Dutta A.C. 2016. Botany for Degree Students. Oxford University Press.

Suggested laboratory exercises:

1. To study the Modifications of root.
2. To study the modifications of stem.

3. To study the modifications of leaf.
4. To study phyllotaxy of leaf.
5. To prepare photo album on root/ stem/leaf.

GUIDELINES FOR PRACTICAL EXAMINATION

Time: 3 Hrs **Max. Marks: 25**
Practical: 22 **Int. Assessment:3**

1. Identify, write illustrated morphological note on Specimen A from root. (4)
2. Identify, write illustrated morphological note on specimen B from stem. (5)
3. Identify, write illustrated morphological note on specimen C from leaf. (4)
4. Practical note-book (3)
5. Viva-voce (3)
6. Photo album (3)

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SEMESTER – III

Course Title: CYTOGENETICS **Course Code: BOT-DSC-4 Maj/Min-302 / Maj-402**

Learning Outcome: This paper with the basic structural unit of life i.e. Cell & its organelles. It provides an insight into structural and cytological basis of functional differentiation in plants and it also deals with various aspects of hereditary trends observed in successive generations.

Max. Marks: 100 **Course Credits** = 3+1=4

1.	Theory: 75	Written Exam: 67	Internal Assessment: 8
2.	Practical: 25	Written Exam: 22	Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration (Two hours)

Instructions for Paper Setters:

Question Paper will have four sections. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY**question consisting of 15 short answers type questions comprising 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teaching methodology includes series of lectures, making use of charts, transparencies, LCD, Models, slides, practical demonstrations, extension lectures from experts, field visits, discussions, quiz competitions etc.

UNIT – I 12 Hours

Cell: Ultrastructure and functions of a typical plant cell and its organelles: Nucleus, Mitochondrion, Plastids, Ribosome, Endoplasmic reticulum, Golgi apparatus, Lysosomes; Structure and functions of cell wall and plasma membrane: fluid mosaic model only.

UNIT – II 11 Hours

Chromosomes: Physical structure of chromosome; Giant chromosomes: Polytene and Lampbrush chromosomes.
Cell division:mitosis and meiosis and its significance.

UNIT – III 11 Hours

Mendelism: Mendel's experiments and results, Mendel's Laws of Dominance, Segregation and Independent assortment; Linkage: complete and incomplete linkage, linkage groups, linkage maps, importance of linkage.

UNIT – IV 11 Hours

Genetic Code: Characteristics, exceptions, Wobble hypothesis; DNA: Structure (Watson and Crick model) and Replication. RNA: Structure and types, transcription, translation.

Suggested Readings:

1. Aggarwal SK (2009) Foundation Course in Biology, 2nd Edition, Ane Books Pvt. Ltd
2. Gardner, E.J., Simmons, M.J., Snustad, D.P. (1991). Principles of Genetics, John Wiley & sons, India. 8th e
3. Snustad, D.P. and Simmons, M.J. (2010). Principles of Genetics, John Wiley & Sons Inc., India.5th edition.
4. Karp, G. (2010). Cell Biology, John Wiley & Sons, U.S.A. 6th edition.
5. Cooper, G.M. and Hausman, R.E. (2009) The Cell: A Molecular Approach. 5th edition. ASM Press & Sunderland, Washington, D.C.; Sinauer Associates, MA.17
6. Becker, W.M., Kleinsmith, L.J., Hardin. J. and Bertoni, G. P. (2009) The World of the Cell. 7th edition. Pearson Benjamin Cummings Publishing, San Francisco
7. Gupta, P.K. (2010). Cytogenetics. Rastogi Publications, Meerut, India.
8. Majumdar, R., Sisodia, R. (2019). Laboratory Manual of Cell Biology, withreference to Plant Cells. New Delhi, Delhi: Prestige Publication

Suggested laboratory exercises:

1. Study of plant cell structure with the help of epidermal peel mount of *Allium/Rhoeo*.
2. Preparation of temporary slides to show different stages of mitosis from root tips of *Allium cepa/A. sativum*.
3. Preparation of temporary slides to show different stages of meiosis from floral buds of *Allium/Brassica*.
4. Study of cell and its organelles with the help of charts/ Photographs.

GUIDELINES FOR PRACTICAL EXAMINATION

Time: 3 Hrs **Max. Marks: 25**
Practical: 22 **Int. Assessment: 3**

1. Prepare a squash mount of specimen A to show the stage of cell division visible in the slide and show it to the examiner. Identify it giving at least one reason. Draw the stage of cell division and show it to the examiner. (06)
2. To prepare slide of peel mount specimen B and draw its diagram. (04)
3. Identify the photograph of specimen C and D (from cell organelles) and write a note on it along with well labeled diagram (06)
4. Viva-voce (03)
5. Practical Note-book. (03)

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SEMESTER – IV

Course Title: REPRODUCTIVE BIOLOGY OF ANGIOSPERMS **Course Code: BOT-DSC-5 Maj/Min-401**

Learning Outcome: This paper with the basic structural unit of life i.e. Cell & its organelles. It provides an insight into structural and cytological basis of functional differentiation in plants and it also deals with various aspects of hereditary trends observed in successive generations.

Max. Marks: 100 **Course Credits**= 3+1=4

1. Theory: 75 Written Exam: 67 Internal Assessment: 8
2. Practical: 25 Written Exam: 22 Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration (Two hours)

Instructions for Paper Setters:

Question Paper will have four sections. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 15 short answers type questions comprising 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teaching methodology includes series of lectures, making use of charts, transparencies, LCD, Models, slides, practical demonstrations, extension lectures from experts, field visits, discussions, quiz competitions etc.

UNIT – I 12 Hours

Flower: As a modified shoot, functions; Symmetry of flower, parts of flower, structure of anther and pistil; Inflorescence (Racemose, Cymose, Compound inflorescence and Special type)

UNIT – II 11 Hours

Pollination: Types of pollination, advantages and disadvantages of self and cross- pollination, Contrivances for self and cross- pollination, Various agencies to bring about cross pollination, characters of flowers pollinated by various agencies.

UNIT -III 11 Hours

Structure and development of male gametophyte, structure and development of female gametophyte (Polygonum type). Structure of dicot and monocot seed, non-endospermic and endospermic seed.

UNIT – IV 11 Hours

General account of asexual means of reproduction (natural with the help of roots, stem (underground, creeping and aerial), leaf, bulbils and turions and artificial means (cutting, layering, grafting, micropropagation). Significance of vegetative propagation.

Suggested Readings:

1. Bhojwani, S.S. and Bhatnagar, S.P. The Embryology of Angiosperms, 5th edition. Vikas Publishing House Pvt. Ltd., Delhi, 2009
2. Goyal J.P. and Saini, A. Angiosperms: Structure Development and Reproduction, Trueman Book Co. Jalandhar & New Delhi, 2006.
3. Singh, V., Pande, P. C. and Jain, D. K., (1997). Embryology of Angiosperms, Rastogi Publications, Meerut.
4. Johri, B. M., Srivastava, P. S., & Singh, N. (2001). Reproductive Biology of Angiosperms. Reproductive Biology of Plants, 237–272

Suggested laboratory exercises:

1. To study various types of inflorescences.
2. To study the dicot and monocot seeds.
3. To study the various forms of asexual means of reproduction (root/stem/leaf).
4. To study various parts of any flower (calyx, corolla, androecium, gynoecium).
5. To prepare project report on pollination types/ inflorescence.

GUIDELINES FOR PRACTICAL EXAMINATION

Time: 3 Hrs **Max. Marks: 25**
Practical : 22 **Int. Assessment: 3**

1. Identify, write illustrated morphological note on specimen A (Dicot/ Monocot seed). (4)
2. Identify, write illustrated morphological note on specimen B (related to asexual means of reproduction (root/stem/leaf). (5)
3. Write a morphological note on specimen C (inflorescence type). (4)
4. Practical note-book (3)
5. Viva-voce (3)
6. Project Report (3)

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SEMESTER – IV

Course Title: PHYCOLOGY **Course Code: BOT-DSC-6 Maj-402**

Learning Outcome: To impart knowledge about Algae on various aspects like Classification, Life cycles, Thallus organization and Reproduction in various groups; Evolution, Habitats and Economic importance of algae.

Max. Marks: 100 **Course Credits=** 3+1=4

1.	Theory: 75	Written Exam: 67	Internal Assessment: 8
2.	Practical: 25	Written Exam: 22	Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration (Two hours)

Instructions for Paper Setters:

Question Paper will have four sections. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 15 short answers type questions comprising 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teaching methodology includes series of lectures, making use of charts, transparencies, LCD, Models, slides, practical demonstrations, extension lectures from experts, field visits, discussions, quiz competitions etc.

UNIT-I 12 Hours

Criteria for algal classification (pigments, reserve food, flagella, chloroplasts, pyrenoids, eye spots, etc.)
Salient features: Cell structure, thallus organization, reproduction and broad classification of i) Cyanophyta ii) Xanthophyta iii) Bacillariophyta iv) Dinophyta.

UNIT-II 11 Hours

Salient features: Cell structure, thallus organization, reproduction and broad classification of v) Chlorophytavi) Phaeophyta vii) Cryptophyta and viii) Rhodophyta.

UNIT-III 11 Hours

Diversity in algal habitats (terrestrial, freshwater and marine).
Thallus organization in algae.
Reproduction (vegetative, asexual and sexual), life cycles.

UNIT-IV 11Hours

Rhythms and bioluminescence in dinoflagellates.
Economic importance of algae (algal biofertilizers, algal blooms, algae as food and feed, uses in industries etc.)
Algae in Biotechnology

SUGGESTED READINGS

1. Ahluwalia, A.S., (Ed.) Phycology: Principles, Processes and Applications, Daya Publishing House, New Delhi, 2003
2. Kumar, H.D., Introductory Phycology, East West Press, New Delhi, 1999
3. Lee, R.E. Phycology. 4th Ed. Cambridge University Press, London. 2008.

Suggested laboratory exercises:

1. Morpholgy and microscopic observation of following genera (through permanent slides/photographs/ Fresh material):
   Cyanophyta: Nostoc,
   Chlorophyta: Spirogyra
   Xanthophyta: Vaucheria.
   Phaeophyta: Dictyota
   Rhodophyta: Polysiphonia
2. Project report on economic importance of algae.

GUIDELINES FOR PRACTICAL EXAMINATION

Time: 3 Hrs **Max. Marks: 25**
Practical: 22 **Int. Assessment: 3**

1. Identify and write illustrated morphological note on specimen A and B. (8)
2. Identify the slides / photographs of specimen C and D giving at least two reasons for each. (6)
3. Project report (2)
4. Viva voce (3)
5. Practical Note Book. (3)

SEMESTER – IV

Course Title: MYCOLOGY **Course Code: BOT-DSC-7 Maj-403**

Learning Outcome: The objective of the paper is to impart knowledge about Classification, Mycelium, and Reproduction in various groups of Fungi.

Max. Marks: 100 **Course Credits** = 3+1=4

1.	Theory: 75	Written Exam: 67	Internal Assessment: 8
2.	Practical: 25	Written Exam: 22	Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration (Two hours)

Instructions for Paper Setters:

Question Paper will have four sections. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 15 short answers type questions comprising 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teaching methodology includes series of lectures, making use of charts, transparencies, LCD, Models, slides, practical demonstrations, extension lectures from experts, field visits, discussions, quiz competitions etc.

UNIT – I 10 Hours

Introduction to Fungi: Characteristics of Fungi; habitats; hyphal structure and their modifications; mode of nutrition; reproduction types and life cycle in Fungi.
Classification of Fungi (comparative statement) with recent trends

UNIT – II 11 Hours

General account (mainly structure, reproduction and life cycle) of the following:
Oomycota – *Phytopthora, Plasmopara*
Chytidriomycota- *Synchytriumendobioticum*; Zygomycota - *Mucor*
Ascomycota- *Clavicepsand**Morchella*

UNIT – III 11 Hours

General account (mainly structure, reproduction and life cycle) of the following:
Basidiomycota-*Puccinia*and *Agaricus*
Anamorphic fungi- *Alternaria*and*Cercospora*
Myxomycota-True slime moulds

UNIT – IV 13 Hours

Mycorrhiza: General characters, types and mode of nutrition, mycorrhizal application in agriculture and plant growth; economic importance to humans, ecosystem and environment; negative aspects of Fungi.

Suggested Readings:

1. Alexopolous, C.J., C.W. Mims and M. Blackwell. Introductory Mycology. 4 th Ed. John Wiley & Sons, New York. 880 pp. 2007.
2. Webster, J. and Webes R. Introduction to Fungi. Cambridge University Press, Cambridge 3rd Edition, 2007.
3. Bos, L. Introduction to Plant Virology. Longman, New York. 160 pp. 1992
4. 4.Bilgrami,K.S. and R.N.Verma. Physiology of Fungi.2 nd Ed. Vikas Publication House, New Delhi
5. Pelczar M.J. Jr., E.C.S.Chan and N.R.Krieg. Microbiology. 5th Ed. Tata McGraw Hill, New Delhi. 2007.
6. Kendrick, B. The Fifth Kingdom. Focus Publishing, Newburyport, M.A. U.S.A, 2000

Suggested laboratory exercises:

1. Study of morphological and reproductive structures of following genera :
   Oomycota – *Phytopthora, Plasmopara*
   Chytidriomycota- *Synchytriumendobioticum*
   Zygomycota - *Mucor*
   Ascomycota- *Claviceps, Morchella*
   Basidiomycota-*Puccinia, Agaricus*
   Anamorphic fungi- *Alternaria,Cercospora*

2. Project report of Mycorrhizal application in agriculture and plant growth

GUIDELINES FOR PRACTICAL EXAMINATION

Time: 3 Hrs **Max. Marks: 25**
Practical:22 **Int. Assessment: 3**

1. Identify and write illustrated morphological note on specimen A and B. (08)
2. Identify the photograph or slide of specimen C and D giving at least two reasons. (06)
3. Viva-voce (03)
4. Practical Note-book. (03)
5. Project report (02)

SEMESTER – IV

Course Title: CELL BIOLOGY **Course Code: BOT-DSC-8 Maj-404**

Learning Outcome: This paper with the basic structural unit of life i.e. Cell & its organelles. It provides an insight into structural and cytological basis of functional differentiation in plants and it also deals with various aspects of hereditary trends observed in successive generations.

Max. Marks: 100 **Course Credits** = 3+1=4

1. Theory: 75 Written Exam: 67 Internal Assessment: 8
2. Practical: 25 Written Exam: 22 Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration (Two hours)

Instructions for Paper Setters:

Question Paper will have four sections. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 15 short answers type questions comprising 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teaching methodology includes series of lectures, making use of charts, transparencies, LCD, Models, slides, practical demonstrations, extension lectures from experts, field visits, discussions, quiz competitions etc.

UNIT – I 12 Hours

Cell: Introduction to cell theory**,** Structure of prokaryotic and eukaryotic cell, structure of bacterial cell.

Structure and biological roles of carbohydrates, lipids, proteins and nucleic acids, ATP an energy currency molecule**.**

UNIT – II 11 Hours

Chromosomal alterations: Deletion, Duplication, Inversion, Translocation and their importance. Variations in chromosome number (aneuploidy and polyploidy) and their importance.

UNIT – III 11 Hours

Non allelic gene interactions: Dominant and recessive epitasis, supplementary genes, complementary genes, quantitative or polygenic inheritance, duplicate genes. Allelic gene interactions: Incomplete dominance, co dominance, multiple alleles, pleiotropic genes.

UNIT – IV 11 Hours

Sex linked Inheritance: Characteristics and examples (haemophilia, color blindness), cytoplasmic or extra nuclear inheritance: Mitochondrial and plastid DNA: Plastid Inheritance in *Mirabilis*, Mitochondrial inheritance in Yeast.

Suggested Readings:

1. Karp, G. (2009). Cell and molecular biology: concepts and experiments. John Wiley & Sons.
2. Singh, S. P., &Tomar, B. S. (2008). Cell biology. Rastogi Publications, Meerut, India.

3. Renu Gupta, SeemaMakhija and Ravi Toteja (2018). Cell Biology Practical Manual,Prestige Publishers, New Delhi
4. Iwasa, J, Marshall, W. (2020). Karps's Cell Biology, 9th edition, New Jersey,U.S.A.: John Wiley & Son
5. Majumdar, R., Sisodia, R. (2019). Laboratory Manual of Cell Biology, withreference to Plant Cells. New Delhi, Delhi: Prestige Publication.
6. Nelson, D.L., Cox, M.M. (2021). Lehninger Principles of Biochemistry, 8thedition. New York, NY: W.H. Freeman and Company.
7. Reven, F.H., Evert, R.F., Eichhorn, S.E. (1992). Biology of Plants. New York,NY: W.H.Freeman and Company.

Suggested laboratory exercises:

1. Study of bacterial cell structure with the help of slide/ photograph/ chart.
2. Study of prokaryotic cell and eukaryotic cell with the help of slide/ photograph/ chart.
3. Problems related to Gene Interactions.
4. Demonstration of the phenomenon of plasmolysis and deplasmolysis
5. Microchemical tests for carbohydrates (reducing, non-reducing sugars and starch).
6. Project report on sex-linked inheritance

GUIDELINES FOR PRACTICAL EXAMINATION

Time: 3 Hrs **Max. Marks: 25**
Practical: 22 **Int. Assessment: 3**

1. Problem related to gene interaction (to be announced by the examiner). (06)
2. Identify the photograph or slide of specimen A (bacterial/ prokaryotic/ eukaryotic cell) and write a note on it along with a well-labeled diagram. (04)
3. Write procedure of plasmolysis and de-plasmolysis or micro chemical tests (to be announced by the examiner). (04)
4. Viva-voce (03)
5. Practical Note-book. (03)
6. Project report (02)

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Chemistry

CURRICULUM AND CREDIT DISTRIBUTION FOR CHEMISTRY AS MAJOR SUBJECT AS PER TABLE 1 AND TABLE 2

Table 1						Table 2					
Sem	**Level**	**Course Code**	**Course Name**	**Credits**	**Total Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits**	**Total Credits**
III	L-200	(CHM-DSC-3 MAJ/MIN-301)	GENERAL CHEMISTRY – 3	4	24	III	L-200	(CHM-DSC-3 MAJ/MIN-301)	GENERAL CHEMISTRY – 3	4	24
		(CHM-DSC-4 MAJ-302)	BASIC CONCEPTS IN CHEMISTRY - 1	4				Major - 3	To be chosen by the student	4	
		Minor – 3	To be chosen by the student	4				Minor – 3	To be chosen by the student	4	
		Minor - 4	To be chosen by the student	4				Minor - 4	To be chosen by the student	4	
		IDC-3	To be chosen from the available options	3				IDC-3	To be chosen from the available options	3	
		(CHM-SEC-3-301)	Hands on chemistry Lab skills-3/ to be chosen from available options	3				(CHM-SEC-3-301)	Hands on chemistry Lab skills-3/ to be chosen from available options	3	
		ENG-3	Compulsory	2				ENG-3	Compulsory	2	
IV	L-200	(CHM-DSC-5 MAJ-401/MIN-302)	GENERAL CHEMISTRY – 4	4	24	IV	L-200	(CHM-DSC-5 MAJ-401/MIN-302)	GENERAL CHEMISTRY – 4	4	24
		(CHM-DSC-6 MAJ-402/MIN-401)	GENERAL CHEMISTRY – 5	4				(CHM-DSC-6 MAJ-402/MIN-401)	GENERAL CHEMISTRY – 5	4	
		(CHM-DSC-7 MAJ-403)	BASIC CONCEPTS IN CHEMISTRY - 2	4				Major - 5	To be chosen by the student	4	
		(CHM-DSC-8 MAJ-404)	BASIC CONCEPTS IN CHEMISTRY - 3	4				Major - 6	To be chosen by the student	4	
		Minor - 5	To be chosen by the student	4				Minor - 5	To be chosen by the student	4	
		ENG-4	Compulsory	2				ENG-4	Compulsory	2	
		VAC-3	To be chosen from the common basket	2				VAC-3	To be chosen from the common basket	2	

CURRICULUM AND CREDIT DISTRIBUTION FOR CHEMISTRY AS MINOR SUBJECT AS PER TABLE 1 AND TABLE 2

Table 1					Table 2				
Sem	**Level**	**Course Code**	**Course Name**	**Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits**
III	L-200	(CHM-DSC-3 MAJ/MIN-301)	GENERAL CHEMISTRY – 3	4	III	L-200	(CHM-DSC-3 MAJ/MIN-301)	GENERAL CHEMISTRY – 3	4
	L-200	(CHM-DSC-5-MIN-302/MAJ-401)	GENERAL CHEMISTRY – 4	4		L-200	(CHM-DSC-5-MIN-302/MAJ-401)	GENERAL CHEMISTRY – 4	4
IV	L-200	(CHM-DSC-6 MIN-401/MAJ-402)	GENERAL CHEMISTRY – 5	4	IV	L-200	(CHM-DSC-6 MIN-401/MAJ-402)	GENERAL CHEMISTRY – 5	4

CURRICULUM AND CREDIT DISTRIBUTION FOR CHEMISTRY AS IDC AS PER TABLE 1AND TABLE 2

Table 1					Table 2				
Sem	**Level**	**Course Code**	**Course Name**	**Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits**
I	L-100	(CHM-IDC-1-101)	Basic Analytical Methods-1	3	I	L-100	(CHM-IDC-1-101)	Basic Analytical Methods-1	3
II	L-100	(CHM-IDC-2-201)	Basic Analytical Methods-2	3	II	L-100	(CHM-IDC-2-201)	Basic Analytical Methods-2	3
III	L-200	(CHM-IDC-3-301)	Colours and Dyes	3	III	L-200	(CHM-IDC-3-301)	Colours and Dyes	3

CURRICULUM AND CREDIT DISTRIBUTION FOR CHEMISTRY AS VAC AS PER TABLE 1 AND TABLE 2									
Table 1					Table 2				
Sem	**Level**	**Course Code**	**Course Name**	**Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits**
I	L-100	(CHM-VAC-1-101)	Understanding Science-I	2	I	L-100	(CHM-VAC-1-101)	Understanding Science-I	2
II	L-100	(CHM-VAC-2-201)	Understanding Science-II	2	II	L-100	(CHM-VAC -2-201)	Understanding Science-II	2
IV	L-200	(CHM-VAC-3-301)	Food poisons	2	IV	L-200	(CHM-VAC -3-301)	Food poisons	2

Note:-

1. In the first semester, studentsare offered three different subjects: A, B & C.
2. At the end of the second semester, each student has the option to choose any one of the two streams, namely Single Major or Multi-Disciplinary. The student choosing the Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student choosing the Multi-Disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in the Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for a 4-year degree B.SC HONOURS with research, on the condition that the Institution is a PU-approved Research Centre
5. The student opting for a Single Major must secure 50% of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for a Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year, securing 48 credits, will be awarded a UG Certificate provided he/she secure 4 credits from summer internship/community work. Further, the student exiting the programme after two years, securing 96 credits, will be awarded a UG Diploma provided he/she secure 4 credits from a summer internship. The student successfully securing 144 credits will be awarded a BA/BSc degree.
7. Language papers chosen as MIL-1 and MIL-2 should be on the same subject.
8. Once Major and Minor papers are chosen in the 3rd semester, the same will continue as Major and Minor subjects in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write a Dissertation of 08 credits in one of the two Major subjects.

Semester	**III**
Code of Paper	**CHM-DSC-3-MAJ/MIN-301**
Title of Paper	**General Chemistry-3**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3Hrs. per week** **Practical: 2 Hrs. per week**

Applicable to the students

1. Those who have opted for chemistry as a Major subject in Table 1 or Table 2
2. Those who have opted for chemistry as a Minor subject in Table 1 or Table 2

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The students will learn the following-

1. Important aspects of s-block elements and noble gases
2. Chemistry of compounds of alkali metals, alkaline earth metals and noble gases
3. Important aspects of Alkenes, Cycloalkenes, alkynes and dienes
4. Chemical reactions of alkenes, alkynes, cycloalkanes, and dienes along with their mechansims.

GENERIC LEARNING OUTCOMES

1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,** Ethical and Professional Responsibility, **Application based Knowledge**
2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.

Semester – 3
Syllabus for Theory Paper
CHM-DSC-3-MAJ/MIN-301
General Chemistry-3

End semester = 67	**Credits - 03**
Internal Marks = 08	**Contact Hours – 3/week**
Examination Time allowed	**3 Hours**

UNIT I: INORGANIC CHEMISTRY (15 Hrs.)

Chemistry of s-Block Elements (7 Hrs.)

Comparative study, diagonal relationships, salient features of hydrides, solvation and complexation tendencies, including their function in biosystems, and an introduction to alkyls and aryls.

Chemistry of Noble Gases (8 Hrs.)

Occurrence and uses, rationalisation of inertness of noble gases, Clathrates; preparation and properties of XeF_2, XeF_4 and XeF_6; Nature of bonding in noble gas compounds (Valence bond treatment and MO treatment for XeF_2), Molecular shapes of noble gas compounds (VSEPR theory).

UNIT II: ORGANIC CHEMISTRY(15 Hrs.)

Alkenes andCycloalkenes (8 Hrs.)

Nomenclature of alkenes, methods of formation, mechanisms of dehydration of alcohols and dehydrohalogenation of alkyl halides, regioselectivity in alcohol dehydration.Saytzeff's Rule, Hofmann elimination, physical properties and relative stabilities of alkenes, Chemical reactions of alkenes–mechanisms involved in hydrogenation, electrophilic and free radical additions, Markownikoff's rule, hydroboration-oxidation, oxymercuration-reduction. Epoxidation, ozonolysis, hydration, hydroxylation and oxidation with $KMnO_4$, Polymerisation of alkenes, Substitution at the allylic and vinylic positions of alkenes, Methods of formation, conformation and chemical reactions of cycloalkenes.

DienesandAlkynes (7 Hrs.)

Nomenclature and classification of dienes: Isolated, conjugated and cumulated dienes. Structure of allenes and butadiene, methods of formation, and polymerisation. Chemical reactions – 1,2 and 1,4 additions, Diels-Alder reaction.Nomenclature, structure and bonding in alkynes. Methods of formation.Chemical reactions of alkynes, acidity of alkynes.Mechanism of electrophilic and nucleophilic addition reactions, hydroboration-oxidation, metal-ammonia reductions, oxidation and polymerisation.

UNIT-III PHYSICAL CHEMISTRY (15 Hrs.)

Thermochemistry (7 Hrs.)

Standard state, standard enthalpy of formation-Hess's Law of constant Heat Summation and its applications. Heat of reaction at constant pressure and at constant volume.Enthalpy of neutralisation.Bond dissociation energy and its calculation from thermochemical data, temperature dependence of enthalpy.Kirchhoff's equation.

Thermodynamics-II (8 Hrs.)

Second Law of Thermodynamics: Need for the law, different statements of the law, Carnot cycle and its efficiency, Carnot theorem. Thermodynamic scale of temperature, Concept of Entropy: Entropy as a state function, entropy as a function of V & T, entropy as a function of P & T, entropy change in physical change, Clausius inequality, entropy as a criterion of spontaneity and equilibrium. Entropy change in ideal gases and mixing of gases.

Books/Literature suggested

1. Cotton,F.A.,Wilkinson,G.,Gaus,P.L.,BasicInorganicChemistry;2nd edition,Pubs:JohnWileyand Sons, 1995.
2. Lee,J.D.,ConciseInorganicChemistry;4th edition,Pubs:Chapman&HallLtd.,1991.
3. Shriver,D.E.,Atkins,P.W.,InorganicChemistry;4th edition,Pubs:OxfordUniversityPress,2006.
4. Douglas, B., Medaniel, D., Atenander, J., Concepts and Models of Inorganic Chemistry; 3rd edition, Pubs: John Wiley and Sons Inc., 1994,
5. Porterfield,W.W.,Wesky,A.,InorganicChemistry;Pubs:Addison-WesleyPublishingCompany,1984.
6. Miessler,G.L.,Tarr,D.A.,InorganicChemistry;3rd edition,Pubs:PearsonEducationInc.,2004.
7. Puri,B.R.,Sharma,L.R.,Kalia,K.K.,PrinciplesofInorganicChemistry;30th edition,Pubs:Mileston es Publisher, 2006-07
8. Atkins, P., Paula, J.de, Atkins Physical Chemistry; 8th edition, Pubs: Oxford University Press, 2008.

9. Puri, B.R., Sharma, L.R., Pathania, M.S., Principles of Physical Chemistry; 43rd edition, Pubs: Vishal Publishing Co., 2008.
10. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw-Hill Inc, 1996.
11. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
12. Albert, R.A., Silbey, R.J., Physical Chemistry; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
13. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs:Wiley Eastern Limited, 1991.
14. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw-Hill Publishing Co. Ltd., 2002.
15. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
16. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book Company, 1989.
17. Morrison,R.T.,Boyd,R.N.,OrganicChemistry;6th edition,Pubs:Prentice-Hall,1992.
18. Solomons,T.W.,Fryhle,C.B.,OrganicChemistry;9th edition,Pubs:WileyIndia,2007.
19. WadeJr.,L.G.,Singh,M.S.,OrganicChemistry;6th edition,Pubs:PearsonEducation,2008.
20. Mukherji,S.M.,Singh,S.P.,Kapoor,R.P.,OrganicChemistry;Pubs:NewAgeInternational,1985.
21. Carey,F.A.,Sundberg,R.J.,AdvancedOrganicChemistryPartB:ReactionsandSynthesis;5th editio n, Pubs: Springer, 2007

Instructions for the examiner

1. The examiner will set the question paper comprising of **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:**the compulsory question and one question each from the UNIT-1, UNIT-2 and UNIT-3.
2. Theuse of scientific calculators is not allowed.

Semester – 3
Syllabus for Practical Paper
CHM-DSC-3-MAJ/MIN-301
General Chemistry-3

End semester = 22	**Credit - 01**
Internal Marks = 03	**Contact Hours – 2/week**
Examination Time allowed	**3 Hours**

I. INORGANIC CHEMISTRY

1. Gravimetric analysis
    a. Ni^{2+} as Ni-DMG complex
    b. Cu^{2+} as copper(I)-thiocyanate
    c. Ba^{2+} as barium(II) sulphate

II. ORGANIC CHEMISTRY

2. Qualitative Analysis:

    a. Investigation and characterisation of simple organic compounds for the detection of elements (N, S and halogens) present in the given organic compound. (Alcohols, carboxylic acids, carbonyl compounds, alkanes, amines, nitro, thio- and halo-compounds) and Identification of an unknown organic compound.

III. PHYSICAL CHEMISTRY

3. Thermochemistry

    a. To determine the solubility of benzoic acid at different temperatures and to determine the ph of the dissolution process.

    b. To determine the enthalpy of neutralisation of a weak acid/weak base versus strong base/strong acid, and determine the enthalpy of ionisation of the weak acid/weak base.

Books/Literature suggested

1. Practical Organic Chemistry, 4TH EDN, F G Mann & B C Saunders,Cambridge. March 1973
2. Vogel's Textbook of Practical Organic Chemistry, 5th Ed., Brian S Furniss, Anthony J Hannaford, Peter W G Smith & Austin R Tatchell,
3. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
4. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.
5. Levitt, B.P., Findlay's Practical Physical Chemistry; 8th edition, Pubs: Longman Group Ltd., London & New York, 1978.

Semester	**III**
Code of Paper	**CHM-DSC-4-MAJ-302**
Title of Paper	**Basic Concepts in Chemistry-1**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week** **Practical: 2 Hrs. per week**
Applicable to the students 1. Those who have opted for chemistry as a Major subject in Table-2	
LEARNING OUTCOMES SPECIFIC TO DISCIPLINE The students will learn the following- 1. Important aspects and principles of acids and bases 2. Chemistry of carbohydrates 3. Basic concepts of quantum chemistry	
GENERIC LEARNING OUTCOMES 1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,** Ethical and Professional Responsibility, **Application based Knowledge.** 2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.	

Semester – 3
Syllabus for Theory Paper
CHM-DSC-4-MAJ-302
Basic Concepts in Chemistry-1

End semester = 67 **Credits - 03**
Internal Marks = 08 **Contact Hours – 3/week**
Examination Time allowed **3 Hours**

UNIT I: INORGANIC CHEMISTRY (15 Hrs.)

Non-aqueous Solvents (5 Hrs.)

Introduction to non-aqueous solvents, their classification, Physical properties of a solvent, types of solvents and their general characteristics, Solvent system concept and its failure, reactions in non-aqueous solvents with reference to liquid NH_3, liquid SO_2, HF and H_2SO_4.

Acids and Bases (5 Hrs.)

Introduction, Arrhenius concept, Bronsted-Lowry, the Lux-Flood, solvent system and Lewis concept of acids and bases.

Hard and Soft Acids and Bases (HSAB) (5 Hrs.)

Classification of acids and bases as hard and soft, Pearson's HSAB concept, acid-base strength and hardness and softness, Symbiosis, theoretical basis of hardness and softness, electronegativity and hardness and softness, Limitation of HSAB principle

UNIT II: ORGANIC CHEMISTRY (15 Hrs.)

Carbohydrates(9 Hrs.)

Carbohydrate Introduction: brief descriptions (structure and properties) of: (i) Monosaccharides: glucose, fructose, and galactose, (ii) Disaccharides: sucrose, lactose, and maltose(iii) Polysaccharides: starch, cellulose, and glycogen. Monosaccharides, mechanism of osazone formation, interconversion of glucose and fructose, chain lengthening and chain shortening of aldoses. Configuration of monosaccharides. Erythro and threodiastereomers. Conversion of glucose in tomannose. Formation of glycosides, ethersandesters. Determination of ring size of monosaccharides. Structure of D (+)-glucose.Mechanism of mutarotation.

Structureofriboseanddeoxyribose(6 Hrs.)

Concept of reducing and non-reducing sugars, analysis of carbohydrates (Molisch test, Benedict's test, and Barfoed's test); carbohydrate metabolism, glycemic index, common diseases related to carbohydrate metabolism (diabetes, galactosemia, lactose intolerance, etc.).

UNIT II: PHYSICAL CHEMISTRY (15 Hrs.)

Elementary Quantum Mechanics-I (15 Hrs.)

Black-body radiation, Planck's radiation law, photoelectric effect, heat capacity of solids, Bohr's model of the hydrogen atom (no derivation) and its defects, Compton effect.De Broglie hypothesis, the Heisenberg's uncertainty principle, Sinusoidal wave equation, Hamiltonian operator, Schrodinger wave equation and its importance, physical interpretation of the wave function, postulates of quantum mechanics, particle in a one dimensional box,Schrodinger wave equation for H-atom, separation into three equations (without derivation), quantum numbers and their importance, hydrogen like wave functions, radial wave functions, angular wave functions.

Books/Literature suggested

1. Cotton,F.A.,Wilkinson,G.,Gaus,P.L.,BasicInorganicChemistry;2nd edition,Pubs:JohnWileyand Sons, 1995.
2. Lee,J.D.,ConciseInorganicChemistry;4th edition,Pubs:Chapman&HallLtd.,1991.
3. Shriver,D.E.,Atkins,P.W.,InorganicChemistry;4th edition,Pubs:OxfordUniversityPress,2006.
4. Douglas, B., Medaniel, D., Atenander, J., Concepts and Models of Inorganic Chemistry; 3rd edition, Pubs: John Wiley and Sons Inc., 1994,
5. Porterfield,W.W.,Wesky,A.,InorganicChemistry;Pubs:Addison-WesleyPublishingCompany,1984.
6. Miessler,G.L.,Tarr,D.A.,InorganicChemistry;3rd edition,Pubs:PearsonEducationInc.,2004.
7. Puri,B.R.,Sharma,L.R.,Kalia,K.K.,PrinciplesofInorganicChemistry;30th edition,Pubs:Mileston es Publisher, 2006-07
8. Atkins, P., Paula, J.de, Atkins Physical Chemistry; 8th edition, Pubs: Oxford University Press, 2008.
9. Puri, B.R., Sharma, L.R., Pathania, M.S., Principles of Physical Chemistry; 43rd edition, Pubs: Vishal Publishing Co., 2008.
10. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw-Hill Inc, 1996.
11. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
12. Albert, R.A., Silbey, R.J., Physical Chemistry; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
13. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs:Wiley Eastern Limited, 1991.

14. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw-Hill Publishing Co. Ltd., 2002.
15. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
16. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book Company, 1989.
17. Morrison, R.T., Boyd, R.N., Organic Chemistry; 6th edition, Pubs: Prentice-Hall, 1992.
18. Solomons, T.W., Fryhle, C.B., Organic Chemistry; 9th edition, Pubs: Wiley India, 2007.
19. Wade Jr., L.G., Singh, M.S., Organic Chemistry; 6th edition, Pubs: Pearson Education, 2008.
20. Mukherji, S.M., Singh, S.P., Kapoor, R.P., Organic Chemistry; Pubs: New Age International, 1985.
21. Carey, F.A., Sundberg, R.J., Advanced Organic Chemistry Part B: Reactions and Synthesis; 5th edition, Pubs: Springer, 2007

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions:**9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:**the**compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 3
Syllabus for Practical Paper
CHM-DSC-4-MAJ-302
Basic Concepts in Chemistry-1

End semester = 22	**Credit - 01**
Internal Marks = 03	**Contact Hours – 2/week**
Examination Time allowed	**3 Hours**

I. INORGANIC CHEMISTRY

1. Preparation of the following compounds
   a. Cuprous Chloride,Cu_2Cl_2
   b. Manganese(III) phosphate, $MnPO_4.H_2O$
   c. Turnbull's Blue and Prussian Blue

II.ORGANIC CHEMISTRY

2. Qualitative Analysis:

   a. Investigation, characterisation and identification of simple carbohydrates. Analysis of carbohydrates (Molisch test, Benedict's test, Fehling's test and Barfoed's test).

   b. Separation of a mixture of 2 sugars by ascending paper chromatography

III. PHYSICAL CHEMISTRY

3. pHmetry

   a. Preparation of buffer solutions of different pH: i. Sodium acetate-acetic acid, ii. Ammonium chloride-ammonium hydroxide

   b. pH metric titration of (i) strong acid vs. strong base, (ii) weak acid vs. strong base.

   c. Determination of the dissociation constant of a weak acid.

Books/Literature suggested

1. Jolly,W.L., ModernInorganicChemistry;2nd edition,Pubs:TataMcGraw-HillPublishingCompany Limited, 1991.
2. Practical Organic Chemistry, 4TH EDN, F G Mann & B C Saunders,Cambridge. March 1973
3. Vogel's Textbook of Practical Organic Chemistry, 5th Ed., Brian S Furniss, Anthony J Hannaford, Peter W G Smith & Austin R Tatchell,
4. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
5. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.
6. Levitt, B.P., Findlay's Practical Physical Chemistry; 8th edition, Pubs: Longman Group Ltd., London & New York, 1978.

Semester	**III**
Code of Paper	**CHM-DSC-5-MIN-302/MAJ-401**
Title of Paper	**General Chemistry-4**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week** **Practical: 2 Hrs. per week**
Applicable to the students 1. Those who have opted for chemistry as a Minor subject in Table 2	
LEARNING OUTCOMES SPECIFIC TO DISCIPLINE The students will learn the following- 1. Important aspects group 13 and 14 elements 2. Chemistry of compounds of B and C and their group elements 3. Chemistry of arenes, and alkyl ad aryl halides 4. Concept of aromaticity 5. Concepts of thermodynamics and chemical equilibrium	
GENERIC LEARNING OUTCOMES 1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,** Ethical and Professional Responsibility, **Application based Knowledge.** 2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.	

Semester – 3
Syllabus for Theory Paper
CHM-DSC-5-MIN-302/MAJ-401
General Chemistry-4

End semester = 67 **Credits - 03**
Internal Marks = 08 **Contact Hours – 3/week**
Examination Time allowed **3 Hours**

UNIT I: INORGANIC CHEMISTRY (15 Hrs.)

Group 13 Elements (8 Hrs)

Comparative study (including diagonal relationship) of groups 13 elements, Study of the following compounds with emphasis on structure, bonding, preparation, properties and uses: Boric acid and borates, boron nitrides, boron hydrides (diborane), boron halides, alumina.

Group 14 Elements (7 Hrs)

Comparative study (including diagonal relationship) of groups 14 elements, , allotropic forms of carbon, graphitic compounds, compounds like hydrides, oxides, oxyacids and halides of groups 14, fullerenes, carbides, fluorocarbons.

UNIT II: ORGANIC CHEMISTRY (15 Hrs.)

Aromaticity and Arenes(8 Hrs.)

Aromaticity: The Huckel's rule, aromatic ions.

Nomenclature of benzene derivatives, aryl group, aromatic nucleus and side chain, Structure of benzene: Molecular formula and Kekule structure. Stability and carbon-carbon bond lengths of benzene, resonance structure, and MO picture.

Aromatic electrophilic substitution-General pattern of the mechanism, role of σ and π-complexes. Mechanism of nitration, halogenation, sulphonation, mercuration and Friedel-Crafts reaction.Energy profile diagrams. Activating and deactivating substituents, orientation and ortho/pararatio.Side chain reactions of benzene derivatives.Methods of formation and chemical reactions of alkylbenzenes, alkynylbenzenes and biphenyl.

Alkyl and Arylhalides(7 Hrs.)

Nomenclature and classes of alkylhalides, methods of formation, chemical reactions. Mechanisms of nucleophilic substitution reactions of alkylhalides, SN2 and SN1 reactions with energy profile diagrams.

Methods of formation of arylhalides, nuclear and side chain reactions.The addition-elimination and the elimination-addition mechanisms of nucleophilicaromatic substitution reactions.

Relative reactivities of alkylhalides vs. allyl, vinylandarylhalides.

UNIT-III PHYSICAL CHEMISTRY (15 Hrs.)

Thermodynamics-III (8 Hrs.)

Third Law of Thermodynamics: Nernst heat theorem, statement and concept of residual entropy, evaluation of absolute entropy from heat capacity data. Gibbs and Helmholtz functions; Gibbs function (G) and Helmholtz functions (A) as thermodynamic quantities, A &G as criteria for thermodynamic equilibrium and spontaneity, and their advantage over entropy change. Variation of G and A with P, V and T.

Chemical Equilibrium(7 Hrs.)

Equilibrium constant and free energy, Thermodynamic derivation of law of mass action, Le-Chatelier's principle, Reaction isotherm and Reaction isochore-Clapeyron equation and Clausius–Clapeyron equation, applications.

Books/Literature suggested

1. Cotton,F.A.,Wilkinson,G.,Gaus,P.L.,BasicInorganicChemistry;2nd edition,Pubs:JohnWileyand Sons, 1995.
2. Lee,J.D.,ConciseInorganicChemistry;4th edition,Pubs:Chapman&HallLtd.,1991.
3. Shriver,D.E.,Atkins,P.W.,InorganicChemistry;4th edition,Pubs:OxfordUniversityPress,2006.
4. Douglas, B., Medaniel, D., Atenander, J., Concepts and Models of Inorganic Chemistry; 3rd edition, Pubs: John Wiley and Sons Inc., 1994,
5. Porterfield,W.W.,Wesky,A.,InorganicChemistry;Pubs:Addison-WesleyPublishingCompany,1984.
6. Miessler,G.L.,Tarr,D.A.,InorganicChemistry;3rd edition,Pubs:PearsonEducationInc.,2004.
7. Puri,B.R.,Sharma,L.R.,Kalia,K.K.,PrinciplesofInorganicChemistry;30th edition,Pubs:Mileston es Publisher, 2006-07
8. Atkins, P., Paula, J.de, Atkins Physical Chemistry; 8th edition, Pubs: Oxford University Press, 2008.

9. Puri, B.R., Sharma, L.R., Pathania, M.S., Principles of Physical Chemistry; 43rd edition, Pubs: Vishal Publishing Co., 2008.
10. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw-Hill Inc, 1996.
11. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
12. Albert, R.A., Silbey, R.J., Physical Chemistry; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
13. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs:Wiley Eastern Limited, 1991.
14. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw-Hill Publishing Co. Ltd., 2002.
15. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
16. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book Company, 1989.
17. Morrison,R.T.,Boyd,R.N.,OrganicChemistry;6th edition,Pubs:Prentice-Hall,1992.
18. Solomons,T.W.,Fryhle,C.B.,OrganicChemistry;9th edition,Pubs:WileyIndia,2007.
19. WadeJr.,L.G.,Singh,M.S.,OrganicChemistry;6th edition,Pubs:Pearson Education,2008.
20. Mukherji,S.M.,Singh,S.P.,Kapoor,R.P.,OrganicChemistry;Pubs:NewAgeInternational,1985.
21. Carey,F.A.,Sundberg,R.J.,AdvancedOrganicChemistryPartB:ReactionsandSynthesis;5th editio n, Pubs: Springer, 2007

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the compulsory question and one question each from the UNIT-1, UNIT-2 and UNIT-3.
2. The use of scientific calculators is not allowed.

Semester – 3
Syllabus for Practical Paper
CHM-DSC-5-MIN-302/MAJ-401
General Chemistry-4

End semester = 22	**Credit - 01**
Internal Marks = 03	**Contact Hours – 2/week**
Examination Time allowed	**3 Hours**

I. INORGANIC CHEMISTRY

1. Iodometry and Iodimetry
    a. Determination of Cu^{2+}
    b. Determination of vitamin C
    c. Determination of chlorine in a sample of bleaching powder

II. ORGANIC CHEMISTRY

2. Preparation of the following compounds via Aromatic electrophilic substitution:-
    a. 1,3-dinitrobenzene from nitrobenzene
    b. 4-nitroacetanilide from acetanilide
    c. 4-iodoaniline from aniline

III. PHYSICAL CHEMISTRY

3. Refractive indices

    a. Determine the Refractive indices of given liquids (water, acetone, methanol, ethylacetate,cyclohexane) by Abbe's refractometer and calculate their specific refractions.

4. Molecular Weight Determination

    a. Determination of molecular weight of a non–volatile solute by the Rast method.

    b. Determinationoftheapparentdegreeofdissociationofanelectrolyte(e.g.NaCl) inaqueous solution of the substance.

Books/Literature suggested

1. Practical Organic Chemistry, 4TH EDN, F G Mann & B C Saunders,Cambridge. March 1973
2. Vogel's Textbook of Practical Organic Chemistry, 5th Ed., Brian S Furniss, Anthony J Hannaford, Peter W G Smith & Austin R Tatchell,
3. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
4. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.

Semester	**IV**
Code of Paper	**CHM-DSC-5-MAJ-401/MIN-302**
Title of Paper	**General Chemistry-4**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week** **Practical: 2 Hrs. per week**
Applicable to the students 1. Those who have opted for chemistry as a Major subject in Table 1 or Table 2 2. Those who have opted for chemistry as a Minor subject in Table 1 or Table 2	
LEARNING OUTCOMES SPECIFIC TO DISCIPLINE The students will learn the following- 1. Important aspects group 13 and 14 elements 2. Chemistry of compounds of B and C and their group elements 3. Chemistry of arenes, and alkyl ad aryl halides 4. Concept of aromaticity 5. Concepts of thermodynamics and chemical equilibrium	
GENERIC LEARNING OUTCOMES 1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,** Ethical and Professional Responsibility, **Application based Knowledge.** 2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.	

Semester – 4
Syllabus for Theory Paper
CHM-DSC-5-MAJ-401/MIN-302
General Chemistry-4

End semester = 67 **Credits - 03**
Internal Marks = 08 **Contact Hours – 3/week**
Examination Time allowed **3 Hours**

UNIT I: INORGANIC CHEMISTRY (15 Hrs.)

Group 13 Elements (8 Hrs)

Comparative study (including diagonal relationship) of groups 13 elements, Study of the following compounds with emphasis on structure, bonding, preparation, properties and uses: Boric acid and borates, boron nitrides, boron hydrides (diborane), boron halides, alumina.

Group 14 Elements (7 Hrs)

Comparative study (including diagonal relationship) of groups 14 elements, , allotropic forms of carbon, graphitic compounds, compounds like hydrides, oxides, oxyacids and halides of groups 14, fullerenes, carbides, fluorocarbons.

UNIT II: ORGANIC CHEMISTRY (15 Hrs.)

Aromaticity and Arenes(8 Hrs.)

Aromaticity: The Huckel's rule, aromatic ions.

Nomenclature of benzene derivatives, aryl group, aromatic nucleus and side chain, Structure of benzene: Molecular formula and Kekule structure. Stability and carbon-carbon bond lengths of benzene, resonance structure, and MO picture.

Aromatic electrophilic substitution-General pattern of the mechanism, role of σ and π-complexes. Mechanism of nitration, halogenation, sulphonation, mercuration and Friedel-Crafts reaction.Energy profile diagrams. Activating and deactivating substituents, orientation and ortho/pararatio.Side chain reactions of benzene derivatives.Methods of formation and chemical reactions of alkylbenzenes, alkynylbenzenes and biphenyl.

Alkyl and Arylhalides(7 Hrs.)

Nomenclature and classes of alkyl halides, methods of formation, chemical reactions.Mechanisms of nucleophilic substitution reactions of alkyl halides, SN2 and SN1 reactions with energy profile diagrams.

Methods of formation of aryl halides, nuclear and side chain reactions.The addition-elimination and the elimination-addition mechanisms of nucleophilic aromatic substitution reactions.

Relative reactivities of alkyl halides vs. allyl, vinyl and aryl halides.

UNIT-III PHYSICAL CHEMISTRY (15 Hrs.)

Thermodynamics-III (8 Hrs.)

Third Law of Thermodynamics: Nernst heat theorem, statement and concept of residual entropy, evaluation of absolute entropy from heat capacity data. Gibbs and Helmholtz functions; Gibbs function (G) and Helmholtz functions (A) as thermodynamic quantities, A &G as criteria for thermodynamic equilibrium and spontaneity, and their advantage over entropy change. Variation of G and A with P, V and T.

Chemical Equilibrium(7 Hrs.)

Equilibrium constant and free energy, Thermodynamic derivation of law of mass action, Le-Chatelier's principle, Reaction isotherm and Reaction isochore-Clapeyron equation and Clausius–Clapeyron equation, applications.

Books/Literature suggested

1. Cotton,F.A.,Wilkinson,G.,Gaus,P.L.,BasicInorganicChemistry;2nd edition,Pubs:JohnWileyand Sons, 1995.
2. Lee,J.D.,ConciseInorganicChemistry;4th edition,Pubs:Chapman&HallLtd.,1991.
3. Shriver,D.E.,Atkins,P.W.,InorganicChemistry;4th edition,Pubs:OxfordUniversityPress,2006.
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6. Miessler,G.L.,Tarr,D.A.,InorganicChemistry;3rd edition,Pubs:PearsonEducationInc.,2004.
7. Puri,B.R.,Sharma,L.R.,Kalia,K.K.,PrinciplesofInorganicChemistry;30th edition,Pubs:Mileston es Publisher, 2006-07

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10. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw-Hill Inc, 1996.
11. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
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13. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs:Wiley Eastern Limited, 1991.
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15. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
16. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book Company, 1989.
17. Morrison,R.T.,Boyd,R.N.,OrganicChemistry;6th edition,Pubs:Prentice-Hall,1992.
18. Solomons,T.W.,Fryhle,C.B.,OrganicChemistry;9th edition,Pubs:WileyIndia,2007.
19. WadeJr.,L.G.,Singh,M.S.,OrganicChemistry;6th edition,Pubs:PearsonEducation,2008.
20. Mukherji,S.M.,Singh,S.P.,Kapoor,R.P.,OrganicChemistry;Pubs:NewAgeInternational,1985.
21. Carey,F.A.,Sundberg,R.J.,AdvancedOrganicChemistryPartB:ReactionsandSynthesis;5th editio n, Pubs: Springer, 2007

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

Instructions for students

3. The students will attempt **four questions in all:**the compulsory question and one question each from the UNIT-1, UNIT-2 and UNIT-3.
4. The use of scientific calculators is not allowed.

Semester – 4
Syllabus for Practical Paper
CHM-DSC-5-MAJ-401/MIN-302
General Chemistry-4

End semester = 22	**Credit - 01**
Internal Marks = 03	**Contact Hours – 2/week**
Examination Time allowed	**3 Hours**

II. INORGANIC CHEMISTRY

1. Iodometry and Iodimetry
    a. Determination of Cu^{2+}
    b. Determination of vitamin C
    c. Determination of chlorine in a sample of bleaching powder

II. ORGANIC CHEMISTRY

2. Preparation of the following compounds via Aromatic electrophilic substitution:-
    a. 1,3-dinitrobenzene from nitrobenzene
    b. 4-nitroacetanilide from acetanilide
    c. 4-iodoaniline from aniline

III. PHYSICAL CHEMISTRY

3. **Refractive indices**

    a. Determine the Refractive indices of given liquids (water, acetone, methanol, ethylacetate,cyclohexane) by Abbe's refractometer and calculate their specific refractions.

4. **Molecular Weight Determination**

    a. Determination of molecular weight of a non–volatile solute by the Rast method.

    b. Determinationoftheapparentdegreeofdissociationofanelectrolyte(e.g.NaCl) inaqueous solution of the substance.

Books/Literature suggested

1. Practical Organic Chemistry, 4TH EDN, F G Mann & B C Saunders,Cambridge. March 1973
2. Vogel's Textbook of Practical Organic Chemistry, 5th Ed., Brian S Furniss, Anthony J Hannaford, Peter W G Smith & Austin R Tatchell,
3. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
4. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.

Semester	**IV**
Code of Paper	**CHM-DSC-6-MAJ-402/MIN-401**
Title of Paper	**General Chemistry-5**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of the Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of the Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week** **Practical: 2 Hrs. per week**
Applicable to the students 1. Those who have opted for chemistry as a Major subject in Table 1 or Table 2 2. Those who have opted for chemistry as a Minor subject in Table 1 or Table 2	
LEARNING OUTCOMES SPECIFIC TO DISCIPLINE The students will learn the following- 1. Important aspects group 15, 16 and 17 elements 2. Chemistry of compounds of N, As, Sb, O, S, Se, F, Cl, and Br 3. Chemistry of alcohols, phenols, ethers and epoxides 4. Concepts of solid state chemistry	
GENERIC LEARNING OUTCOMES 1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,** Ethical and Professional Responsibility, **Application based Knowledge.** 2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.	

Semester – 4
Syllabus for Theory Paper
CHM-DSC-6-MAJ-402/MIN-401
General Chemistry-5

End semester = 67 **Credits - 03**
Internal Marks = 08 **Contact Hours – 3/week**
Examination Time allowed **3 Hours**

UNIT I: INORGANIC CHEMISTRY (15 Hrs.)

Group 15 Elements (5 Hrs.)

Types of covalence in nitrogen, stereochemistry, chemical reactivity, dinitrogen complexes (basic idea only), hydrides of nitrogen, nitrogen halides, oxides and oxoacids, Phosphorus, As, Sb& Bi: Stereochemistry of their compounds, production of elemental P and its allotropic forms, hydrides, halides, oxides and oxy-acids

Group 16 Elements (5 Hrs.)

Chemical Reactivity, group trends and stereochemistry of Oxygen, S, Se and Te Family, dioxygen as a ligand (basic idea only), structure of O_3 and H_2O_2, clathrate hydrates allotropic forms of S and Se, structures of halides, oxides and oxyacids of S, Se and Te, S-N compounds (neutral) Polyatomic cations of S, Se and Te.

Group 17 Elements (5 Hrs.)

Comparative study of group 17 elements, basic properties of halogens, Oxides and oxoacids of Chlorine, interhalogen compounds, polyhalide ions, and pseudohalogens.

UNIT II: ORGANIC CHEMISTRY (15 Hrs.)

Alcohols and Phenols(9 Hrs.)

Classification and nomenclature, Monohydric alcohols-Nomenclature, methods of formation by reduction of aldehydes, ketones, carboxylic acids and esters. Hydrogen bonding, Acidic nature, Reactions of alcohols, Dihydric and Trihydric alcohols- Nomenclature, methods of formation, chemical reactions of vicinal glycols and glycerol,Preparation of phenols, physical properties and acidic character, Comparative acidic strengths of alcohols and phenols, resonance stabilization of phenoxide ion, Reactions of phenols-electrophilic aromatic substitution, acylation and carboxylation, Mechanisms of Fries rearrangement, Claisen rearrangement, Gattermann synthesis, and Reimer-Tiemann reaction.

Ethers and Epoxides(6 Hrs.)

Nomenclature of ether and methods of their formation, physical properties, Chemical reaction-cleavage and autoxidation, Ziesel'smethod,Synthesisof epoxides, Acid and base-catalyzed ring opening of epoxides, orientation of epoxide ring opening, reactions of Grignard and organolithium reagents with epoxides.

UNIT-III PHYSICAL CHEMISTRY (15 Hrs.)

Solid State(15 Hrs.)

Definition of space lattice, unit cell and Miller Indices

Laws of Crystallography – (i) Law of Constancy of Interfacial Angles, (ii) Law of Rationality of Indices, (iii) Law of Symmetry. Symmetry elements in crystals.

X-ray diffraction by crystals.Derivation of the Bragg equation.Determination of the crystal structure of NaCl, KCl and CsCl (Laue's method and powder method).
Applications of Powder diffraction for structure determination, Thermal and photochemical reaction in the solid state

Books/Literature suggested

1. Cotton,F.A.,Wilkinson,G.,Gaus,P.L.,BasicInorganicChemistry;2nd edition,Pubs:JohnWileyand Sons, 1995.
2. Lee,J.D.,ConciseInorganicChemistry;4th edition,Pubs:Chapman&HallLtd.,1991.
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4. Douglas, B., Medaniel, D., Atenander, J., Concepts and Models of Inorganic Chemistry; 3rd edition, Pubs: John Wiley and Sons Inc., 1994,
5. Porterfield,W.W.,Wesky,A.,InorganicChemistry;Pubs:Addison-WesleyPublishingCompany,1984.
6. Miessler,G.L.,Tarr,D.A.,InorganicChemistry;3rd edition,Pubs:PearsonEducationInc.,2004.
7. Puri,B.R.,Sharma,L.R.,Kalia,K.K.,PrinciplesofInorganicChemistry;30th edition,Pubs:Mileston es Publisher, 2006-07

8. Atkins, P., Paula, J.de, Atkins Physical Chemistry; 8th edition, Pubs: Oxford University Press, 2008.
9. Puri, B.R., Sharma, L.R., Pathania, M.S., Principles of Physical Chemistry; 43rd edition, Pubs: Vishal Publishing Co., 2008.
10. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw-Hill Inc, 1996.
11. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
12. Albert, R.A., Silbey, R.J., Physical Chemistry; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
13. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs:Wiley Eastern Limited, 1991.
14. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw-Hill Publishing Co. Ltd., 2002.
15. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
16. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book Company, 1989.
17. Morrison,R.T.,Boyd,R.N.,OrganicChemistry;6th edition,Pubs:Prentice-Hall,1992.
18. Solomons,T.W.,Fryhle,C.B.,OrganicChemistry;9th edition,Pubs:WileyIndia,2007.
19. WadeJr.,L.G.,Singh,M.S.,OrganicChemistry;6th edition,Pubs:PearsonEducation,2008.
20. Mukherji,S.M.,Singh,S.P.,Kapoor,R.P.,OrganicChemistry;Pubs:NewAgeInternational,1985.
21. Carey,F.A.,Sundberg,R.J.,AdvancedOrganicChemistryPartB:ReactionsandSynthesis;5th edition, Pubs: Springer, 2007

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:**the compulsory question and one question each from the UNIT-1, UNIT-2 and UNIT-3.
2. The use of scientific calculators is not allowed.

Semester – 4
Syllabus for Practical Paper
CHM-DSC-6-MAJ-402/MIN-401
General Chemistry-5

End semester = 22	**Credit - 01**
Internal Marks = 03	**Contact Hours – 2/week**
Examination Time allowed	**3 Hours**

I. INORGANIC CHEMISTRY

1. Preparation of the crystals of the following compounds
   a. Potash alum
   b. Chrome alum
   c. Mohr's salt

II. ORGANIC CHEMISTRY

2. Williamson ether synthesis using cresol and chloroacetic acid.
3. Determination of the amount of phenol in the given sample by acetylation and bromination methods, and perform acid-base and iodimetric titrations.
4. To observe the solubility of alcohols relative to their chemical structure
5. Tests to distinguish primary, secondary and tertiary alcohols and a colour test for phenol

III. PHYSICAL CHEMISTRY

a) Calibration of Thermometer: 80-82° (Naphthalene), 113-114° (acetanilide), 132.5-133^{o}(Urea), 100° (distilled Water)

b) Determination of Critical Micelle Concentration (CMC) of a Cationic Surfactant by Conductivity Measurements

c) Determination of the Critical Micelle Concentration (CMC) of an Anionic Surfactant by Conductivity Measurements

Books/Literature suggested

1. Jolly,W.L., ModernInorganicChemistry;2ndedition,Pubs:TataMcGraw-HillPublishingCompany Limited, 1991.
2. Practical Organic Chemistry, 4TH EDN, F G Mann & B C Saunders,Cambridge. March 1973
3. Vogel's Textbook of Practical Organic Chemistry, 5th Ed., Brian S Furniss, Anthony J Hannaford, Peter W G Smith & Austin R Tatchell,
4. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
5. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.

Semester	**IV**
Code of Paper	**CHM-DSC-7-MAJ-403**
Title of Paper	**Basic Concepts in Chemistry-2**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week** **Practical: 2 Hrs. per week**
Applicable to the students 1. Those who have opted for chemistry as a Major subject in Table 1	
LEARNING OUTCOMES SPECIFIC TO DISCIPLINE The students will learn the following- 1. Important aspects oxidation and reduction 2. Chemistry methods and techniques used in metallurgy 3. Chemistry of organic polymers 4. Concepts of liquid state crystals and colloids	
GENERIC LEARNING OUTCOMES 1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,** Ethical and Professional Responsibility, **Application based Knowledge.** 2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.	

Semester – 4
Syllabus for Theory Paper
CHM-DSC-7-MAJ-403
Basic Concepts in Chemistry-2

End semester = 67 **Credits - 03**
Internal Marks = 08 **Contact Hours – 3/week**
Examination Time allowed **3 Hours**

UNIT I: INORGANIC CHEMISTRY (15 Hrs.)

OxidationandReduction (7 Hrs)

Introduction, oxidation-reduction reactions (electronic and oxidation number concept), Balancing of redox reactions, Use of redox potential data – analysis of redox cycle, redox stability in water – Frost, Latimer and Pourbaixdiagrams.

Metallurgy (8 Hrs)

General Principles of Metallurgy: Chief modes of occurrence of metals based on standard electrode potentials. Ellingham diagrams for the reduction of metal oxides using carbon and carbon monoxide as reducing agents. Electrolytic Reduction, Hydrometallurgy. Methods of purification of metals: Electrolytic process, Parting process, van Arkel-de Boer process and Mond's process, Zone refining

UNIT II: ORGANIC CHEMISTRY (15 Hrs.)

Polymer Chemistry(15 Hrs.)

Basic characteristics of polymers, melting point of polymers, glass transition temperature, molecular weight and molecular weight distribution.

Organic polymers: Synthetic polymers and their syntheses via free radical polymerisation, polymerisation by ionic initiators, Ziegler-Natta type polymerisation, atom transfer radical polymerisation, ring opening polymerisation, olefin metathesis and polymerisation, group transfer polymerisation, condensation polymerisation.

Polyesters,polyamides,phenolformaldehyderesins,ureaformaldehyde resins, epoxy resins and polyurethanes.

Naturalandsyntheticrubbers.

UNIT II: PHYSICAL CHEMISTRY (15 Hrs.)

Liquid state and liquid crystals(7 Hrs.)

Introduction, Intermolecular forces, Structure of Liquids, Theories of Liquid state, Structural differences between Solids, Liquids and Gases.Liquid crystals, classification and structure of thermotropic phases (Nematic, Smectic and Cholesteric phases), applications of liquid crystals.

Colloidal State (8 Hrs.)

Definition of colloids, classification of colloids, Solids in liquids (sols): Properties –kinetic, optical and electrical; stability of colloids, protective action, Hardy-Schulze rules, gold number, Liquids in liquids (emulsions): Types of emulsions, preparation. Emulsifier, Liquids in solids (gels): Classification, preparation and properties, inhibition, general applications of colloids.

Books/Literature suggested

1. Cotton,F.A.,Wilkinson,G.,Gaus,P.L.,BasicInorganicChemistry;2nd edition,Pubs:JohnWileyand Sons, 1995.
2. Lee,J.D.,ConciseInorganicChemistry;4th edition,Pubs:Chapman&HallLtd.,1991.
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9. Puri, B.R., Sharma, L.R., Pathania, M.S., Principles of Physical Chemistry; 43rd edition, Pubs: Vishal Publishing Co., 2008.
10. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw-Hill Inc, 1996.
11. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
12. Albert, R.A., Silbey, R.J., Physical Chemistry; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
13. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs:Wiley Eastern Limited, 1991.
14. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw-Hill Publishing Co. Ltd., 2002.

15. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
16. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book Company, 1989.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions:**9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

3. The students will attempt **four questions in all:**the**compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
4. The use of scientific calculators is not allowed.

Semester – 4
Syllabus for Practical Paper
CHM-DSC-7-MAJ-403
Basic Concepts in Chemistry-2

End semester = 22	**Credit - 01**
Internal Marks = 03	**Contact Hours – 2/week**
Examination Time allowed	**3 Hours**

I. INORGANIC CHEMISTRY

1. Precipitation Titration: Volhard's method (for Cl^- ions)
2. Gravimetric analysis
   a. Ag^+ as AgCl
   b. Fe^{2+} as Iron(III) oxide
   c. Ba^{2+} as barium(II) sulphate

II. ORGANIC CHEMISTRY

3. Investigate and identify a variety of polymers used in everyday materials by testing their density.
4. Study of properties of cross-linked polymers.
5. To study the effect of varying concentrations of the poly(vinyl alcohol) and borax solution on slime

III. PHYSICAL CHEMISTRY

Determination of boiling points; Ethanol 78°, Cyclohexane 81.4°, Toluene 110.6°, Benzene 80°.

Books/Literature suggested

1. Jolly,W.L., Modern Inorganic Chemistry; 2nd edition, Pubs: Tata McGraw-Hill Publishing Company Limited, 1991.
2. Practical Organic Chemistry, 4TH EDN, F G Mann & B C Saunders,Cambridge. March 1973
3. Vogel's Textbook of Practical Organic Chemistry, 5th Ed., Brian S Furniss, Anthony J Hannaford, Peter W G Smith & Austin R Tatchell,
4. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
5. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.
6. Levitt, B.P., Findlay's Practical Physical Chemistry; 8th edition, Pubs: Longman Group Ltd., London & New York, 1978.

Semester	**IV**
Code of Paper	**CHM-DSC-8-MAJ-404**
Title of Paper	**Basic Concepts in Chemistry-3**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week** **Practical: 2 Hrs. per week**
Applicable to the students 1. Those who have opted for chemistry as a Major subject in Table 1	
LEARNING OUTCOMES SPECIFIC TO DISCIPLINE The students will learn the following- 1. Important aspects chain like inorganic compounds 2. Chemistry of inorganic polymers 3. Chemistry of amino acids 4. Chemistry of peptides and proteins 5. Chemistry of nucleic acids	
GENERIC LEARNING OUTCOMES 1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,** Ethical and Professional Responsibility, **Application based Knowledge.** 2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.	

Syllabus for Theory Paper

CHM-DSC-8-MAJ-404

Basic Concepts in Chemistry-3

End semester = 67 **Credits - 03**
Internal Marks = 08 **Contact Hours – 3/week**
Examination Time allowed **3 Hours**

UNIT I: INORGANIC CHEMISTRY (15 Hrs.)

Inorganic Chains (8 Hrs.)

Chains: Catenation including heterocatenation, intercalation compounds, silicate minerals, aluminate minerals, one-dimensional sulphur-nitrogen conductor.

SiliconesandPhosphazenes(7 Hrs.)

Structure and bonding in siloxanes, triphosphazenes,phosphazenes, Synthesis, properties and applications of polysiloxanes (Silicones), polysilanes and polyphosphazenes

UNIT II: ORGANIC CHEMISTRY (15 Hrs.)

Chemistry of AminoAcids (5 Hrs.)

Classification, structure and stereochemistry of amino acids, acid-base behaviour, isoelectric point and electrophoresis, Preparation and reactions of α-amino acids.

Chemistry of Peptides and Proteins (5 Hrs.)

Structure and nomenclature of peptides and proteins, Classification of proteins, Peptide structure determination, end group analysis, selective hydrolysis of peptides, Classical peptide synthesis, solid–phase peptide synthesis, Structures of peptides and proteins, Levels of protein structure, Protein denaturation/renaturation.

Chemistry of Nucleic Acids (5 Hrs.)

Introduction, Constituents of nucleic acids, Ribonucleosides and ribonucleotides, the double helical Structure of DNA.

UNIT III: PHYSICAL CHEMISTRY (15 Hrs.)

Elementary Quantum Mechanics-II (15 Hrs.)

Molecular orbital theory, basic ideas – criteria for forming M.O. from A.O., construction of M.O.'s by LCAO – H_2^+ ion. Calculation of energy levels from wave functions, physical picture of bonding and antibonding wave functions, concept of σ, σ*, π, π* orbitals and their characteristics. Hybrid orbitals – sp, sp^2, sp^3, calculation of coefficients of A.O.'s used in these hybrid orbitals. Introduction to the valence bond model of H_2, comparison of M.O. and V.B. models.

Books/Literature suggested

1. Cotton,F.A.,Wilkinson,G.,Gaus,P.L.,BasicInorganicChemistry;2nd edition,Pubs:JohnWileyand Sons, 1995.
2. Lee,J.D.,ConciseInorganicChemistry;4th edition,Pubs:Chapman&HallLtd.,1991.
3. Shriver,D.E.,Atkins,P.W.,InorganicChemistry;4th edition,Pubs:OxfordUniversityPress,2006.
4. Douglas, B., Medaniel, D., Atenander, J., Concepts and Models of Inorganic Chemistry; 3rd edition, Pubs: John Wiley and Sons Inc., 1994,
5. Porterfield,W.W.,Wesky,A.,InorganicChemistry;Pubs:Addison-WesleyPublishingCompany,1984.
6. Miessler,G.L.,Tarr,D.A.,InorganicChemistry;3rd edition,Pubs:PearsonEducationInc.,2004.
7. Puri,B.R.,Sharma,L.R.,Kalia,K.K.,PrinciplesofInorganicChemistry;30th edition,Pubs:Mileston es Publisher, 2006-07
8. Atkins, P., Paula, J.de, Atkins Physical Chemistry; 8th edition, Pubs: Oxford University Press, 2008.
9. Puri, B.R., Sharma, L.R., Pathania, M.S., Principles of Physical Chemistry; 43rd edition, Pubs: Vishal Publishing Co., 2008.
10. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw-Hill Inc, 1996.
11. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
12. Albert, R.A., Silbey, R.J., Physical Chemistry; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
13. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs:Wiley Eastern Limited, 1991.
14. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw-Hill Publishing Co. Ltd., 2002.
15. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.

16. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book Company, 1989.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions:**9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:**the**compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Syllabus for Practical Paper
CHM-DSC-8-MAJ-404
Basic Concepts in Chemistry-3

End semester = 22 | **Credit - 01**

Internal Marks = 03 | **Contact Hours – 2/week**

Examination Time allowed | **3 Hours**

I. INORGANIC CHEMISTRY

1.Preparation of crystals of ferrous oxalate and estimation of iron and oxalate.

2. Preparation of double salt $CoSO_4(NH_4)_2SO_4.6H_2O$ and estimation of copper and sulphate.

3. Preparation of tris (thiourea) Copper (I) sulphate (estimation of copper and sulphate.

II.ORGANIC CHEMISTRY

4. Qualitative Analysis:
    a. Separation of a mixture of 2 amino acids by radial &ascending paper chromatography and to determine the R_f values.
    b. Titration curve of an amino acid & determination of its pK_a and isoelectric point.
    c. To determine the acid-base behaviour of amino acids by the volumetric method.

III. PHYSICAL CHEMISTRY

5. Chemical Kinetics
    a. Study the kinetics of the following reactions.
        i. Acid hydrolysis of methyl acetate with hydrochloric acid.
        ii. Saponification of ethyl acetate.

Books/Literature suggested

1. Jolly,W.L., Modern Inorganic Chemistry; 2nd edition, Pubs: Tata McGraw-Hill Publishing Company Limited, 1991.
2. Practical Organic Chemistry, 4TH EDN, F G Mann & B C Saunders,Cambridge. March 1973
3. Vogel's Textbook of Practical Organic Chemistry, 5th Ed., Brian S Furniss, Anthony J Hannaford, Peter W G Smith & Austin R Tatchell,
4. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
5. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.
6. Levitt, B.P., Findlay's Practical Physical Chemistry; 8th edition, Pubs: Longman Group Ltd., London & New York, 1978.

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Computer Science and Application

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR (Table 1*) and STUDENTS OPTING FOR MULTI-DISCIPLINARY/INTER-DISCIPLINARY(Table 2*)

Table 1*

Sem	Level	Course Code	Course Name	Credits (Th+Pr)		Total Credits
III	L-200	CSA-DSC-3(Maj/Min)-301	Programming in C++	2+2	4 (x2)	24
		CSA-DSC-3(Maj/Min)-302	Web Designing	2+2	4 (x2)	
		CSA-MDC-101/201/301	E-Commerce	3+0	3	
		CSA-MDC-102/202/302	Management Information System	3+0		
		CSA-MDC-103/203/303	Web Development/ Designing	0+3		
		CSA-MDC-104/204/304	Introduction to Information Technology	0+3		
		CSA-SEC-101/201/301	Computer Fundamentals and Personal Computing Software	0+3	3	
		CSA-SEC-102202/302	Desktop Publishing using GIMP	0+3		
		CSA-SEC-103/203/303	Backend Web Development	0+3		
		CSA-SEC-104/204/304	Business Intelligence and Data Visualization	0+3		
		CSA-SEC-105/205/305	Digital Marketing	0+3		
		CSA-SEC-106/206/306	Computer Literacy	0+3		
		CSA-SEC-107/207/307	Desktop Publishing Using Photoshop	0+3		
		CSA-SEC-108/208/308	Fundamentals of Web Designing	0+3		
		CSA-SEC-109/209/309	Application Software for Office Automation	0+3		
		CSA-SEC-110/210/310	Python Programming	0+3		
		(AEC-3-301)	Eng-3/MIL-1	2		

Table 2*

Sem	Level	Course Code	Course Name	Credits (Th+Pr)		Total Credits
III	L-200	CSA-DSC-3(Maj/Min)-301	Programming in C++	2+2	4 (x3)	24
		CSA-DSC-3(Min)-302	Web Designing	2+2	4	
		CSA-MDC-101*	E-Commerce	3+0	3	
		CSA-MDC-102*	Management Information System	3+0		
		CSA-MDC-103*	Web Development/ Designing	0+3		
		CSA-MDC-104*	Introduction to Information Technology	0+3		
		CSA-SEC-101*	Computer Fundamentals and Personal Computing Software	0+3	3	
		CSA-SEC-102*	Desktop Publishing using GIMP	0+3		
		CSA-SEC-103*	Backend Web Development	0+3		
		CSA-SEC-104*	Business Intelligence and Data Visualization	0+3		
		CSA-SEC-105*	Digital Marketing	0+3		
		CSA-SEC-106*	Computer Literacy	0+3		
		CSA-SEC-107*	Desktop Publishing Using Photoshop	0+3		
		CSA-SEC-108*	Fundamentals of Web Designing	0+3		
		CSA-SEC-109*	Application Software for Office Automation	0+3		
		CSA-SEC-110*	Python Programming	0+3		
		(AEC-3-301)	Eng-3/MIL-1	2		

<table>
<tr><td rowspan="11">IV</td><td rowspan="11">L-200</td><td>CSA-DSC-5(Maj/Min)-401</td><td>Software Engineering</td><td>4+0</td><td>4 (x2)</td><td rowspan="11">24</td><td rowspan="11">IV</td><td rowspan="11">L-200</td><td>CSA-DSC-5(Maj/Min)-401</td><td>Software Engineering</td><td>4+0</td><td>4 (x3)</td><td rowspan="11">24</td></tr>
<tr><td>CSA-DSC-6(Maj)-402</td><td>Database Management System using Oracle</td><td colspan="2">2+2=4</td><td>CSA-DSC-6(Maj)-402</td><td>Database Management System using Oracle</td><td>2+2</td><td>4 (x2)</td></tr>
<tr><td>CSA-DSC-7(Maj)-403</td><td>Data Structure</td><td colspan="2">2+2=4</td><td>CSA-VAC-101/201/301/401</td><td>Introduction to Blockchain Technology</td><td>2+0</td><td rowspan="6">2</td></tr>
<tr><td>CSA-DSC-8(Maj)-404</td><td>Introduction to Artificial Intelligence</td><td colspan="2">4+0=4</td><td>CSA-VAC-102/202/302/402</td><td>Introduction to Computer Hardware</td><td>2+0</td></tr>
<tr><td>CSA-VAC-101/201/301/401</td><td>Introduction to BlockchainTechnology</td><td>2+0</td><td rowspan="4">2</td><td>CSA-VAC-103/203/303/403</td><td>Digital Empowerment</td><td>2+0</td></tr>
<tr><td>CSA-VAC-102/202/302/402</td><td>Introduction to Computer Hardware</td><td>2+0</td><td rowspan="3">CSA-VAC-104/204/304/404</td><td rowspan="3">Ethics of Artificial Intelligence</td><td rowspan="3">2+0</td></tr>
<tr><td>CSA-VAC-103/203/303/403</td><td>Digital Empowerment</td><td>2+0</td></tr>
<tr><td>CSA-VAC-104/204/304/404</td><td>Ethics of Artificial Intelligence</td><td>2+0</td></tr>
<tr><td>(AEC-4-401)</td><td>Eng-4/MIL-2</td><td colspan="2">2</td><td>(AEC-4-401)</td><td>Eng-4/MIL-2</td><td colspan="2">2</td></tr>
<tr><td colspan="14">The student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secures 4 credits from summer internship</td></tr>
</table>

***Note: Student need to select one MDC, VAC, SEC out of given options, respectively.**
NO MDC, SEC, VAC will be repeated. All MDC, SEC and VAC are Level-100

***Note: The candidate will opt for either Template 1 or 2 from Semester III onwards.**

Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

Semester-III

Paper Title : Programming in C++
Paper Code:CSA-DSC-3(Maj/Min)-301

Th	Pr	Cr
2	2	4

Theory External Marks: 45
Theory Internal Marks: 05
Number of Th. Hrs. : 30
Practical Ext. Marks : 50
Number of Pr. Hrs. : 60

Time Duration: 3 Hrs.
Level: L-200

Objectives: By the end of the course, students will be able to write C++ programs using the more esoteric language features, utilize Object Oriented techniques to design C++ programs, use the standard C++ library, and explore advanced C++ techniques.

Learning Outcome:
On successful completion of the course, students will be able to:

i. Understand the features and importance of object oriented programming concepts.
ii. Use the concepts like inheritance, polymorphism, and dynamic binding to develop codes.
iii. Gets better understanding of developing reusable components using C++ programming.
iv. Develop of large software that needs bottom up and better modular approach.

Note :

i. The Question Paper will consist of Four Units.
ii. Examiner will set total of NINE questions comprising TWO questions from each Unit and ONE compulsory question of short answer type covering whole syllabi.
iii. The students are required to attempt ONE question from each Unit and the Compulsory question.
iv. All questions carry equal marks unless specified.

UNIT - I

Principles of Object-Oriented Programming (OOP): Introduction to OOP, Difference between OOP and Procedure Oriented Programming; Concepts: Object, Class, Encapsulation, Abstraction, Polymorphism and Inheritance, Applications of OOP. Special operators: scope resolution operator, Member Dereferencing operators, Memory management operators, Manipulators and Type cast operator.

Structure of a C++ Program and Classes and Objects: Class Declaration: Data Members, Member Functions, Private and Public members, Creating Objects, accessing class data members, Accessing member functions; Class Function Definition: Member Function definition inside the class declaration and outside the class declaration.

UNIT - II

Friend function, inline function, Static data members, Function Overloading, Arrays within a class. Arrays of Objects; Objects as function arguments: Pass by value, Pass by reference, Pointers to Objects.

Constructors: Declaration and Definition, Types of Constructors, (Default, Parameterized, Copy Constructors). Destructors: Definition and use.

Operator Overloading: Concept of Operator Overloading, Overloading unary and binary operators.

UNIT - III

Inheritance: Extending Classes Concept of inheritance, Base class, defining derived classes, Visibility modes: Public, Private, Protected; Types of Inheritance: Single inheritance:
Privately derived, publicly derived; Making a protected member inheritable, multilevel inheritance, multiple Inheritance and ambiguity of multiple inheritance, Hierarchal Inheritance, Hybrid, Nesting of classes.
Polymorphism: Definition, Application and demonstration of Data Abstraction, Encapsulation and Polymorphism, Static and Dynamic Polymorphism, Virtual Functions, pure virtual functions.

UNIT - IV

Exception Handling: Definition, Exception Handling Mechanism: Throwing mechanism and Catching Mechanism, Rethrowing an Exception
File Processing: Opening and closing of file, Binary file operations, structures and file operations, classes and file operations, Random file processing.

Suggested Readings :

Essential :

1. E. Balaguruswamy, 2008 : Object Oriented Programming with C++, TMH.

Further Reading :

2. Bjarne Stroustrup, 2009 : The C++ Programming Language, Addison-Wesley Publishing Company.
3. Robert Lafore, 2003 : Object Oriented Programming in Turbo C++, Galgotia Pub.
4. Salaria, R. S. : Object Oriented Programming Using C++, Khanna Book Publishing Co. (P.) Ltd., New Delhi.

Title : Practical Based on Paper – CSA-DSC-3(Maj/Min)-301 (Programming in C++)
Paper Code: CSA-DSC-3(Maj/Min)-301P

Time Duration: 3 Hrs. **Number of Hours: 60**

Note : Paper will be set at the time of examination. Due weight-age may be given to the practical note-book and Assignments in evaluation.

Paper Title : Web Designing
Paper Code :CSA-DSC-3(Maj/Min)-302

Th	Pr	Cr
2	2	4

Theory External Marks: 45
Theory Internal Marks: 05
Number of Th. Hrs. : 30
Practical Ext. Marks : 50
Number of Pr. Hrs. : 60

Time Duration: 3 Hrs.
Level: L-200
Objective: This course enables students to create web pages using HTML, CSS, Java script and dream weaver.

Learning Outcome:
After completing this course students will be able to:

i. Get familiar with the terms related to World Wide Web.
ii. Use the front end web site developing technologies like HTML and CSS
iii. Know about the basics of JavaScript.
iv. Use Dreamweaver for creating Web Pages

Note :

i. The Question Paper will consist of Four Units.
ii. Examiner will set total of NINE questions comprising TWO questions from each Unit and ONE compulsory question of short answer type covering whole syllabi.
iii. The students are required to attempt ONE question from each Unit and the Compulsory question.
iv. All questions carry equal marks unless specified.

UNIT-I

Web Terminology: Web Server; Web Client /Browser, Understanding how a Browser communicates with a Web Server, Internet, Intranet, Extranet, WWW, URL

Introduction to HTML : Structure of an HTML program, Paragraph Breaks, Line Breaks; Emphasizing Material in a Web Page (Heading Styles, Drawing Lines); Text Styles (Bold, Italics, Underline); Other Text Effects (Centering (Text, Images etc.) ;Lists: Unordered List, Ordered Lists, Definition lists; Adding Images: Img element using Border, Width, Height, Align, ALT Attributes ;Tables: Caption Tag, Width, Border, Cell padding, Cell spacing, BGCOLOR, COLSPAN and ROWSPAN Attribute

UNIT-II

Linking Documents: Anchor tag, External Document References, Internal Document References and Image Maps

Frames: understanding frames, creating frames, Targeting Named Frames

Cascading style sheets (CSS): Style tag, Link tag, Types of CSS: In-Line, Internal, External Forms: Attributes of Form element, Input element: Text Element, Password, Button, Submit Button, Reset Button, The Checkbox, Radio, Text Area, Select and Option

UNIT-III

Java Script: Features, tokens, data types, variables, operations, control constructs, strings arrays, functions, core language objects, client side objects, event handling. Applications related to client side form validation

Other Built-In Objects in Java Script: The String Object, The Math Object, and The Date Object; User Defined Objects: Creating a User Defined Object, Instances, Objects within Objects

UNIT-IV

Creating Web Pages using Dream weaver: Introduction to Dreamweaver, Understanding Workspace Layout, Managing Websites, Creating a Website, Using Dream weaver Templates, Adding New Web Pages, Text and Page Format, Inserting Tables, Lists, Images, Adding Links.

Suggested Reading:

Essential :

1. Wanger&Wyke : Java Script Unleased, Pearson, Latest reprint New Delhi.
2. Bayross, Ivan : HTML, DHTML, JAVA SCRIPT by BPB, Latest reprint

Further Reading :

3. Joseph Lowery : Adobe Dream weaver CS6 Bible by Wiley India
4. Kogent Learning Solutions Inc DreamweaverCS5inSimple Steps by Wiley India

Title : Practical Based on Paper – CSA-DSC-3(Maj/Min)-302 (Web Designing)
Paper Code: CSA-DSC-3(Maj/Min)-302P

Time Duration: 3 Hrs. **Number of Hours: 60**

Note : Paper will be set at the time of examination. Due weight-age may be given to the practical note-book and Assignments in evaluation.

Semester-IV

Paper Title: Software Engineering
Paper Code: CSA-DSC-5(Maj/Min)-401

Th	Pr	Cr	
4	-	4	**Theory External Marks: 90**
			Theory Internal Marks: 10

Time Duration: 3 Hrs. **Number of Hours: 60**
Level: L-200
Objectives: This course enables students to understand Software Engineering Tools and Techniques.

Learning Outcome:
After the completion of this paper, student will be able to: -

(i) Use principles, concepts, methods, and techniques of the software engineering approach to produce quality software.
(ii) Apply software engineering principles and practices in the planning and development of an actual software product.

Note:
The Question Paper will consist of Four Units.

(i) Examiner will set total of **NINE** questions comprising **TWO** questions from eachUnit and **ONE** compulsory question of short answer type covering whole syllabi.
(ii) The students are required to attempt **ONE** question from each Unit and the Compulsory question.
(iii) All questions carry equal marks unless specified.

UNIT – I

Introduction to Software Engineering: Definition, Software Engineering goals, Characteristics of well-engineered software.
Software Process Models: Waterfall Model, Prototyping Model, Spiral Model.
Software Requirement Specification (SRS): Software Requirements, Definition of SRS, Characteristics of SRS, Components of SRS, Designing of SRS.

System Analysis: Principles of Structures Analysis, DFDs, E-R Diagrams, Data Dictionary.

UNIT – II

Software Design: Design Objectives, Design Principles, Concepts, Design Process, Design Methodologies: Structured Design, Modular Design, Object Oriented Design, User Interface Design and its elements and its Characteristics.

Software Project Planning & Scheduling: Objectives, Decomposition techniques, Planning and Scheduling Tools: GANTT Chart, PERT Chart, Critical Path Method and Work Breakdown Structure; Cost estimation, Cost estimation Models: Single Variable Model, COCOMO Model; Software Risks, Risk Assessment and management.

UNIT – III

Software Metrics: Role of Metrics and Measurement, Types of Software Metrics: Product Metrics, Software Size Metrics: LOC and Function Points, Process Metrics, People Metrics.

System Maintenance and Reliability: Software Quality and features, Maintenance and its types; Factors Affecting Software Reliability, Software Reliability vs Hardware Reliability, Software Reliability Metrics.

UNIT-IV

Software Testing Techniques: Introduction to Software Testing Process, Objectives of Software Testing, BBT & its Techniques, WBT & its Techniques. Integration Testing, Functional Testing, Object Oriented Testing, Alpha and Beta Testing.

Hardware and Software Selection: Types of Software, Procedure for Hardware/Software selection: Major phases in selection, Evaluation and Validation, Vendor Selection, Post – Installation Review. Software selection- Criteria for Software Selection, the evaluation process.

REFERENCES

1. Roger S. Pressman and Bruce R. Maxim: Software Engineering, Edition 2023, Tata-McGraw Hill Publishing House
2. Ali Behforooz and Frederick J. Hudson: Software Engineering Fundamentals, Edition 2021, Oxford University Press
3. Rajib Mall: Fundamentals of Software Engineering, Edition 2018, PHI Learning Pvt. Limited
4. Sommerville, I: Software Engineering, Edition 2017, Pearson Education.

Paper Title: Database Management System using Oracle
Paper Code: CSA-DSC-6(Maj)-402

Th	Pr	Cr
2	2	4

Theory External Marks:45
Theory Internal Marks: 05
Time Duration : 3 Hrs **Number of Hours: 30**

Level: L-200
Objective: This course aims at giving the students the insight of the underlying concepts of database management system and implement them using Database software.

Learning Outcome:
On successful completion of the course, students will be able to:
(i) Gain a solid grasp of fundamental database concepts, Including database models and schemas.
(ii) Understand the basic elements of a relational database management system
(iii) Master the basics of SQL and construct queries using SQL
(iv) Basic programming concepts using PL/SQL

Note:
(i) The Question Paper will consist of Four Units.
(ii) Examiner willset total of **NINE** questions comprising **TWO** questions from each Unit and **ONE** compulsory question of short answer type covering whole syllabi.
(iii) The students are required to attempt **ONE** question from each Unit and the Compulsory question.
(iv) All questions carry equal marks unless specified.

UNIT – I
Database Concepts: Introduction to databases, need, database architecture, Terminology: Tuple, Degree, attributes, Domain, Primary key, foreign keys, candidate keys; The 12 Rules (Codd's Rule) for an RDBMS, Normalization: First, second and third Normal Form, Boyce Codd Normal Form.

UNIT – II
Understanding SQL-1: Data Types, Creating Tables, Creating a Table with data from Another table, Inserting Values into a Table, Updating Column(s) of a Table, Deleting Row(s) from a Table, Dropping a Column, Querying database tables, Conditional retrieval of rows, Working with Null Values, Matching a pattern from a table, ordering the result of a Query Aggregate Functions, Grouping the Result of a Query, creation and deletion of Views

UNIT – III
Understanding SQL-II: Managing privileges with Grant and Revoke Command, COMMIT and ROLLBACK, Functions: Arithmetic Functions, Character Functions, Date Functions, Group Functions, Querying Multiple Tables using Equi-Joins, Cartesian Joins, Outer Joins, Self-Joins, SET Operators: Union, Intersect, Minus.

UNIT – IV
PL/SQL: Introduction to PL/SQL, The Advantage of PL/SQL, PL/SQL Block Structure, PL/SQL Architecture, Fundamentals of PL/SQL, PL/SQL Data Types, Variables and Constants, Scope and Visibility of a Variable, Assignments and Expressions, Operator Precedence, Conditional and Iterative Control, Cursor Management in PL/SQL, Implicit/explicit Cursor Attributes, Exception Handling in PL/SQL; Predefined Exceptions, User Defined Exceptions.

Suggested Readings:

1. Elmasri & Navathe: Fundamentals of Database systems, 2017, 7th Edition, AddisonWesley, New Delhi.
2. Ivan Bayross: SQL, PL/SQL, 2020, Fourth Edition, BPB Publication, NewDelhi.
3. C.J.Date: An Introduction to Databases Systems 8th Edition, 2003, Addison Wesley, New Delhi.
4. Korth & Silberschatz: Database System Concept, 2010, 6th Edition, McGraw Hill International Edition
5. Bipin C.Desai: An Introduction to Database System, 2015, Galgotia Publication,NewDelhi
6. Raghu Ramakrishnan & Johannes Gehrke: Database ManagementSystems, 2002, 3rd edition, Mcgraw Hill International Edition.

Paper Code: CSA-DSC-6 (Maj)-402P
(Practical Based on paper CSA-DSC-6(Maj)-402)

Practical External Marks: 50
Number of Practical Hours :60

Time Duration: 3 Hrs.

List of Practicals:

Implementation of SQL concepts
DDL, DML, DCL, TCL, Data Retrieval, Functions, Aggregate Functions, Views, Joins, Set Operators

Implementation of Programsin PL/SQL
1. Write PL/SQL blocks to show the declaration of variables.
2. Write a PL/SQL program to check whether a number is even or odd
3. Write a PL/SQL program to find greatest of three numbers
4. Write a PL/SQL program to display the grade of a student using else-if ladder.
5. Write a PL/SQL program to display the description against a grade using CASE statement
6. Write a program in PL/SQL to print 1st n numbers.
7. Write a program in PL/SQL to find factorial of a number.
8. Write a program in PL/SQL to print table of a given number.
9. Write a program in PL/SQL to show implicit/explicit cursor.
10. Write a program in PL/SQL to include implicit/explicit exception.

Paper Title: Data Structure
Paper Code: CSA-DSC-7(Maj)-403

Th	Pr	Cr
2	2	4

Theory External Marks: 45
Theory Internal Marks: 05
Number of Hours :30

Level: L-200
Time Duration: 3 Hrs.

Objectives: The objective of this course is to familiarize students with Basic Data Structure and its implementation and to develop skills to apply appropriate Data Structure in Problem Solving.

Learning Outcome:
On successful completion of the course, students will be able to:
(i) Understand the Basic types of Data Structure, implementation, Applications.
(ii) Implement Linear and Non-Linear Data Structure.
(iii) Analyze and apply various searching and sorting techniques.
(iv) Develop Programming skills which helps in problem solving.

Note:
(i) The Question Paper will consist of Four Units.
(ii) Examiner will set total of **NINE** questions comprising **TWO** questions from each Unit and **ONE** compulsory question of short answer type covering whole syllabi.
(iii) The students are required to attempt **ONE** question from each Unit and the Compulsory question.
(iv) All questions carry equal marks unless specified.

UNIT – I

Data Structure: Introduction and Classification of Data Structure, Data Structure operations. Applications of Data Structure, Basic data Structures;
Arrays: Introduction, Types of Arrays, Memory representation, Applications and operations.

UNIT – II

Stacks: Introduction, representing stack in memory, Applications and operations.
Linked List: Introduction, Operations: -traversing, searching, inserting, deleting, Types of Linked List, Memory representation.

UNIT – III

Queue: Introduction, Types, Memory Representation.
Trees: Definition and Basic concepts, Representation in Contiguous Storage, Binary Tree, Binary Tree Traversal, Binary Search tree.

UNIT – IV

Graphs: Introduction, Memory Representation, Breadth First Search, Depth First Search.
Searching: Binary and Linear Search; **Sorting**: Bubble Sort, Merge Sort, Quick Sort.

Suggested Readings:
1. Lipschuitz L. Seymour, 2007: Data Structure, Schaum Outline Series, TMH, New Delhi.
2. Tannenbaum, Aaron M., 2015: Data Structure Using C & C++ 2e, Pearson.
3. Jean-Paul Tremblay & Paul G. Sorenson,2017: An Introduction to Data Structures with Applications, Publisher-Tata McGraw Hill.

Paper Code: CSA-DSC-7(Maj)-403P
(Practical Based on paper CSA-DSC-7(Maj)-403)

Practical External Marks: 50
Number of Practical Hours :60

Time Duration: 3 Hrs.

Implement Data Structures Operations using C / C++.

LIST OF PRACTICALS

1. Write a program to check whether a string is palindrome or not
2. To read n numbers and display it.
3. To demonstrate the concept of one-dimensional array.
4. Implementation of Program to perform the following operations on an Array
   - Traversal
   - Insertion
   - Deletion
5. Program to Implement Stack
   - Push an Element
   - Pop an Element
6. Write a program to implement linear and binary search algorithm.
7. Write a program to create a linked list and perform following operations:
   - Insertion
   - Deletion
   - Transversing
   - Searching
8. Program to implement Linear Queue using an array
9. Program to Implement Circular Queue using an array
10. Program to implement Linear Linked List
11. Program to implement Doubly Linked List
12. Program to perform Insertion in a Binary tree
13. Program to perform Deletion in a Binary tree
14. Program to perform Searching in a Binary tree
15. Program to perform Traversal in Graph
16. Write a program to sort a given list of number using following algorithm.
    - Bubble sort
    - Merge sort
    - Quick sort

Paper Title: Introduction to Artificial Intelligence
Paper Code: CSA-DSC-8(Maj)-404

Th	Pr	Cr	Theory External Marks: 90
4	-	4	Theory Internal Marks: 10
			Number of Hours: 60

Time Duration: 3 Hrs.

Objective: The course aims at introducing basic concepts of artificial intelligence, understanding of search techniques. To learn working of expert system and natural language processing.

Learning Outcome:
On successful completion of the course, students will be able to:
(i) Understanding of artificial intelligence, intelligent agents and state-space representation.
(ii) Understanding of working of searching techniques and its algorithms.
(iii) Understand basics of game playing, knowledge representation and expert systems.
(iv) Obtain insight about natural language processing and its components.

Note :
(i) The Question Paper will consist of Four Units.
(ii) Examiner will set total of **NINE** questions comprising **TWO** questions from each Unit and **ONE** compulsory question of short answer type covering whole syllabi.
(iii) The students are required to attempt **ONE** question from each Unit and the Compulsory question
(iv) All questions carry equal marks unless specified.

UNIT-I
Introduction: What is AI?, Intelligent Agents: Agents and environment, the concept of Rationality, the nature of the environment, the structure of agents, Introduction to AI technique, Turing test, History and developments in AI, Applications of AI, State space representation, Production systems and problem characteristics.

UNIT-II
Searching techniques: Breadth First Search, Depth First Search, Generate and test, Hill climbing, Best First Search, A* and AO* algorithms, Constraint Satisfaction Problem, Means-End Analysis.

UNIT-III
Game Playing, Knowledge Representation and Expert Systems: Introduction to Game Playing, Minimax Search, Alpha-Beta pruning. Propositional Logic, First Order Predicate Logic, Resolution Principle, Architecture and role of expert systems, atleast two case studies of Expert Systems.

UNIT-IV
Natural Language Processing: Introduction to Natural Language Processing, Components of Natural Language Processing, Tokenisation and Sentence splitting, Lexical Analysis, Syntactic Analysis, Semantic Analysis, Discourse Integration and Pragmatic Analysis.

Suggested Readings:
1. Elaine Rich and Kevin Knight, "Artificial Intelligence," Tata McGraw Hill, 2018
2. Patterson, Dan W.: Introduction to Artificial Intelligence and Expert systems, Pearson Education, 2015
3. Sharma, A., Essentials of AI and Soft Computing, PHI Learning, 2024
4. Russell, Stuart, J. and Norvig, Peter, Artificial Intelligence - A Modern Approach, Pearson, 4th edition, 2020.

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Environment Conservation

SEMESTER-III

MINOR COURSE

ENVIRONMENTALCONSERVATION:EMERGING ISSUES ON ENVIRONMENT

Course Code: MINC3 Credits=4(T=3+ P=1)

Course Title: Environmental Conservation: Emerging Issues on Environment
Total Hours = 60 (Theory=45+Practical=15)
Total Marks=100(Theory=75+Practical=25)
Internal Assessment=20%

Course Outcomes

1. *The students will be able to understand about natural resources, their types and realize the importance of resources.*
2. *The students will be able to acquire deep knowledge about land and forest resources.*
3. *The students will be able to describe energy resources.*
4. *The students will be able to analyse Indian as well as global status of natural resources.*
5. *The students will be able to study conservation methods of natural resources.*

THEORY

Unit-I(15Hrs)
Environmental Pollution

1.1 Concept of Sound and Noise and Measurement
1.2 Noise Pollution; Anthropogenic and Natural Sources
1.3 Effects of Noise Pollution
1.4 Control of Noise pollution
1.5 Radiation Pollution ; Concept of Sources
1.6 Effects of Radiation Pollution and Control Measures

Unit-II(15Hrs)
Mineral Extraction and Its Impacts

2.1 Concept of Minerals and Ores
2.2 Process of Mineral Extraction
2.3 Mining and Types of Mines
2.4 Impact of Mining on Environment
2.5 Reclamation of Mines

UnitII(15Hrs)
Solid Waste Management

3.1 Concept of Solid Waste and Its types
3.2 Rising Solid Waste; Reasons and Issues
3.3 Impact of Solid Waste Accumulation on Environment
3.4 Solid Waste Management and Treatment
3.5 Concept of Vermicomposting and Vermiculture
3.6 Legislatives Measures on Solid Waste

PRACTICALS(15Hrs)

1. To prepare a report on Noise Pollution, its sources and effects in a given area
2. Tostudy and do survey on major minerals and their ores.
3. TodoaHousehold Waste Audit and give recommendations
4. Tostudy and compare different composting methods
5. To draw analysis of types of Solid waste and their management techniques
6. To visit Solid Waste Handling Site / Biomedical Incinerator/ landfills in the surrounding area.

Instructions for paper setter for Theory Paper Environment Conservation BA Semester IV (Theory

Paper: EMERGING ISSUES ON ENVIRONMENT
Total Marks: 75
Theory Marks: 65
Internal Assessment 10

1. There will be two sections.
2. Section A will be a total of 20 marks while section B will carry 45 marks
3. Section A will consist of 10 compulsory questions covering the whole syllabus carrying 2 Marks each (Total Marks=2*10= 20)
4. Section B will be divided into three units.

Unit 1 will carry two questions from unit 1 of the syllabus. Students need to attempt one question out of these two.(15 Marks)
Unit 2 will carry two questions from unit 2 of the syllabus. Students need to attempt one question out of these two.(15 Marks)
Unit 3 will carry two questions from unit 3 of the syllabus. Students need to attempt one question out of these two.(15 Marks)
(Total Marks=15*3=45)

SEMESTER-IV
MINORCOURSE

ENVIRONMENTAL CONSERVATION: EIA AND ENVIRONMENT AUDITING

Course Code: MINC4 Credits=4(T=3+ P=1)

Course Title: Environmental Conservation: EIA and Environment Auditing
Total Hours=60(Theory=45+Practical=15)
Total Marks=100(Theory=75+Practical=25)
Internal Assessment=20%

Course Outcomes

1. *The students will be able to understand about the importance of EIA.*
2. *The student will be able to understand the impacts of different projects on environment.*
3. *The students will be acquiring knowledge about environment auditing.*
4. *The students will get knowledge about the career prospects in the field of EIA and Environment Auditing.*

THEORY

Unit-I (15Hrs)

1.1 EIA origin, development, purpose and aims
1.2 Principles of Environment Impact Assessment
1.3 EIA Methodology
1.4 EIA processes: Project screening, scoping, base-line data, impact identification; prediction, evaluation, valuation of environmental impacts, mitigation.
1.5 Public participation, presentation, review and decision making, monitoring and auditing.
1.6 Environmental Management Plan, Environmental components of EIA.

Unit-II(15Hrs)

2.1 Environmental Appraisal procedures in India
2.2 Impact identification methods.
2.3 Environmental impacts of mining industry; nuclear power plant, textile industry; pulp and paper industry; petroleum refining; pesticide manufacturing industry; fertilizer industry.
2.4 Case studies of EIA – Hydroelectric dam and river valley projects; thermal power plants and petroleum exploration.

Unit-III (15Hrs)

3.1 Types of environmental audits: Assessment and compliance audit
3.2 Occupation health and safety
3.2 Energy audits. Green audit, Water audit
3.4 Environmental Management systems in India
3.5 Environmental audit and EIA, Vocational prospects in the field of EIA

PRACTICALS (15Hrs)

1. To prepare a EIA report
2. To prepare an energy audit report of college campus
3. 3. To survey and prepare an Green audit report of college campus
4. To prepare a water audit report of college campus
5. Internship with some organization in the field of EIA or Environment Auditing

Instructions for paper setter for Theory Paper Environment Conservation BA Semester IV (Theory)

Paper: EIA AND ENVIRONMENT AUDITING

Total Marks: 75
Theory Marks: 65
Internal Assessment 10

1. There will be two sections.
2. Section A will be a total of 20 marks while section B will carry 45 marks
3. Section A will consist of 10 compulsory questions covering the whole syllabus carrying 2 Marks each (Total Marks=2*10= 20)
4. Section B will be divided into three units.

Unit 1 will carry two questions from unit 1 of the syllabus. Students need to attempt one question out of these two.(15 Marks)
Unit 2 will carry two questions from unit 2 of the syllabus. Students need to attempt one question out of these two.(15 Marks)
Unit 3 will carry two questions from unit 3 of the syllabus. Students need to attempt one question out of these two.(15 Marks)
(Total Marks=15*3=45)

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Industrial Chemistry

CURRICULUM AND CREDIT DISTRIBUTION FOR INDUSTRIAL CHEMISTRY AS **MAJOR** SUBJECT AS PER TABLE 2					
Table 2					
Sem	**Level**	**Course Code**	**Course Name**	**Credits**	**Total Credits**
III	L-200	(ICH-DSC-3 MAJ/MIN-301)	Industrial Chemistry and Environmental Sustainability	4	24
		Major - 3	To be chosen by the student	4	
		Minor – 3	To be chosen by the student	4	
		Minor - 4	To be chosen by the student	4	
		IDC-3	To be chosen from the available options	3	
		SEC-3	To be chosen from available options	3	
		ENG-3	Compulsory	2	
IV	L-200	(ICH-DSC-5 MAJ-401/MIN-302)	Chemical & Industrial Elemental Process	4	24
		(ICH-DSC-6 MAJ-402/MIN-401)	Heat and Mass Transfer Processes	4	
		Major - 5	To be chosen by the student	4	
		Major - 6	To be chosen by the student	4	
		Minor - 5	To be chosen by the student	4	
		ENG-4	Compulsory	2	
		VAC-3	To be chosen from the common basket	2	

CURRICULUM AND CREDIT DISTRIBUTION FOR INDUSTRIAL CHEMISTRY AS **MINOR** SUBJECT AS PER TABLE 1 AND TABLE 2				
Table 2				
Sem	**Level**	**Course Code**	**Course Name**	**Credits**
III	L-200	(ICH-DSC-3 MIN/MAJ-301)	Industrial Chemistry and Environmental Sustainability	4
	L-200	(ICH-DSC-5-MIN-302/MAJ-401)	Chemical & Industrial Elemental Process	4
IV	L-200	(ICH-DSC-6 MIN-401/MAJ-402)	Heat and Mass Transfer Processes	4

Note:-

1. In the first semester, students are offered three different subjects: A, B & C.
2. At the end of the second semester, each student has the option to choose any one of the two streams, namely Single Major or Multi-Disciplinary. The student choosing the Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student choosing the Multi-Disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in the Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for a 4-year degree B.SC HONOURS with research, on the condition that the Institution is a PU-approved Research Centre
5. The student opting for a Single Major must secure 50% of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for a Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year, securing 48 credits, will be awarded a UG Certificate provided he/she secure 4 credits from summer internship/community work. Further, the student exiting the programme after two years, securing 96 credits, will be awarded a UG Diploma provided he/she secure 4 credits from a summer internship. The student successfully securing 144 credits will be awarded a BA/BSc degree.
7. Language papers chosen as MIL-1 and MIL-2 should be on the same subject.
8. Once Major and Minor papers are chosen in the 3rd semester, the same will continue as Major and Minor subjects in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write a Dissertation of 08 credits in one of the two Major subjects.

Semester	**III**
Code of Paper	**ICH-DSC-3-MAJ/MIN-301**
Title of Paper	**Industrial Chemistry and Environmental Sustainability**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week** **Practical: 3 Hrs. per week**
Applicable to the students **1. Those who have opted for Industrial chemistry as a Major subject in Table 2.** **2. Those who have opted for Industrial chemistry as a Minor subject in Table 1 or Table 2.**	
LEARNING OUTCOMES SPECIFIC TO DISCIPLINE By studying this applied course, the students will be able to: • Explain the different sources and causes of air pollution, their control measures and equipment used for pollution control. • The student shall be able to give the details of water pollutants, the source, characterization and methods used in water purification. • Explain and discuss various aspects of waste management and strategies required for its minimization. • Explain the heavy metal ions in the environment, its monitoring, assessment and remediation techniques. • Emphasis is given to developing practical skills that will aid the student to understand various test involved in testing water samples, industrial and domestic.	
GENERIC OUTCOMES The second year of the course is designed to deepen the chemical temperament amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out data analysis, evaluation and draw logical conclusions. They work is supported by both theoretic and practical skill development.	

For practical work, groups shall be constituted, with student strength not exceeding 20 students per group.

Semester – 3
Syllabus for Theory Paper
ICH-DSC-3-MAJ/MIN-301
Industrial Chemistry and Environmental Sustainability

End semester = 67	**Credits - 03**
Internal Marks = 08	**Contact Hours – 3/week**
Examination Time allowed	**3 Hours**

UNIT-1 (15Hrs.)

Air Pollution: Introduction, Sources of air pollution,Causes of air pollution – types & sources from chemical industries, Classification of air pollutants: based on the origin, chemical composition and state of matter, Important air pollutants: oxides of Sulphur, CO, NOx, hydrocarbons, Suspended particulate matter, Effect of air pollution in human health, Effect of air pollution on animals, Effect of air pollution on plants, Effects of air pollution on aquatic life, Different types of plume behaviour: looping, coning, fanning, lofting, fumigating, and trapping plumes, Key factors influencing plume behavior, Importance of plumes for industries.

Air Pollution Control Measures & Equipment: Control of Particulate Emission – Control of Gaseous Emission: Absorption, adsorption, Combustion or incineration and condensation, Removal of particulate matter, Removal of Sulphur dioxide, Removal of oxides of nitrogen, Control of CO_2 and CO, Construction, working and principle of Spray tower, packed tower, Gravitational settling chambers, Cyclone separators, Fabric filters, Electrostatic precipitators, Wet scrubbers, Advantages and disadvantages.

UNIT-2 (15Hrs.)

Water pollution: Definition of water pollution, Categorization of pollutants by source (e.g., industrial, domestic, agricultural) and by type (e.g., physical, chemical, biological), Sources of water pollution, Analysis of the impacts on human health, ecosystems, and aquatic life, Techniques for measuring water pollution, Water purification methods. Effluent treatment plants (primary, secondary and tertiary treatment).

Water Quality Parameters for waste water, industrial water and domestic water: Physical Parameters: Turbidity, temperature, color, and total dissolved solids (TDS), Chemical Parameters: pH, acidity, alkalinity, hardness, chlorides, salinity, dissolved oxygen (DO), biochemical oxygen demand (BOD), chemical oxygen demand (COD), nitrates, phosphates, heavy metals, Biological Parameters, Water sampling procedures, Sample containers, Sample preservation and preparation, physicochemical analysis of water: pH, acidity, alkalinity and hardness.

UNIT-3 (15Hrs.)

Solid Waste Management: Introduction, Sources of solid waste, Types of solid waste, Composition of solid waste and its determination, Waste characterization, Classification of hazardous waste, Impacts of solid waste on human health and the environment, Methods of solid waste management: Sanitary landfill: Site selection, Landfilling methods, Design and operation of landfills, Incineration: Types of incinerators, Factor affecting incineration, Methods of incineration and Concept of recycling, Strategies for minimizing waste generation at the source, including waste reduction, reuse, and recycling.

Heavy metal ions in the environment: Sources of heavy metal pollution, Heavy metal behavior in the environment, Toxicity of specific heavy metals (Mercury, Copper, Chromium, Cadmium, Cobalt, lead, Arsenic), Mechanism of toxicity, Health effects on humans, Effects on ecosystems: impact on plant and animal life, Remediation techniques, Monitoring and assessment of heavy metal contamination: sampling techniques, analytical methods, and risk assessment.

Books Recommended

1. Rao, C. S.: Environmental Pollution Control Engineering, New Age Int. (P) Ltd. Publishers, 2006.
2. Rao, M.N. and Rao, H.V.N.: Air Pollution, Tata McGraw-Hill Publishing Company Limited, New Delhi, 2003.
3. Zajic, J.E.: Water pollution, Marcel Dekker, Inc. New York, 1971.
4. Davis, M.L. and Cornwell, D.A.: Introduction to Environmental Engg. McGraw-Hill International Editions, 1991.
5. Hazardous Waste minimization by Harry Freeman (1990).

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the compulsory question and one question each from UNIT-1, UNIT-2 and UNIT-3.
2. The use of scientific calculators is not allowed.

Semester-3
Syllabus for Practical Paper
ICH-DSC-3-MAJ/MIN-301
Industrial Chemistry and Environmental Sustainability

End semester = 22	**Credit - 01**
Internal Marks = 03	**Contact Hours – 3/week**
Examination Time allowed	**3 Hours**

Industrial Chemistry Lab

1. To determine the pH of given samples using (a) pH paper and (b) digital pH meter.
2. To determine the acidity of the given sample of water.
3. To determine the alkalinity of the given sample of water.
4. To determine the hardness of a water sample by measuring the amount of calcium present.
5. To perform the Limit test for Sulphate.
6. To perform the Limit test for chloride.
7. Calibration of conductivity meter and determination of conductivity of given water sample.
8. Determination of heavy metal ion impurities from given waste water sample using titration method.
9. Power point presentations.

Books Recommended

1. Vogel: Practical Organic Chemistry, 3rd Edition, ELBS, 1989.
2. Vogel: Textbook of quantitative analysis, 5th Edition, ELBS, 1989.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (i.e. half hour for written and next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written+5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.

Semester	**III**
Code of Paper	**ICH-DSC-5-MIN-302**
Title of Paper	**Chemical & Industrial Elemental Process**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of the Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of the Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week** **Practical: 3 Hrs. per week**
Applicable to the students **1. Those who have opted for Industrial chemistry as a Major subject in Table 2.** **2. Those who have opted for Industrial chemistry as a Minor subject in Table 1 or Table 2.**	
LEARNING OUTCOMES SPECIFIC TO DISCIPLINE By studying this applied course, the students will be able to: • Explain that out of thedifferent sources of organic compounds, how petroleum, natural gas and coal play a substantial role in providing an array of organic compounds. • The student shall be able to give the details of different physiochemical processes required for extraction of metals. • Explain and discuss various types of inorganic material of industrial importance such as Alumina, Silicates and Zeolites. • Emphasis is given to developing practical skills that will aid the student to understand various test involved in adsorption of heavy metals. And how to estimate the different components of coal.	
GENERIC OUTCOMES The second year of the course is designed to deepen the chemical temperament amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out data analysis, evaluation and draw logical conclusions. They work is supported by both theoretic and practical skill development.	

For practical work, groups shall be constituted, with student strength not exceeding 20 students per group.

Semester-3
Syllabus for Theory Paper
ICH-DSC-5-MIN-302
Chemical & Industrial Elemental Process

End semester = 67	**Credits - 03**
Internal Marks = 08	**Contact Hours – 3/week**
Examination Time allowed	**3 Hours**

UNIT-1 (15Hrs.)

Raw materials for organic compounds: Sources of organic chemicals, Petroleum: crude oil, refining, petrochemicals, Fractionation of crude oilcomposition using fractional distillation,Understanding the properties and uses of different fractions like refinery gases, gasoline, kerosene, diesel, fuel oil, and lubricating oils, purpose and mechanisms of cracking, types of cracking, Reforming processes, hydroforming, isomerization.

Natural gas:Hydrocarbon gases (Methane, Ethane, Propane, Butane, and Pentane), Non-hydrocarbon gases (Carbon dioxide, Nitrogen, Hydrogen sulfide, Water Vapor).

Coal: Types (Anthracite, Bituminous, Subbituminous, Lignite), Coal structure and composition, Properties of coal, distillation of Coal, advantages and disadvantages of coal.

UNIT-2 (15Hrs.)

Surface chemistry and Interfacial phenomena: Introduction to surface and interfacial phenomena, Surface and interfacial tension, Absorption and adsorption, Adsorption isotherm: Freundlich and Langmuir adsorption isotherms.

Catalysis: Introduction to Catalysis, Fundamental Concepts, Characteristics of Catalysts, Specificity of Catalyst, Types of Catalysis-Homogeneous and Heterogeneous, Catalytic Steps in Heterogeneous Catalysis, Activation Energy, their mechanism of action and their applications, Catalytic poisoning, Catalyst Promoter, Catalyst properties, Activity of a catalyst, Autocatalysis, Negative catalysis.

Basic metallurgical operations: Pulverizations, Calcinations, Roasting, Refining, Types of refining: Fire Refining, Electrolytic Refining, Zone Refining.

UNIT-3 (15Hrs.)

Physicochemical principles of extraction of Metals: Fundamental principles and methods used to extract metals from their ores, Occurrence of Metals, Extraction Methods, Ore Concentration, Extraction of specific metals like – Iron: (Concentration of the Ore, Blast Furnace, Reduction and Collection), Lead: (Ore Concentration, Roasting, Smelting, Refining), Silver:(leaching), Aluminum: (Bayer Process).

Inorganic materials of industrial importance - Availability, Forms, Structure, and Modifications of Key Inorganic Materials, Characteristics of materials like Alumina: Definition and Formation, Structure and Properties, Industrial Applications, Silicates, Zeolites: structure, properties, and applications of zeolites.

Books Recommended

1. Admson, A.W., Physical Chemistry of Surfaces, 4th edition, Pubs: John Wiely& Sons, New York, 1982.
2. Austin H. T., Shreve's Chemical Process Industries, Pubs: McGraw Hill Book Company,New York (1984).

3. Kent James A. (ed.), Reigel's Handbook of Industrial Chemistry, Pubs: Van Nostrand inhold Company, London (1983).
4. Pandey G.N., Text Book of Chemical Technology, Vol. I & II, Pubs: Vikas Publishing House, Pvt. Ltd., New Delhi (1999).
5. Buchner V., Sohliebs P., Winter G. & Buchel K.H., Industrial Inorganic Chemistry, Pubs:V. Ch. Publishers, New York (1989).
6. Badger W.L. and Bancher J.T., Introduction to Chemical Engineering, Pubs: McGraw-HillCo., U.S.A,(1986).
7. McCabe W.L. and Smith J.C., Unit Operations in Chemical Engineering, Pubs:McGraw-HillBook Company, New York,(1984).
8. Perry J.H., Chemical Engineering Handbook, Pubs: McGraw-Hill Book Company, NewYork, (1993).

Instructions for the examiner

4. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
5. **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
6. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the compulsory question and one question each from UNIT-1, UNIT-2 and UNIT-3.
2. The use of scientific calculators is not allowed.

Semester-3
Syllabus for Practical Paper
ICH-DSC-5-MIN-302
Chemical & Industrial Elemental Process

End semester = 22	**Credit - 01**
Internal Marks = 03	**Contact Hours – 3/week**
Examination Time allowed	**3 Hours**

Industrial facets of Chemistry Lab

1. Adsorption of heavy metal ions from given waste water sample using various adsorbents:
   (a) silica (b) nanoparticles
2. Study of Freundlich and Langmuir adsorption isotherm models and determination of isotherm constants.
3. Determination of surface tension of NaCl/Sugar at different concentrations.
4. Limit tests for heavy metals
   (a) lead (b) Iron
5. To determine the kinematic viscosity and absolute viscosity of the given oil at different temperatures using Redwood Viscometer.
6. Determination of volatile content in coal.
7. Determination of ash content in coal.
8. Power point presentations.

Books Recommended

1. Vogel: Practical Organic Chemistry, 3rd Edition, ELBS, 1989.
2. Vogel: Textbook of quantitative analysis, 5th Edition, ELBS, 1989.
3. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11thedition.
4. Pubs: R. Chand & Co., New Delhi, 2002.
5. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw HillPublishing Co. Ltd., 1998.
6. Levitt, B.P., Findlays Practical Physical Chemistry; 8th edition, Pubs: Longman GroupLtd., London & New York, 1978.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (i.e. half hour for written and next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written+5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.

Semester	**IV**
Code of Paper	**ICH-DSC-5-MAJ-401**
Title of Paper	**Chemical & Industrial Elemental Process**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week** **Practical: 3 Hrs. per week**
Applicable to the students **2. Those who have opted for Industrial chemistry as a Major subject in Table 2.** **3. Those who have opted for Industrial chemistry as a Minor subject in Table 1 or Table 2.**	
LEARNING OUTCOMES SPECIFIC TO DISCIPLINE By studying this applied course, the students will be able to: • Explain that out of thedifferent sources of organic compounds, how petroleum, natural gas and coal play a substantial role in providing an array of organic compounds. • The student shall be able to give the details of different physiochemical processes required for extraction of metals. • Explain and discuss various types of inorganic material of industrial importance such as Alumina, Silicates and Zeolites. Emphasis is given to developing practical skills that will aid the student to understand various test involved in adsorption of heavy metals. And how to estimate the different components of coal.	
GENERIC OUTCOMES The second year of the course is designed to deepen the chemical temperament amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out data analysis, evaluation and draw logical conclusions. They work is supported by both theoretic and practical skill development.	

For practical work, groups shall be constituted, with student strength not exceeding 20 students per group.

4th semester: TRAINING/WORKSHOP/INDUSTRIAL VISITS

1.Visit to a local industry during the year and students to submit a report on the same.

2. Students will also submit a project report on water-effluent analysis/soil analysis/ food adulteration etc.

3. Student to submit a report on one small scale unit: Safety matches, Naphthalene balls, Wax candles, Shoe Polish, Gum Paste, Pen Ink, Chalk, Plaster of Paris

4. Each Student to present a seminar relevant to the syllabus.

Semester – 4
Syllabus for Theory Paper
ICH-DSC-5-MAJ-401
Chemical & Industrial Elemental Process

End semester = 67	**Credits - 03**
Internal Marks = 08	**Contact Hours – 3/week**
Examination Time allowed	**3 Hours**

UNIT-1 (15Hrs.)

Raw materials for organic compounds: Sources of organic chemicals, Petroleum: crude oil, refining, petrochemicals, Fractionation of crude oilcomposition using fractional distillation,Understanding the properties and uses of different fractions like refinery gases, gasoline, kerosene, diesel, fuel oil, and lubricating oils, purpose and mechanisms of cracking, types of cracking, Reforming processes, hydroforming, isomerization.

Natural gas:Hydrocarbon gases (Methane, Ethane, Propane, Butane, and Pentane), Non-hydrocarbon gases (Carbon dioxide, Nitrogen, Hydrogen sulfide, Water Vapor).

Coal: Types (Anthracite, Bituminous, Subbituminous, Lignite), Coal structure and composition, Properties of coal, distillation of Coal, Advantages and disadvantages of coal.

UNIT-2 (15Hrs.)

Surface chemistry and Interfacial phenomena: Introduction to surface and interfacial phenomena, Surface and interfacial tension, Absorption and adsorption, Adsorption isotherm: Freundlich and Langmuir adsorption isotherms.

Catalysis: Introduction to Catalysis, Fundamental Concepts, Characteristics of Catalysts, Specificity of Catalyst, Types of Catalysis-Homogeneous and Heterogeneous, Catalytic Steps in Heterogeneous Catalysis, Activation Energy, their mechanism of action and their applications, Catalytic poisoning, Catalyst Promoter, Catalyst properties, Activity of a catalyst,Autocatalysis, Negative catalysis.

Basic metallurgical operations: Pulverizations, Calcinations, Roasting, Refining, Types of refining: Fire Refining, Electrolytic Refining, Zone Refining.

UNIT-3 (15Hrs.)

Physicochemical principles of extraction of Metals: Fundamental principles and methods used to extract metals from their ores, Occurrence of Metals, Extraction Methods, Ore Concentration, Extraction of specific metals like – Iron: (Concentration of the Ore, Blast Furnace, Reduction and Collection), Lead: (Ore Concentration, Roasting, Smelting, Refining), Silver:(leaching), Aluminum: (Bayer Process).

Inorganic materials of industrial importance - Availability, Forms, Structure, and Modifications of Key Inorganic Materials, Characteristics of materials like Alumina: Definition and Formation, Structure and Properties, Industrial Applications, Silicates, Zeolites: structure, properties, and applications of zeolites.

Books Recommended

1. Admson, A.W., Physical Chemistry of Surfaces, 4th edition, Pubs: John Wiely& Sons, New York, 1982.
2. Austin H. T., Shreve's Chemical Process Industries, Pubs: McGraw Hill Book Company,New York (1984).
3. Kent James A. (ed.), Reigel's Handbook of Industrial Chemistry, Pubs: Van Nostrand inhold Company, London (1983).

4. Pandey G.N., Text Book of Chemical Technology, Vol. I & II, Pubs: Vikas Publishing House, Pvt. Ltd., New Delhi (1999).
5. Buchner V., Sohliebs P., Winter G. & Buchel K.H., Industrial Inorganic Chemistry, Pubs:V. Ch. Publishers, New York (1989).
6. Badger W.L. and Bancher J.T., Introduction to Chemical Engineering, Pubs: McGraw-HillCo., U.S.A,(1986).
7. McCabe W.L. and Smith J.C., Unit Operations in Chemical Engineering, Pubs:McGraw-HillBook Company, New York,(1984).
8. Perry J.H., Chemical Engineering Handbook, Pubs: McGraw-Hill Book Company, NewYork, (1993).

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the compulsory question and one question each from UNIT-1, UNIT-2 and UNIT-3.
2. The use of scientific calculators is not allowed.

Semester-4
Syllabus for Practical Paper
ICH-DSC-5-MAJ-401
Chemical & Industrial Elemental Process

End semester = 22	**Credit - 01**
Internal Marks = 03	**Contact Hours – 3/week**
Examination Time allowed	**3 Hours**

Industrial facets of Chemistry Lab

1. Adsorption of heavy metal ions from given waste water sample using various adsorbents:
   (b) silica (b) nanoparticles
2. Study of Freundlich and Langmuir adsorption isotherm models and determination of isotherm constants.
3. Determination of surface tension of NaCl/Sugar at different concentrations.
4. Limit tests for heavy metals
   (b) lead (b) Iron
5. To determine the kinematic viscosity and absolute viscosity of the given oil at different temperatures using Redwood Viscometer.
6. Determination of volatile content in coal.
7. Determination of ash content in coal.
8. Power point presentations.

Books Recommended

1. Vogel: Practical Organic Chemistry, 3rd Edition, ELBS, 1989.
2. Vogel: Textbook of quantitative analysis, 5th Edition, ELBS, 1989.
3. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11thedition.
4. Pubs: R. Chand & Co., New Delhi, 2002.
5. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw HillPublishing Co. Ltd., 1998.
6. Levitt, B.P., Findlays Practical Physical Chemistry; 8th edition, Pubs: Longman GroupLtd., London & New York, 1978.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (i.e. half hour for written and next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written+5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.

Semester	**IV**
Code of Paper	**ICH-DSC-6-MAJ-402/MIN-401**
Title of Paper	**Heat and Mass Transfer Processes**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of the Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of the Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week** **Practical: 3 Hrs. per week**
Applicable to the students 3. **Those who have opted for Industrial chemistry as a Major subject in Table 2.** 4. **Those who have opted for Industrial chemistry as a Minor subject in Table 1 or Table 2.**	
LEARNING OUTCOMES SPECIFIC TO DISCIPLINE By studying this applied course, the students will be able to: • Explain the different modes of heat transfer from convection to conduction. And types of heat exchangers. • The student shall be able to give the details of different types of distillation and drying techniques. • Explain and discuss various types of evaporation and filtration techniques, their advantages and disadvantages. • Emphasis is given to developing practical skills that will aid the student to understand various test involved in heat transfer methods. Rate of evaporation and filtrations.	
GENERIC OUTCOMES The second year of the course is designed to deepen the chemical temperament amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out data analysis, evaluation and draw logical conclusions. They work is supported by both theoretic and practical skill development.	

For practical work, groups shall be constituted, with student strength not exceeding 20 students per group.

Semester – 4
Syllabus for Theory Paper
ICH-DSC-6-MAJ-402/MIN-401
Heat and Mass Transfer Processes

End semester = 67	**Credits - 03**
Internal Marks = 08	**Contact Hours – 3/week**
Examination Time allowed	**3 Hours**

UNIT-1 (15Hrs.)

Heat transfer through Conduction and Convection: Introduction to Heat Transfer, Modes of Heat Transfer: Conduction, Convection, and Radiation, Fundamentals: Heat transfer mechanisms, definitions, and units, Heat Transfer Mechanisms, Conduction Heat Transfer: Steady state conduction, conduction through a plain wall, composite wall, cylindrical wall, Convective Heat Transfer: Natural convection and forced convection.

Heat exchangers: Types of Heat exchangers- Construction Types: 1,1 and 1,2 shell and Tube type; Difference between single pass and multipass shell and tube heat exchanger, finned tube heat exchangers, plate heat exchangers, Double pipe heat exchanger, Fluid Flow Configurations: Parallel Flow, Crossflow, Counterflow, Applications of Heat exchangers in various chemical processes.

UNIT-2 (15Hrs.)

Distillation-Introduction, Applications, Vapour liquid equilibrium, Boiling point diagram, Ideal solutions, Raoult's law, Dalton's law, Henry's law, Volatility, Relative volatility, Batch and continuous distillation, Classification of distillation methods: working and construction of Simple distillation or differential distillation, Rayleigh equation, working and construction of Flash distillation, Minimum boiling azeotropic solutions, Maximum boiling azeotropic distillation, working and construction of Azeotropic distillation and Extractive distillation with examples, working and construction of Steam distillation, Advantages of steam distillation, Rectifying section and stripping sections of distillation column.

Drying- Introduction, Definition and Importance of Drying, Objectives of Drying, Applications of drying operation, Theory of drying, Bound water, Unbound water, Mechanism of drying process, Diffusion theory, capillarity theory, Pressure gradient theory, Equilibrium moisture content (EMC), Free moisture content (FMC), Percentage loss on drying, Percentage moisture content, Drying rate, Analysis of factors influencing the drying rate, including air velocity, temperature, and moisture content. Graphical representation of drying rate vs moisture content using FMC curve, Classification of drying equipment: Principle, working, construction and uses of tray dryer, rotary dryer, flash dryer, fluidized bed dryer, drum dryer, spray dryer, Advantages and Disadvantages.

UNIT-3 (15Hrs.)

Evaporation- Introduction, Definition, Importance, Distinguishing feature of evaporation with other heat processes, Applications, Factors Affecting Evaporation: Temperature, vapour pressure, surface area, moisture content of the feed, types of product required, time of evaporation, film and deposits, economic factor, Classification of Evaporators, Principle, working, construction and uses of Horizontal tube evaporator, Vertical tube evaporator, short tube (standard) Evaporator, forced circulation evaporators, falling film evaporators, climbing film (upward flow) evaporators or rising film evaporator, Advantages and Disadvantages, Economy of a single effect evaporator.

Filtration-Introduction, Definition, Process of filtration, Clarification, Applications, Driving forces, Mechanism of filtration: Straining, Impingement, Entanglement, Attractive forces, Types of filtration, Surface filtration (Screen filtration), Depth filtration, Theories of filtration, Rate of filtration, Factors influencing filtration process, filter media and filter aids, Classification of filtration equipment, Principle, working, construction and uses of Plate and frame filter press, rotary drum filter, bag filter, Advantages and Disadvantages.

Books Recommended

1. McCabe, W.L., Smith, J.C. and Harriot H.P.: Unit Operations of Chemical Engineering, 7th Edition, McGraw Hill,2005.
2. Holman, J.P.: Heat Transfer, McGraw Hill Publication Co.Ltd., New Delhi, 9th Edition, 2008.
3. Chapman, A.J. : Heat Transfer, McMillan Publishing Co., 4th Edition, 1984
4. Kern, D.Q. : Process heat Transfer, Tata McGraw Hill Publishing Co., New Delhi, Edition 1997, 2006
5. Kreith, F.: Principles of Heat Transfer, Harper & Row Pub., London.
6. Geankoplis, C.J.: Transport Processes and Unit Operations, Prentice Hall of India Pvt. Ltd., 3rd Edition, 1999.
7. Treybal, Robert E.: Mass Transfer Operations, 3rd Edition, McGraw-Hill, 1981.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the compulsory question and one question each from UNIT-1, UNIT-2 and UNIT-3.
2. The use of scientific calculators is not allowed.

Semester – 4
Syllabus for Practical Paper
ICH-DSC-6-MAJ-402/MIN-401
Heat and Mass Transfer Processes

End semester = 22	**Credit - 01**
Internal Marks = 03	**Contact Hours – 3/week**
Examination Time allowed	**3 Hours**

Heat and Mass Transfer Operation Lab

1. Determination of heat transfer coefficient under forced convection.
2. Determination of heat transfer coefficient under natural convection.
3. To separate a mixture of water and alcohol using simple distillation.
4. To study the factors affecting the rate of evaporation.
5. To do the effect of thickness of filter medium and pressure difference across the filter medium using (a) Simple Filtration (b) Vacuum Filtration
6. To determine the porosity of the given packing material for packed column.
7. To determine the % loss on drying, % moisture content and drying rate of given sample.
8. Power point presentations.

Books Recommended

1. McCabe, W.L., Smith, J.C. and Harriot H.P.: Unit Operations of Chemical Engineering, 7th Edition, McGraw Hill,2005.
2. Holman, J.P.: Heat Transfer, McGraw Hill Publication Co.Ltd., New Delhi, 9th Edition, 2008.
3. Chapman, A.J.: Heat Transfer, McMillan Publishing Co., 4th Edition, 1984.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (i.e. half hour for written and next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written+5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.

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Mathematics

AS PER NEP (2020)

Syllabus of **Mathematics** for UG Classes

Table-1 (SCIENCE TEMPLET)

SINGLE MAJOR, SINGLE / DOUBLE MINOR. **BSc, BSc HONOURS AND BSc HONOURS WITH RESEARCH**

Sem.	Level	MAJOR (4 Cr)	MINOR (4 Cr)	MDC (3 Cr) Any paper in each of the first three semesters	VAC (2 Cr) Required three papers, one for each in 1st, 2nd, & 4th semesters	SEC (3 Cr) Required three papers, one for each of the first three semesters	AEC (2 Cr)	PBI / HCP (2 Cr)
I	100	• MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry	• MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry	• MAT-MDC-101: Basic Mathematics • MAT-MDC-102: Calculus • MAT-MDC-103: Algebra	• MAT-VAC-101: Vedic Mathematics • MAT-VAC-102: Vedic Algebra • MAT-VAC-103: Theory of Numbers • MAT-VAC-104: Theory of Numbers with html	• MAT-SEC-101: Maths Based Practicals • MAT-SEC-102: Maths Based Programming And Practicals • MAT-SEC-103: Latex And Html • MAT-SEC-104: Quantitative Aptitude • MAT-SEC-105: Discrete Mathematics	Eng -1	Pbi-1/ HCP-1
II	100	• MAT-DSC2 (MAJ/MIN)-201: Calculus-I	• MAT-DSC2 (MAJ/MIN)-201: Calculus-I				Eng -2	Pbi-2/ HCP-2
III	200	• MAT-DSC3 (MAJ/MIN)-301: Theory of Equations • MAT-DSC3 (MAJ)-302: Differential Equations-I	• MAT-DSC3 (MAJ/MIN)-301: Theory of Equations • MAT-DSC3 (MAJ)-302: Differential Equations-I				Eng-3/ MIL-1	
IV	200	• MAT-DSC4 (MAJ/MIN)-401: Calculus-II • MAT-DSC4 (MAJ)-402: Discrete Mathematics and Graph Theory • MAT-DSC4 (MAJ)-403: Coordinate Geometry-I • MAT-DSC4 (MAJ)-404: Numerical Analysis	• MAT-DSC4 (MAJ/MIN)-401: Calculus-II				Eng-4/ MIL-2	
V	300	• MAT-DSC5 (MAJ/MIN)-501: Linear Algebra • MAT-DSC5 (MAJ)-502: Sequence and Series of real numbers • MAT-DSC5 (MAJ)-503: Differential Equations-II • MAT-DSC5 (MAJ)-504: Solid Geometry • MAT-DSC5 (MAJ)-505: Statics			Internship / community work			
VI	300	• MAT-DSC6 (MAJ/MIN)-602: Analysis • MAT-DSC6 (MAJ)-603: Linear Programming • MAT-DSC6 (MAJ)-604: Probability Theory • MAT-DSC6 (MAJ)-605: Dynamics • MAT-DSC6 (MAJ)-606: Modern Algebra	MAT-DSC6 (MAJ/MIN)-601: Analytical Geometry					

Table-2 (SCIENCE TEMPLET)
MULTI-DISCIPLINARY / INTER-DISCIPLINARY**. BSc, BSc HONOURS AND BSc HONOURS WITH RESEARCH**

Sem.	Level	MAJOR (4 Cr)	MINOR (4 Cr)	MDC (3 Cr) Any paper in each of the first three semesters	VAC (2 Cr) Required three papers, one for each in 1st, 2nd, & 4th semesters	SEC (3 Cr) Required three papers, one for each of the first three semesters	AEC (2 Cr)	PBI / HCP (2 Cr)
I	100	• MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry	• MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry	• MAT-MDC-101: Basic Mathematics • MAT-MDC-102: Calculus • MAT-MDC-103: Algebra	• MAT-VAC-101: Vedic Mathematics • MAT-VAC-102: Vedic Algebra • MAT-VAC-103: Theory of Numbers • MAT-VAC-104: Theory of Numbers with html	• MAT-SEC-101: Maths Based Practicals • MAT-SEC-102: Maths Based Programming And Practicals • MAT-SEC-103: Latex And Html • MAT-SEC-104: Quantitative Aptitude • MAT-SEC-105: Discrete Mathematics	Eng -1	Pbi-1/ HCP-1
II	100	• MAT-DSC2 (MAJ/MIN)-201: Calculus-I	• MAT-DSC2 (MAJ/MIN)-201: Calculus-I				Eng -2	Pbi-2/ HCP-2
III	200	• MAT-DSC3 (MAJ/MIN)-301: Theory of Equations	• MAT-DSC3 (MAJ/MIN)-301: Theory of Equations • MAT-DSC3 (MAJ)-302: Differential Equations-I				Eng-3/ MIL-1	
IV	200	• MAT-DSC4 (MAJ/MIN)-401: Calculus-II • MAT-DSC4 (MAJ)-402: Differential Equations-I	• MAT-DSC4 (MAJ/MIN)-401: Calculus-II				Eng-4/ MIL-2	
V	300	• MAT-DSC5 (MAJ/MIN)-501: Linear Algebra • MAT-DSC5 (MAJ)-502: Sequence and Series of real numbers	• MAT-DSC5 (MAJ/MIN)-501: Linear Algebra		Internship / community work			
VI	300	• MAT-DSC6 (MAJ/MIN)-601: Analytical Geometry • MAT-DSC6 (MAJ)-602: Analysis	• MAT-DSC6 (MAJ/MIN)-601: Analytical Geometry • MAT-DSC6 (MAJ)-602: Analysis					

Notes.

1. Members of the Board of Studies unanimously decided not to offer mathematics in Table-3 of Arts stream.

2. Practicals (wherever included in the syllabi) may be conducted into small groups (say 15 – 20 students) to facilitate the smooth conduct of practical sessions.

3. Scientific Calculator is allowed in the Numerical Analysis Paper.

IMPORTANT NOTE: All papers in VAC / SEC / MDC would be floated in each of the first three semesters. However, the students are not allowed to repeat any paper.

Semester - III
MAT-DSC3 (MAJ/MIN)-301: THEORY OF EQUATIONS

Credits = 4
Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.
Exam Time: 3 hours
Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 6 parts covering the entire syllabus. Two questions will be set from each Unit.
2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.
3. The 20 marks of internal assessment has following division
Mid Semester Exam = 10 marks,
Assignment / Project work = 5 marks,
Classroom participation = 3 marks**,**
Attendance = 2 marks for more than 85% and 1 mark for more than 75% attendance.

Objective: The objective of this course is to provide a comprehensive understanding of polynomials and polynomial equations. Students will explore methods for solving these equations, the relationship between roots and coefficients, and various techniques for transforming polynomial equations. Additionally, they will discover key theorems and interesting properties related to polynomials, deepening their problem-solving skills in this area.

Learning Outcomes: After attending this course students will be able to

(i) Analyze the relationship between the roots and coefficients of a polynomial,

(ii) Solve polynomial equations up to degree 4 by learning certain classical methods.

(iii) Apply transformations to polynomial equations and understand their implications.

(iv) Solve complex problems involving polynomial equations and transformations.

Unit-I

Polynomials (definition and examples), Euclid's algorithm, synthetic division, common divisors, G.C.D. of polynomials. Roots of a polynomial equation, repeated roots and their multiplicity, common roots. Fundamental theorem of Algebra, Factor theorem, Complex roots of real polynomials occur in conjugate pairs with same multiplicity. Irrational roots of polynomials over rationals occur in conjugate pairs with same multiplicity.

Unit-II

Relation between roots and coefficients, Vieta's formulae, symmetric functions, diminishing roots of a polynomial equation by h and its application, solution of a cubic when its roots are in A.P. / G.P., solution of a biquadratic when its roots are in A.P. (respect. in G.P.) and sum (resp. product) of two roots is given, Descartes' Rule of Signs, Newton's method of divisors for integral roots.

Unit-III

Transformation of equations: Transform the given polynomial equation into another such that signs of the roots changed, roots multiplied by a constant, roots are symmetric functions of the roots of the original equation. Solutions of cubic and bi-quadratic equations when their roots are in H.P.

Unit-IV

Cardan's method and trigonometric methods for solving cubic equations. Discriminant and nature of roots, of a real cubic equation. Descartes' and Ferrari's method of solving a bi-quadratic equation.

References:

1. **S. R. Knight and H.S. Hall** : Higher Algebra, H. M. Publications, 1994.
2. **Chandrika Prasad**: Text Book on Algebra and Theory of Equations, Pothishala Private Ltd., Allahabad.
3. **William S Burnside and A W Panton**: The theory of equations, Dublin, Hodges, Figgis and co. Dublin University Press Series (1881).

Semester - III
*MAT-DSC3 (MAJ/MIN)-302: DIFFERENTIAL EQUATION-I

Credits = 4
Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.
Exam Time: 3 hours
Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.
2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.
3. The 20 marks of internal assessment has following division
Mid Semester Exam = 10 marks,
Assignment / Project work = 5 marks,
Classroom participation = 3 marks**,**
Attendance = 2 marks for more than 85% and 1 mark for more than 75% attendance.

Objective: The objective of this course is to equip students with essential techniques for solving different types of differential equations. It also introduces the fundamentals of calculus involving two and three variables, which are crucial for understanding and solving differential equations effectively.

Learning Outcomes: By the end of this course, students will be able to:
1. Apply appropriate methods to solve various types of differential equations.
2. To understand calculus concepts for functions of two and three variables.

Unit-I

Limit and continuity of functions of two and three variables. Partial differentiation up to second order. Total differential, differentiation of composite and implicit functions. Euler's Theorem on homogeneous functions, differentiability of real-valued functions of two and three variables. Schwarz and Young's theorems (without proof).

Unit-II

Exact differential equations; Necessary and sufficient condition for a differential equation of the type M(x, y) dx + N(x ,y) dy = 0 to be exact. Integrating factor of a D.E. and methods to find it. First order and higher degree differential equations solvable for x, y, $p = \frac{dy}{dx}$. Clairaut's form and equations reducible to Clairaut's form. Singular solution as an envelope of general solutions.

Unit-III

Geometrical meaning of a differential equation, orthogonal trajectories. Homogeneous linear differential equations with constant coefficients and its solutions.

Theorems for finding particular integrals $\frac{1}{\mathrm{D}-\mathrm{a}}f(x)$, $\frac{1}{f(D)}e^{ax}$, $\frac{1}{f(D)}e^{ax}\mathrm{V}(x)$, $\frac{1}{f(D^2)}sin\ x$, $\frac{1}{f(D^2)}\cos x$, $\frac{1}{f(D)}x\ V(x)$, $\frac{1}{f(x)}\ x^n$; n is non negative integer. Non-homogeneous linear differential equations with constant coefficients and its solutions.

* Not for those students who opt mathematics as a major subject in table-2.

Unit-IV

Solution of Homogenous and non Homogeneous Linear differential equations with variable coefficients-Cauchy and Legendre Equations. Exactness of Linear differential equations of nth order and solution of linear differential equations that can be made exact using integrating factor.

References

1. **Erwin Kreyszig**: Advanced Engineering Mathematics, John Wiley & Sons Inc., New York, 1999.
2. **D.A. Murray**: Introductory Course on Differential Equations, Orient Long men, (India), 1967.
3. **A.R. Forsyth**: A Treatise on Differential Equations, Macmillan and Co. Ltd., London.
4. **Ross, S. L**.: Differential Equations, John Willey & Sons, 2004.

Semester - IV

MAT-DSC4 (MAJ/MIN)-401: Calculus-II

Credits = 4
Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.
Exam Time: 3 hours
Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.
2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.
3. The 20 marks of internal assessment has following division
Mid Semester Exam = 10 marks,
Assignment / Project work = 5 marks,
Classroom participation = 3 marks**,**
Attendance = 2 marks for more than 85% and 1 mark for more than 75% attendance.

Objective: The objective of this course is to analyse plane curves by looking at certain key parameters such as concavity, convexity, points of inflection, multiple points, and asymptotes. Students will also explore concepts of curvature and radius of curvature for various curves. The course will cover integration techniques for hyperbolic and inverse hyperbolic functions, along with reduction formulae for essential integrals.

Learning Outcomes: By the end of this course, students will be able to:

1. Analyse and classify plane curves based on parameters like concavity, convexity, points of inflection, multiple points, and asymptotes.
2. Understand and compute the curvature and radius of curvature for different curves.
3. Perform integration involving hyperbolic and inverse hyperbolic functions.
4. Apply reduction formulae to solve integrals of varying complexity.
5. Implement numerical methods for approximate computation of definite integrals.

Unit-I

Concavity, convexity, points of inflexion, multiple points, double points and its types for curves in a plane. Tangents at origin, asymptote and its types, methods for finding asymptotes of rational algebraic curves. Special methods for finding oblique asymptotes of rational algebraic curves. Intersection of a curve and its asymptotes.

Unit--II

Introduction to the polar coordinate system, tracing of curves represented by equations in Cartesian coordinates, Polar coordinates, and in parametric forms.
Curvature and radius of curvature, at a point, of curves in Cartesian and Polar Co-ordinates including parametric forms as well as curves represented by equation $f(x, y) = 0$ implicitly.

Unit-III

Integral calculus: Integration of hyperbolic and inverse hyperbolic functions. Reduction Formulae for following integrals.
$\int x^n e^x dx, \int x^m (\log x)^n dx, x > 0$ *m and n natural numbers.* $\int \sin^n x\, dx, \int \cos^n x\, dx,$
$\int \tan^n x\, dx, \int \cot^n x\, dx, \int \sec^n x\, dx, \int \operatorname{cosec}^n x\, dx, \int \sin^n x\, e^x\, dx, \int \cos^n x\, e^x\, dx,$
$\int x \sin^n x\, dx, \int x \cos^n x\, dx, \int x^m \sin nx\, dx, \int x^m \cos nx\, dx,$
$\int x^m \sin nx\, dx, \int \cos^m x \cos nx\, dx, \int \cos^m x \sin nx\, dx, \int \cos^m x \sin^n x\, dx$

$\int e^{ax} sin^n bx\, dx\ n \geq 2$ integer, $\int e^{ax} cos^n bx\, dx\ n \geq 2$ integer, $\int_0^{\frac{\pi}{2}} sin^n x\, dx\ \ n \in N$
$\int_0^{\frac{\pi}{2}} cos^n x\, dx\ \ n \in N$, $\int \frac{\sin nx}{\sin x}\, dx$, $\int x^m\ (a + bx^n)^p\, dx$.

Unit-IV

Numerical Integration: Trapezoidal, Prismoidal and Simpson Rules.

Application of definite integral: Summation of Series, Quadrature, rectification, volumes and surfaces of solids of revolution (Cartesian co-ordinates only)

References:

1. **G. B. Thomas & R. L. Finney** : Calculus and Analytic Geometry (Ninth edition), Pearson Publication.
2. **Gabriel Klambauer** : Mathematical Analysis, Marcel Dekkar, Inc. New York, 1975.
3. **N. Piskunov** : Differential and Integral Calculus, Peace Publishers, Moscow.
4. **P. K. Jain and S. K. Kaushik** : An Introduction to Real Analysis, S. Chand & Co. New Delhi, 2000.

Semester - IV
[1]MAT-DSC4 (MAJ)-402: DISCRETE MATHEMATICS AND GRAPH THEORY

Credits = 4
Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.
Exam Time: 3 hours
Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.
2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.
3. The 20 marks of internal assessment has following division
Mid Semester Exam = 10 marks,
Assignment / Project work = 5 marks,
Classroom participation = 3 marks**,**
Attendance = 2 marks for more than 85% and 1 mark for more than 75% attendance.

Objective: To provide basic knowledge about mathematical logic, lattices, Boolean algebra, graph theory and coding theory.

Learning Outcomes: This course will develop logical reasoning and rational thinking of the students.

UNIT – I
Mathematical Logic: Introduction, Conjunction, Disjunction & negation, Propositions and truth table, Tautologies and contradictions, Equivalence of formulas, Duality law, Normal forms: Disjunctive Normal form, Conjunctive Normal form,
Predicate Calculus: Predicates, the statement function, variables and quantifiers, predicate formulas. Methods of proof (Inference Theory).

UNIT – II
Ordered Sets, Latices, Boolean Algebra : Partially Ordered Sets, External elements of POSET, HASSE Diagrams of POSETS, Lattices, Finite Boolean Algebra, Functions on Boolean algebra, Circuit Designs.

UNIT – III
Graphs: Incidence and Degree, Isomorphism, Connectedness, Walk, Path and Circuits, Shortest Path algorithm between two Vertices, Eulerian graph, Directed graphs, Kuratowski's graphs; Detection of planarity.
Trees: Properties of Trees, Rooted and Binary Trees; Directed Tree, Spanning Tree and tree traversals (In-order, Pre-order and Post-order)

UNIT – IV
Functions: Composite functions & their range, domain, Functions for computer science like characteristic function, Hashing function, Growth of a function.
Coding Theory: Recursion, Recurrence Relation, Coding of binary Information & Error detection, Decoding & error correction.

[1] For students who have opted mathematics as a major subject in table-1.

Suggested Readings:

1. Kolman, Busby, Ross: discrete Mathematical Structure, PEARSON Education, 5th Edition.
2. Schaum Series: Theory and Problems of Essential Computer Mathematics, McGraw Hill,New York, 3rd Edition.
3. Sengadir,T. : Discrete Mathematics and Combinatorics, PEARSON Education.
4. Kenneth.H.Rosen: Discrete Mathematics and its Applications, Mc GRAW Hill, International Edition.
5. Deo, N. : Graph Theory with Applications to Engineering and Computer Science, PHI.
6. Clark, John & Hetlan: A First Look at Graph Theory, Allied Publishers Limited.

Semester - IV
[2*] MAT-DSC4 (MAJ)-403: COORDINATE GEOMETRY -I

Credits = 4
Teaching hours = 6 hrs / week including tutorials / practicals etc. in a semester of 15 weeks.
Exam Time: 3 hours
Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.
2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.
3. The 20 marks of internal assessment has following division
Mid Semester Exam = 10 marks,
Assignment / Project work = 5 marks,
Classroom participation = 3 marks**,**
Attendance = 2 marks for more than 85% and 1 mark for more than 75% attendance.

Objective: The main goal of this course is to deliver basics of plane geometry. The course introduces the concept of pair of straight lines, circles and other conics. The objective is to understand the geometrical concepts related to standard conic sections and their properties.

Learning Outcomes:
The students will be able to determine the joint equation of a pair of lines and angle bisectors, equation of a circle in different forms, equation of tangents, normal and chord of contact. The students will also be able to identify a conic from a general second degree equation.

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Unit-I

Transformation of axes in two dimensions: Shifting of origin, rotation of axes, invariants. **Pair of Straight lines**: Joint equation of pair of straight lines and angle between them, Condition of parallelism and perpendicularity, Joint equation of the angle bisectors, Joint equation of lines joining origin to the intersection of a line and a curve.

Unit-II

Circle : General equation of circle, Circle through intersection of two lines, tangents, normals, chord of contact, pole and polar, pair of tangents from a point, equation of chord in terms of mid-point, angle of intersection and orthogonality.

Unit-III

Conic : General equation of conics in Cartesian and Polar coordinate system, parametric forms of conics, equations of tangents, normal, chords of contact, pole and polar, pair of tangents from a point, equation of chord in terms of mid-point.

Unit-IV

Diameter and conjugate diameters of the conics. Special properties of Parabola, Hyperbola and Ellipse. Conjugate hyperbola, asymptotes to the hyperbola, rectangular hyperbola. Identification of conic in general second degree equations.

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[*] For students who have opted mathematics as a major subject in table-1.

References :

1. **S. L. Loney** : *The Elements of Coordinate Geometry,* Macmillan and Company, London, 2nd Edition 2007.
2. **P. K. Jain and Khalil Ahmad** : *A Text Book of Analytical Geometry of two dimensions, Wiley Eastern Ltd., 1999.*
3. **Erwin Kreyszig** : *Advanced Engineering Mathematics, John Wiley & Sons,1999.*
4. **Gorakh Prasad and H.C. Gupta** : *Text Book on Coordinate Geometry, Pothishala Pvt. Ltd., Allahabad, 1955.*
5. **G. B. Thomas and R. L. Finney:** Calculus and Analytic Geometry, Addison-Wesley Publishing Company.

Semester - IV
* MAT-DSC4 (MAJ)-404: NUMERICAL ANALYSIS

Credits = 4
Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.
Exam Time: 3 hours
Max. Marks: 100 (External: 80 + Internal Assessment: 20)
Imp. Remark: Normal / Scientific calculator is allowed in the examination.

Note: **1.** The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.
2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.
3. The 20 marks of internal assessment has following division
Mid Semester Exam = 10 marks,
Assignment / Project work = 5 marks,
Classroom participation = 3 marks**,**
Attendance = 2 marks for more than 85% and 1 mark for more than 75% attendance.

Objective: The objective of this course is to equip students with essential techniques for solving different types of mathematical problems that arise in various fields. It also introduces the fundamentals of calculus involving two and three variables, which are crucial for understanding and solving ordinary differential equations effectively.

Learning Outcomes: By the end of this course, students will be able to:

1. Apply appropriate methods to solve various types of differential equations.
2. To understand calculus concepts for functions of two and three variables.
3. Apply direct methods like Gauss Elimination, LU Decomposition, and Cholesky Decomposition, as well as iterative methods such as Jacobi, Gauss-Seidel, and relaxation methods, to solve systems of linear equations.

Unit-I

Solution of Equations: Bisection Methods, Regula Falsi Method, Secant Method, Newton-Raphson Method, Roots of Polynomials.
Interpolation: Lagrange and Hermite Interpolation, Newton's Divided Differences, Difference Schemes, Interpolation Formulas using Differences.

UNIT II.

Numerical Differentiation Numerical Quadrature: Newton-Cote's Formulas, Gauss Quadrature Formulas , Chebyshev's Formulas.

Unit-III

Linear Equations: Direct Methods for Solving Systems of Linear Equations (Gauss Elimination, LU Decomposition, Cholesky Decomposition), Iterative Methods (Jacobi, Gauss-Seidel, Relaxation Methods).

Unit-IV

The Algebraic Eigenvalue problem: Jacobi's Method, Givens' Method, Householder's Method, Power Method, QR Method, Lanczos' Method.
Ordinary Differential Equations: Euler Method, Single-step Methods, Runge-Kutta's Method, Multi-step Methods.

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* For students who have opted mathematics as a major subject in table-1.

References

1. **C. E. Froberg**: *Introduction to Numerical Analysis* (Second Edition), Addison-Wesley, 1979.
2. **Melvin J. Maron**: *Numerical Analysis: A Practical Approach,* Macmillan Publishing Co., New York, 1982.
3. **M. K. Jain, S. R. K. Iyengar and R.K. Jain**: *Numerical Methods for Scientific and Engineering Computation,* New Age International (P.) Ltd., 1999.
4. **R. Y. Rubistein**: *Simulation and the Monte Carlo Methods,* John Wiley, 1981.
5. **D. J. Yakowitz**: *Computational Probability and Simulation,* Addison-Wesley, 1977.
6. **S. S. Sastry**: *Introductory Methods of Numerical Analysis,* 3rd Edition (2000), Prentice Hall of India Pvt. Ltd., New Delhi.

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Semester - IV
* MAT-DSC4 (MAJ)-405: DIFFERENTIAL EQUATION-I

Credits = 4
Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.
Exam Time: 3 hours
Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.
2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.
3. The 20 marks of internal assessment has following division
Mid Semester Exam = 10 marks,
Assignment / Project work = 5 marks,
Classroom participation = 3 marks**,**
Attendance = 2 marks for more than 85% and 1 mark for more than 75% attendance.

Objective: The objective of this course is to equip students with essential techniques for solving different types of differential equations. It also introduces the fundamentals of calculus involving two and three variables, which are crucial for understanding and solving differential equations effectively.

Learning Outcomes: By the end of this course, students will be able to:

1. Apply appropriate methods to solve various types of differential equations.
2. To understand calculus concepts for functions of two and three variables.

Unit-I

Limit and continuity of functions of two and three variables. Partial differentiation up to second order. Total differential, differentiation of composite and implicit functions. Euler's Theorem on homogeneous functions, differentiability of real-valued functions of two and three variables. Schwarz and Young's theorems (without proof).

Unit-II

Exact differential equations; Necessary and sufficient condition for a differential equation of the type $M(x, y)\,dx + N(x, y)\,dy = 0$ to be exact. Integrating factor of a D.E. and methods to find it. First order and higher degree differential equations solvable for x, y, $p = \frac{dy}{dx}$. Clairaut's form and equations reducible to Clairaut's form. Singular solution as an envelope of general solutions.

Unit-III

Geometrical meaning of a differential equation, orthogonal trajectories. Homogeneous linear differential equations with constant coefficients and its solutions.

Theorems for finding particular integrals $\frac{1}{D-a}f(x)$, $\frac{1}{f(D)}e^{ax}$, $\frac{1}{f(D)}e^{ax}V(x)$, $\frac{1}{f(D^2)}\sin x$, $\frac{1}{f(D^2)}\cos x$. Non-homogeneous linear differential equations with constant coefficients and its solutions.

* Only for those students who have opted math as a major subject in table-2.

Unit-IV

Solution of Homogenous and Non Homogeneous Linear differential equations with variable coefficients-Cauchy and Legendre Equations. Exactness of Linear differential equations of nth order and solution of linear differential equations that can be made exact using integrating factor.

References

2. **Erwin Kreyszig**: Advanced Engineering Mathematics, John Wiley & Sons Inc., New York, 1999.
3. **D.A. Murray**: Introductory Course on Differential Equations, Orient Long men, (India), 1967.
4. **A.R. Forsyth**: A Treatise on Differential Equations, Macmillan and Co. Ltd., London.
5. **Ross, S. L**.: Differential Equations, John Willey & Sons, 2004.

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Microbiology

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR (Table 1*) and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY (Table 2*)

B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH IN MICROBIOLOGY

Table1*						Table2*					
Sem	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**
III	L-200	(MIC-DSC-3 MIN -301)	Microbial genetics and molecular Biology	3+1=4	24	III	L-200	(MIC-DSC-3 MIN -301)	Microbial genetics and molecular Biology	3+1=4	24
		(MIC-DSC-4 MIN -302)	Bacteriology and Virology	3+1=4				(MIC-DSC-4 MIN -302)	Bacteriology and Virology	3+1=4	
								DSC- MAJ	As opted by the student	3+1=4	
		DSC- MAJ	As opted by the student	3+1=4				DSC- MAJ	As opted by the student	3+1=4	
		DSC- MAJ	As opted by the student	3+1=4							
		(MIC-MDC-3-301)	Entrepreneurship in Microbial biotechnology	3				(MIC-MDC-3-301)	Entrepreneurship in Microbial biotechnology	3	
		(MIC-SEC-3-301)	Infield aspects and Entrepreneurship Microbiology	3				(MIC-SEC-3-301)	Infield aspects and Entrepreneurship Microbiology	3	
		(AEC-3-301)	Eng-3/MIL-1	2				(AEC-3-301)	Eng-3/MIL-1	2	
IV	L-200	(MIC-DSC-5 MIN -401)	Immunology	3+1=4	24	IV	L-200	(MIC-DSC-5 MIN -401)	Immunology	3+1=4	24
		DSC- MAJ	As opted by the student	3+1=4				DSC- MAJ	As opted by the student	3+1=4	
		DSC- MAJ	As opted by the student	3+1=4				DSC- MAJ	As opted by the student	3+1=4	
		DSC- MAJ	As opted by the student	3+1=4				DSC- MAJ	As opted by the student	3+1=4	
		DSC- MAJ	As opted by the student	3+1=4				DSC- MAJ	As opted by the student	3+1=4	
		(VAC-3-401)	To be opted by the student	2				(VAC-3-401)	To be opted by the student	2	
		(AEC-4-401)	Eng-4/MIL-2	2				(AEC-4-401)	Eng-4/MIL-2	2	
The student exiting the programme after two year ssecuring 96 credits will be awarded UG Diploma provided he/she secures 4 credits from summer internship											

***Note: The candidate will opt for either Template 1 or 2 from Semester III onwards.**

Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

SEMESTER-III
MIC-DSC-3 MIN – 301: Microbial Genetics and Molecular Biology

Total marks: 75
Theory: 67
Internal assessment: 08
Time: 3 hours
Credits: 3

Note: The question paper will consist of Four Units. There will be nine questions, out of which five questions have to be attempted. Question I will span the complete syllabus and will be compulsory. Rest of the eight questions will be from different Units of the syllabus. There will be two questions from each Unit and one is to be attempted from each unit.

Objective students should be able to the learn the inheritance materail and molecular mechanisms of genetic materaial.

Learning outcome: By the end of sessions students will be able to understand how the inhertance material formed and perform the all molecular activities inside the host cell. How the mutation formed and repaired.Function of plasmids and vectors.

Unit-I

NO. OF HOURS : 15

Microbial Genetics: Experimental proof of DNA and RNA as genetic material, DNA denaturation, DNA melting, Tm value, Cot Value, C-value Paradox. Genetic code, central dogma, reverse transcriptase. DNA replication in prokaryotes (DNA polymerases, initiation, elongation and termination) , Gene transcription (RNA polymerase, initiation, elongation and termination, post transcriptional modifications), Translation (initiation, elongation and termination).

Unit-II

NO. OF HOURS : 15

Microbial Mutations: concept of mutation, spontaneous and induced, Forward and reverse mutation. Methods for the selection of mutants, direct selection methods for resistant mutants, penicillin selection technique, replica plating technique, other technique for mutant selection, conditional lethality and its use in mutant selection. Methods for DNA repair

Unit-III

NO. OF HOURS : 15

Molecular basis of recombination in bacteria: Conjugation, Transformation, Transduction , isolation of auxotrophs Recombinant Proteins: Production of recombinant proteins in bacteria and yeast (Applications and limitations)

Unit-IV

NO. OF HOURS : 15

Genomic Amplification and Cloning:- Brief account of vectors (Plasmids, cosmid, BAC, YAC and shuttle vectors) PCR.Genomic library and construction .Identification of desired clone.

MIC-DSC-3 MIN – 301: Microbial Genetics and Molecular Biology (Practicals)

Total marks: 25
Practical: 22
Internal assessment: 03
Time: 3 hours
Credits: 1

List of Experiments:

1. Isolation of genomic DNA from bacteria (E.coli) and human blood.
2. Quantification of DNA using spectrophotometric method
3. Quantitative estimation of DNA by diphenylamine reaction using colorimeter.
4. Quantitative estimation of RNA by orcinol method using colorimeter.
5. Study of absorption spectra of DNA and protein using UV-Visible spectrophotometer.
6. Demonstartion of DNA/RNA using agarose gel electrophoresis.
7. Amplification of known DNA sequences by Polymerase Chain Reaction.
8. Study of different types of DNA and RNA using micrographs and model / schematic representations

Recommended Books:

1. General Microbiology (5th Edition): by Roger Y Stanier, John L Ingraham, Mark L Wheelis, Page R Painter : MacMillan Publishers
2. Presscots Microbiology (12th Edition, 2023) by Dorothy Wood, Joanne Willey, Kathleen Sandman MacGraw -Hill Publishers
3. Microbiology (5th Edition, 2023) by Michale J. Pelczar Jr., E.C.S. Chan, Noel R. Krieg: Tata-MacGraw-Hill Publishers
4. Lewin's Genes Twelve By Jocelyn E. Krebs, Elliott S. Goldstein, Stephen T. Kilpatrick · 2017
5. Molecular Biology of the GENE: 8 th edition (2024) by James D Watson .
6. Molecular Biotechnology: Principles and Applications of Recombinant DNA: 6th Edition (2022) by Bernard R. Glick and Cheryl L Patten.ASM press.
7. Gene Cloning and DNA analysis: 8th Edition (2020) by T.A.Brown. ASM press.
8. Principles of Gene Manipulation and Genomics: 7th Edition (2014) Sandy Primrose. Black Well Publishers
9. Microbiology: An Introduction (13th Global Edition, 2021) by Gerard J. Tortora, Berdell R. Funke, Christine L. Case, Derek Weber, Warner Bair: Pearson Publication

SEMESTER-III
MIC-DSC-4 MIN – 302: Bacteriology and Virology

Total marks: 75
Theory: 67
Internal assessment: 08
Time: 3 hours
Credits: 3

Note: The question paper will consist of Four Units. There will be nine questions, out of which five questions have to be attempted. Question I will span the complete syllabus and will be compulsory. Rest of the eight questions will be from different Units of the syllabus. There will be two questions from each Unit and one is to be attempted from each unit.

Objective: students should be able to the learn the Organisation and Fundamental concepts of Bacteria and viruses, their origin, isolation, cultivation, characterization of viruses.

Learning outcome:- by the end of course students will be able to understand the different bacterial and viral organisms ,their orgnization,metabolisms,how microganisms causes disesaes. Study deep about the oncology.

Unit-I

NO. OF HOURS : 15

Important archaeal and eubacterial groups

Archaebacteria: General characteristics of genera belonging to Nanoarchaeota (Nanoarchaeum), Crenarchaeota (Sulfolobus, Thermoproteus) and Euryarchaeota [Methanogens (Methanobacterium, Methanocaldococcus), thermophiles (Thermococcus, Pyrococcus, Thermoplasma), and Halophiles (Halobacterium, Halococcus)]

Eubacteria: Morphology, metabolism, ecological significance and economic importance of following groups: Gram Negative: General characteristics:Non proteobacteria, Alpha proteobacteria: Beta proteobacteria, Delta proteobacteria, Epsilon proteobacteria, Zeta proteobacteria.

Endospore: Structure, formation, stages of sporulation

UNIT-II

NO. OF HOURS : 15

Basics in Medical microbiology : Human diseases caused by Gram postive Bacteria (*Staphylococcus aureus* , *Streptococcus pneumoniae*) Gram negative Bacteria (*Escherichiae coli*) , other bacteria Chlamydiae, Rickettsiaceae -Etiology, Epidemiology ,route of transmission, Pathogenicity and virulence, laboratory dignosis and Treatments .

Human diseases caused by fungi and protozoa :-–*Candida albicans*, *Plasmodium falciparum* Etiology, Epidemiology ,route of transmission, Pathogenicity and virulence, laboratory dignosis and Treatments.

Unit-III

NO. OF HOURS : 15

Nature and Properties of Viruses

Introduction: Discovery of viruses, nature and definition of viruses, general properties. Concept of viroids, virusoids, satellite viruses and Prions. Structure of Viruses: Capsid symmetry, enveloped and non-enveloped viruses Isolation, purification and cultivation of viruses.

General concepts of structures and classification of viruses. General characteristics and mode of replication of Hepatitis B virus, Influenza virus. HIV. Prevention & control of viral diseases.

UNIT-IV

NO. OF HOURS : 15

Viruses and Cancer Introduction to oncogenic viruses Types of oncogenic DNA and RNA viruses: Concepts of oncogenes and proto-oncogenes.

Cultivation and purification of viruses: Different in vivo, in vitro and in ovo growth systems for animal viruses.

MIC-DSC-4 MIN – 302: Bacteriology and Virology (Practicals)

Total marks: 25
Practical: 22
Internal assessment: 03
Time: 3 hours
Credits: 1

List of Experiments:

1. Isolation of different bacteria and phages from water.
2. Isolation of bacteria and phages from soil.
3. Simple staining and Negative staining.
4. Gram's staining
5. Acid fast staining and Endospore staining.
6. Isolation of pure cultures of bacteria by streaking method.
7. Preservation of bacterial cultures (slant / stab).
8. Estimation of CFU count by spread plate method/pour plate method.

Recommended Books:

1. Ananthanarayan And Paniker (2017). Textbook Of Microbiology 10th Edition ,
2. Dimmock, NJ, Easton, AL, Leppard, KN (2016). Introduction to Modern Virology. 7th edition, Blackwell Publishing Ltd.
3. Murray PR, Rosenthal KS, Kobayashi GS, Pfaller MA.(2020) Medical Microbiology. 9th edition, Mosby, Inc
4. Carter J and Saunders V (2007).Virology: Principles and Applications. John Wiley and Sons.
5. Flint SJ, Enquist, LW, Krug, RM, Racaniello, VR, Skalka, AM (2004).
6. Flint, S. Jane, et al. Principles of virology,pathogenesis and control. Vol. 2. John Wiley & Sons, 2020.
7. Willey JM, Sherwood LM, and Woolverton CJ. (2013). Prescott's Microbiology. 9th edition. McGraw Hill Higher Education.
8. Gillespie, S. H., & Hawkey, P. M. (Eds.). (2006). *Principles and practice of clinical bacteriology*. John Wiley & Sons. 2nd edition .
9. Cappucino J and Sherman N. (2010). Microbiology: A Laboratory Manual. 9th edition. Pearson Education Limited

SEMESTER-IV
MIC-DSC-5 MIN – 401: Immunology

Total marks: 75
Theory: 67
Internal assessment: 08
Time: 3 hours
Credits: 3

Note: The question paper will consist of Four Units. There will be nine questions, out of which five questions have to be attempted. Question I will span the complete syllabus and will be compulsory. Rest of the eight questions will be from different Units of the syllabus. There will be two questions from each Unit and one is to be attempted from each unit.

Objective: To understand how the immune system functions, both in health and disease, and to develop effective strategies for diagnosing, treating, and preventing immune-related disorders.

Learning outcome : by the end of course Students will be able to the learn the essential of antigens and antibodies ,Type of immunty, phagocytosis, complement systems,hypersensitivity, auto immunity and transplantation. and provide knowledge on immune deficiencies and several immunological techniques.

Unit-I

NO. OF HOURS : 15

Introduction: Historical background of Immunology, Cells and Organs of Immune system :- Hematopoiesis, cells of immune system, Organs of immune system. Inflammation.
Immunity:- Innate and Adaptive immunty.Cell mediated immunity,Humoral immunity, Antigenic determinats, Adjuvants: Freund's complete and incomplete.

Unit-II

NO. OF HOURS : 15

Antigens and Antibodies:-Immunogenicity versus Antigenicity, Factors affecting the Immunogenicity, Epitopes,Haptens. Basic structure of Antibodies, Antigenic determinats of Immunoglobulins, Classification of Immunoglobulins: Types –IgG (G1, G2, G3 & G4), IgM, IgA, IgD and IgE (Origin, structural functions). Monoclonal Antobodies.
Antigen-Antibody Interactions : Precipitation reaction, Agglutination reaction, Compliment fixation tests Radioimmnuoassay.

Unit-III

NO. OF HOURS : 15

Immune effector mechanisms : Cytokines, Lymphokines Colony stimulating factors ,Tumour necrosis factors, Interferons,Accessory cells (Macrophages).The Complement system ,Classical and aleternative pathway,HLA,MHC-I and II
T Cells and B cells:- T cells Receptors , B cells . T Cells and B cells maturation, activation and differentation.

Unit-IV

NO. OF HOURS : 15

Vaccines:- History of vaccination.Types of vaccines.
Immunopathology :- Immunology of Transplantation,Immunology of Malignant diseases, Immunologic reaction in Skin diseases.

MIC-DSC-5 MIN – 401: Immunology (Practicals)

Total marks: 25
Practical: 22
Internal assessment: 03
Time: 3 hours
Credits: 1

List of Experiments:

1. Demonstration of collection and handling of specimens for microbiological examination.
2. Isolation of microbial flora from the human hands and dental carries.
3. Testing the antimicrobial activity of the skin on bacteria.
4. To perform Total Leukocyte Count of the given blood sample
5. To perform tube agglutination test-The Widal test.
6. Studies on histopathology of normal and infected tissue.
7. Antigen –antibody interactions –Blood grouping Immunodiffusion.

Recommended Books:

1. Delves, Peter J., et al. *Roitt's essential immunology*. John Wiley & Sons, 2017.
2. Cellular and Molecular Immunology (2007) Abbas AK, Lichtman AH, Pillai S. 6th edition Saunders Publication, Philadelphia.
3. Roitt's Essential Immunology (2006) Delves P, Martin S, Burton D, Roitt IM. 11th edition WileyBlackwell Scientific Publication, Oxford.
4. Kuby's Immunology (2007) Goldsby RA, Kindt TJ, Osborne BA. 6th edition W.H. Freeman and Company, New York.
5. Janeway's Immunobiology (2008) Murphy K, Travers P, Walport M. 7th edition Garland Science Publishers, New York.
6. Basic and Clinical Immunology (2009) Peakman M, and Vergani D. 2nd edition Churchill Livingstone Publishers, Edinberg.
7. Immunology (2009) Richard C and Geiffrey S. 6th edition. Wiley Blackwell Publication.
8. Immunology –k.R.joshi and N.O.Osamo, 5th edition.

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Physics

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR(Table 1*) and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY(Table 2*)

B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH

Table 1*						Table 2*					
Sem	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**
III	L-200	(PHY-DSC-3 Maj/Min-301)	Optics and Lasers	3+1=4 (x2)	24	III	L-200	(PHY-DSC-3 Maj/Min-301)	Optics and Lasers	3+1=4 (x3)	24
		(PHY-DSC-4 Maj/Min-302)	Vibrations, Waves & E.M. Theory	3+1=4(x2)				(PHY-DSC-4 Min-302)	Vibrations, Waves & E.M. Theory	3+1=4	
		(PHY-MDC-3-301)	Basic Computational Tools	3				(PHY-MDC-3-301)	Basic Computational Tools	3	
		(PHY-SEC-3-301)	Physics instrumentation Skills-III	3				(PHY-SEC-3-301)	Physics instrumentation Skills-III	3	
		(AEC-3-301)	Eng-3/MIL-1	2				(AEC-3-301)	Eng-3/MIL-1	2	
IV	L-200	(PHY-DSC-5 Maj/Min-401)	Quantum Mechanics	3+1=4	24	IV	L-200	(PHY-DSC-5 Maj/Min-401)	Quantum Mechanics	3+1=4 (x3)	24
		(PHY-DSC-6 Maj-402)	Statistical Physics and Thermodynamics	3+1=4 (x2)				(PHY-DSC-6 Maj-402)	Vibrations, Waves & E.M. Theory	3+1=4 (x2)	
		(PHY-DSC-7 Maj-403)	Condensed Matter Physics I	3+1=4				(PHY-VAC-3-401)	Basics of Communication System	2	
		(PHY-DSC-8 Maj-404)	Electronics and Solid-State Devices – I	3+1=4							
		(PHY-VAC-3-401)	Basics of Communication System	2							
		(AEC-4-401)	Eng-4/MIL-2	2				(AEC-4-401)	Eng-4/MIL-2	2	
The student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secures 4 credits from summer internship											

Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attainatleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

B.Sc. Semester 3
(PHY-DSC-3 Maj/Min-301)
(OPTICS AND LASERS)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Practical 25 marks [20 marks+05 marks (internal assessment 20%)]
Teaching hours: Theory-45 Hrs. / 3 Hrs. per week
Total Credits: 3(Theory)+1(Practical)

Learning Outcomes

- To understand the principles and conditions required for interference phenomena, including spatial and temporal coherence, and explore applications like Young's double-slit experiment, Newton's rings, and interferometry.
- To study the theory and applications of diffraction, including Huygens-Fresnel principles, resolving power, and diffraction gratings. To gain insight into polarization phenomena, double refraction, and the tools for producing and analyzing polarized light, such as Nicol prisms and retardation plates.
- To explore the fundamentals of lasers, their working principles, types, and applications, with a focus on coherence, population inversion, and different laser systems.
- To understand the construction, properties, and applications of optical fibers, including their design, numerical aperture, and light transmission losses.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Interference: Concept of coherence, spatial and temporal coherence, coherence time, coherence length, area of coherence. Conditions for observing interference fringes. Interference by wavefront division and amplitude division. Young's double slit experiment. Lloyd's mirror and Fresnel's biprism, phase change on reflection. Newton's rings,Michelson interferometer—working, principle and nature of fringes. Interference in thin films, Role of interference in anti-reflection. Multiple beam interference, Fabry-Perot interferometer.

Diffraction: Huygen-Fresnal theory half period zones, zone plates. resolving power of microscope and telescope, diffraction grating, resolving power,

Polarization: Concept and analytical treatment of unpolarised, plane polarized and elliptically polarized light. Double refraction, Nicol prism, sheet polarisers, retardation plates. Production and analysis of polarized light (quarter and half wave plates).

UNIT-II

Laser Fundamentals: Interaction of light with matter: Absorption, spontaneous emission, stimulated emission, Wave mechanical explanation, Properties of Spectral Lines, Temporal and spatial coherence, Charatertistics of stimulated emission, Einstein coefficients and their relations, Light amplification and threshold condition,Population inversion, Doppler broadening & Natural broadening (qualitative account only), Principal pumping schemes,Three level and four level laser schemes.

Laser Systems: Types of lasers, Construction and working of Ruby, Nd: YAG, He–Ne, Dye and CO_2 lasers. Semiconductor laser (Qualitative idea), Applications of lasers—a general outline, Holography: Principle, recording of hologram and reconstruction of image. (qualitative only).

Fiber Optics: Photonics, Optical fibre, Construction, Numerical aperture, acceptance angle, skip distance, Step index fibre – single mode and multimode, Graded index fibre, Losses in optical fibre, Applications of optical fibre.

Books Suggested:

Essential Readings:

1. Optics, Jenkins and White, McGraw Hill.
2. Optics, Ajoy Ghatak, McMillan India.
3. Physics for Degree Students, C. L. Arora and P .S. Hemne, S. Chand & Co., 2014.
4. Laser Fundamentals, W.T. Silfvast, Foundation Books.
5. Lasers and Non-linear Optics, B.B. Laud, Wiley Eastern Limited.
6. A text book of Optics, Subtahmanyam N., Lal B. and Avadhanulu, M. N., S. Chand & Co., 2012.
7. Optical Fiber Communication, Keiser, MH.

Further Readings:

1. Optics, Born and Wolf, Pergamon.
2. Lasers, SveltoPergmon.

PHYSICS PRACTICALS (PHY-DSC-3 Maj/Min-301)

Max. Marks: 25 (20+5 internal assessment) **Total Credit: 1**
Teaching hours: 30 Hrs. / 2 Hrs. per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i) One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error).	09
(ii) One exercise based on experiment	04
(iii) Viva-Voce	04
(iv) Record (Practical file)	03
(v) Internal Assessment	05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

List of Practicals:

1. To measure the height of inaccessible objects using sextant.
2. To determine the refractive index of a liquid using spectrometer.
3. To determine the Cauchy's constants.
4. To study the refractive index of a doubly refracting prism.
5. Study of rotation of plane of polarization with a polarimeter.
6. To determine the wave length and dispersive power using plane diffraction grating (use Hg source).
7. To determine the resolving power of a grating.
8. Set up Newton's rings to determine wave length of sodium light.
9. To determine the principal points of a lens system.

Texts and Reference Books :

1. "B.Sc. Practical Physics by C.L. Arora, S.Chand& Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal.
3. "Programming with C, Schaum series" by Byron Gottfried&JitenderChhabra.

B.Sc. Semester 3
(PHY-DSC-4 Maj/Min-302)
(VIBRATIONS, WAVES & E.M. THEORY)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Practical 25 marks [20 marks+05 marks (internal assessment 20%)]
Teaching hours: Theory-45 Hrs. / 3 Hrs. per week
Total Credits: 3(Theory)+1(Practical)

Learning Outcomes

- To understand SHM and analyze damped and forced oscillations in mechanical and electrical systems.
- To apply differential equations to study oscillatory motion and interpret damping, Q-factor, and resonance.
- To explain wave propagation in media, including energy transfer, reflection, and group velocity.
- To use Maxwell's equations to describe EM waves, energy flow, and wave behavior in various media.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Simple harmonic motion, energy of a SHM, Compound Pendulum, Torsional Pendulum, Electrical Oscillations, Transverse Vibrations of a mass on a string, Decay of free vibrations due to damping, differential equation of motion, types of damping, determination of damping co-efficient; Logarithmic decrement, relaxation time and Q- Factor. Electromagnetic damping (Electrical oscillator).

Differential equation for forced mechanical and electrical oscillators, Transient and steady state behaviour. Displacement and velocity variation with driving force frequency, variation of phase with frequency, resonance. Power supplied to an oscillator and its variation with frequency. Q-value and band width.

UNIT-II

Waves in physical media, Wave equation and its solution, Types of waves, particle velocity, acceleration and energy in progressive waves. Transverse waves on a string, characteristic impedance of a string, Reflection and Transmission of transverse waves on a string at discontinuity, Reflection and transmission of energy. Reflection and transmission of longitudinal waves at a boundary. Impedance matching, Energy of vibrating string. Wave and group velocity.

Maxwell's equations, E.M. waves and wave equation in a medium having finite permeability, permittivity and conductivity. Energy flow due to EM wave - Poynting vector, Impedance of a dielectric to EM waves. EM waves in a conducting medium and skin depth. Impedance and Refractive index of a dielectric and a conductor.

Books Suggested:

Essential Readings:

1. Text Book of Vibrations and Waves by S.P. Puri (Macmillan India Ltd.).
2. Physics of Vibrations and Waves by H.J. Pain, ELBS & John Wiley, London.

Further Readings:

1. Vibrations and Waves by A.P. French (Arnold Heinemann India, New Delhi).
2. The Mathematics of Waves and Vibrations by P.K. Ghosh (Mcmillan India).
3. Waves and Oscillations by N. Subrahmanayam& B. Lal (Vikas Pub., Delhi).

PHYSICS PRACTICALS (PHY-DSC-4 Maj/Min-302)

Max. Marks: 25 (20+5 internal assessment) **Total Credit: 1**

Teaching hours: 30 Hrs. / 2 Hrs. per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i) One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error). 09
(ii) One exercise based on experiment 04
(iii) Viva-Voce 04
(iv) Record (Practical file) 03
(v) Internal Assessment 05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

List of Practicals:

1. Find the value of g by Bar pendulum.
2. To study the dependence of moment of inertia on distribution of mass (by noting time periods of oscillations using objects of various geometrical shapes but of same mass).
3. Measure time period of oscillation of a Maxwell needle and determine modulus of rigidity of the material of a given wire.
4. To determine the frequency of a tuning fork using sonometer.
5. To verify the laws of transverse vibrations of stretched string using sonometer.
6. To find the density of the given wireusing sonometer.
7. To compare mass per unit length of two strings by Melde's experiment.
8. To find the velocity of sound in the material of the given material of the given rod with a Kundt's tube given that velocity of sound in air at 0 deg. C is 332 m/s.
9. To determine the velocity of ultrasonic waves in a given liquid.

Texts and Reference Books:

1. "B.Sc. Practical Physics by C.L. Arora, S.Chand& Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal

B.Sc. Semester 4
(PHY-DSC-5 Maj/Min-401)
(Quantum Mechanics)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Practical 25 marks [20 marks+5 marks (internal assessment 20%)]
Teaching hours: Theory-45 Hrs. / 3 Hrs. per week
Total Credits: 3(Theory)+1(Practical)

Learning Outcomes

- Understand the concept of wave-particle duality and its experimental demonstrations through phenomena like the photoelectric effect, Compton effect, and De Broglie waves.
- Learn the fundamentals of quantum mechanics, including the Schrödinger equation (time-dependent and time-independent), Born's interpretation of wave functions, and concepts such as eigenfunctions, eigenvalues, and operators.
- Explore applications of the Schrödinger equation to physical systems like particles in a box, potential steps, tunneling, harmonic oscillators, and the hydrogen atom's energy levels and spectra. Analyze atomic structure through quantum numbers, selection rules, Zeeman effects, spin-orbit coupling, and angular momentum.
- Examine the production, properties, and applications of X-rays, including X-ray spectra, Moseley's law, and the Auger effect, providing insights into their theoretical and practical implications.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Wave-particle duality – Photoelectric effect, Compton effect, De Brogile waves, wave packet, Phase velocity and Group velocity, Uncertainty principle and its applications,Time-dependent and Time independent Schrodinger wave equation, Born's interpretation of Wave function, normalization, orthogonality condition, Probability current, Probability conservation equation,Eigen function and eigen value, Operator formalism, Position, momentum and energy operators, Hermitian operator expectation values, Ehrenfest theorem.

Solution of Schrodinger wave equation for particle in one-dimension box, Potential step (E<V and E> V), Tunnel effect (Qualitative), Linear harmonic oscillator.SchrÖdingerequation for spherically symmetric potential, spherical harmonics, hydrogen atom energy levels,Principal, Orbital and Magnetic quantum numbers

UNIT-II

Selection rules, Spectrum of hydrogen atom. Normal Zeeman effect, Degeneracy of H-atom energy levels, fine structure, Electron orbital and spin angular momentum and magnetic moments, Larmor's frequency, spin-orbit coupling, total angular momentum, examples of one electron systems, Anomalous Zeeman effect and its application to sodium D-lines.

Symmetric and Antisymmetric wave functions, exclusion principle, Many electron atoms, electronic configurations, Hund's rule, Spin-Orbit coupling, L-S coupling, J-J couplings, term symbols. Atomic spectra of Na and He.

Production and properties of X-rays, Types of X-rays spectra, Absorption of X-rays, Mosley law, Auger effect.

Essential Readings :

1. Concept of Modern Physics, A. Beiser, S. Mahajan and S. R. Choudhury, Tata McGraw Hill, 6th Edition.
2. Quantum Physics of Atoms, Molecular, R. Eisberg &R.Resnick, Second Edition, John Wiley, 2002.
3. Physics for Degree Students, C.L. Arora and P.S. Hemne, S. Chand & Co., 2014
4. Elements of Modern Physics, S.H. Patil, McGraw Hill, 1998.

Further Readings :

1. Modern Physics, J. Bernstein, P.M. Fishbane, S.G. Gasiorowicz, Pearson, 2000.
2. Elements of Modern Physics, S.H. Patil, McGraw Hill, 1998.
3. Quantum Mechanics, E. Merzbacaher, II Ed., John Wiley.
4. Fundamentals of Molecular Spectroscopy, C.N. Banwell, T.M.H., 1972
5. Atomic and Molecular Spectra, Rajkumar (Kedarnath Ramnath Prakashan, Meerut).
6. Atomic Spectra, H.G. Kuhn, Academic Press, N.Y.

PHYSICS PRACTICALS (PHY-DSC-5 Maj/Min-401)

Max. Marks: 25 (20+5 internal assessment) **Total Credit: 1**

Teaching hours: 30 Hrs. / 2 Hrs. per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i) One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error). 09

(ii) One exercise based on experiment 04

(iii) Viva-Voce 04

(iv) Record (Practical file) 03

(v) Internal Assessment 05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

List of Practicals

1. To determine the ionization potential of mercury.
2. To study the Photoelectric effect and determine the value of Planck's constant.
3. To study the gas discharge spectrum of hydrogen.
4. To study the absorption spectra of iodine vapours.
5. Study of variation of light intensity using photovoltaic cell/inverse square law.
6. To verify uncertainty principle using laser.
7. To determine the resolving power of a telescope.
8. To determine the divergence and wave length of a given laser source.
9. To study interference using young's double slit experiment using laser.

Texts and Reference Books :

1. "B.Sc. Practical Physics by C.L. Arora, S.Chand& Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal.

B.Sc. Semester 4
(PHY-DSC-6 Maj-402)
(VIBRATIONS, WAVES & E.M. THEORY)
(for Table-2)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Practical 25 marks [20 marks+05 marks (internal assessment 20%)]
Teaching hours: Theory-45 Hrs. / 3 Hrs. per week
Total Credits: 3(Theory)+1(Practical)

Learning Outcomes

- To understand SHM and analyze damped and forced oscillations in mechanical and electrical systems.
- To apply differential equations to study oscillatory motion and interpret damping, Q-factor, and resonance.
- To explain wave propagation in media, including energy transfer, reflection, and group velocity.
- To use Maxwell's equations to describe EM waves, energy flow, and wave behavior in various media.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Simple harmonic motion, energy of a SHM, Compound Pendulum, Torsional Pendulum, Electrical Oscillations, Transverse Vibrations of a mass on a string, Decay of free vibrations due to damping, differential equation of motion, types of damping, determination of damping co-efficient; Logarithmic decrement, relaxation time and Q- Factor. Electromagnetic damping (Electrical oscillator).

Differential equation for forced mechanical and electrical oscillators, Transient and steady state behaviour. Displacement and velocity variation with driving force frequency, variation of phase with frequency, resonance. Power supplied to an oscillator and its variation with frequency. Q-value and band width.

UNIT-II

Waves in physical media, Wave equation and its solution, Types of waves, particle velocity, acceleration and energy in progressive waves. Transverse waves on a string, characteristic impedance of a string, Reflection and Transmission of transverse waves on a string at discontinuity, Reflection and transmission of energy. Reflection and transmission of longitudinal waves at a boundary. Impedance matching, Energy of vibrating string. Wave and group velocity.

Maxwell's equations, E.M. waves and wave equation in a medium having finite permeability, permittivity and conductivity. Energy flow due to EM wave - Poynting vector, Impedance of a dielectric to EM waves. EM waves in a conducting medium and skin depth. Impedance and Refractive index of a dielectric and a conductor.

Books Suggested:

Essential Readings:

1. Text Book of Vibrations and Waves by S.P. Puri (Macmillan India Ltd.).
2. Physics of Vibrations and Waves by H.J. Pain, ELBS & John Wiley, London.

Further Readings:

1. Vibrations and Waves by A.P. French (Arnold Heinemann India, New Delhi).
2. The Mathematics of Waves and Vibrations by P.K. Ghosh (Mcmillan India).
3. Waves and Oscillations by N. Subrahmanayam& B. Lal (Vikas Pub., Delhi).

PHYSICS PRACTICALS (PHY-DSC-6 Maj-402)

Max. Marks: 25 (20+5 internal assessment) **Total Credit: 1**

Teaching hours: 30 Hrs. / 2 Hrs. per week

General Guidelines for Physics Practical Examinations:

The distribution of marks is as follows:

(i) One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error).	09
(ii) One exercise based on experiment	04
(iii) Viva-Voce	04
(iv) Record (Practical file)	03
(v) Internal Assessment	05

Note for Examiners:

1. The marks scored under each head must be clearly written on the answer sheet.
2. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
3. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
4. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
5. Number of candidates in a group for practical examination should not exceed 12.
6. In a single group, no experiment to be allotted to more than three examinees in the group.

List of Practicals:

1. Find the value of g by Bar pendulum.
2. To study the dependence of moment of inertia on distribution of mass (by noting time periods of oscillations using objects of various geometrical shapes but of same mass).
3. Measure time period of oscillation of a Maxwell needle and determine modulus of rigidity of the material of a given wire.
4. To determine the frequency of a tuning fork using sonometer.
5. To verify the laws of transverse vibrations of stretched string using sonometer.
6. To find the density of the given wireusing sonometer.
7. To compare mass per unit length of two strings by Melde's experiment.
8. To find the velocity of sound in the material of the given material of the given rod with a Kundt's tube given that velocity of sound in air at 0 deg. C is 332 m/s.
9. To determine the velocity of ultrasonic waves in a given liquid.

Texts and Reference Books:

1."B.Sc. Practical Physics by C.L. Arora, S.Chand & Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal

B.Sc. Semester 4
(PHY-DSC-6 Maj-402)
(Statistical Physics and Thermodynamics)
(for Table-1)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Teaching hours: 45 Hrs. / 3 Hrs. per week
Total Credits: 3 (Theory)+1 (Practical)

Learning Outcomes

- Understand fundamental probability concepts and explore the distribution of particles in compartments, macrostates, microstates, and thermodynamic probability.
- Analyze phase space, elementary cell division, and the key approaches of Maxwell-Boltzmann (M.B.), Bose-Einstein (B.E.), and Fermi-Dirac (F.D.) statistics, with applications to gases and radiation laws.
- Develop an understanding of entropy, its statistical and thermodynamic definitions, laws of thermodynamics, entropy changes, and implications for natural processes and the universe's heat death.
- Derive and apply Maxwell's thermodynamic relations to various processes like adiabatic changes and the Joule-Thomson effect, and examine the statistical basis of the ideal gas equation and related phenomena.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Basic ideas about probability, distribution of four distinguishable particles in two compartments of equal size. Concept of macrostates, microstates, thermodynamic probability, effects of constraints on the system, Stirling formula, distribution of n particles in two compartments, deviation from the state of maximum probability, distribution of distinguishable n particles in k compartments of unequal sizes.

Phase space and its division into elementary cells, three kinds of statistics. The basic approach in the three statistics. Maxwell-Boltzman statistics applied to an ideal gas in equilibrium, Maxwell-Boltzman's law of distribution of molecular speeds.

Need of quantum statistics--B.E. statistics, derivation of Planck's law of radiation, deduction of Wien's displacement law and Stefan's law from Planck's law, F.D. statistics, Comparison of M.B., B.E. and F.D. statistics.

UNIT-II

Statistical definition of entropy, change of entropy of a system, additive nature of entropy, law of increase of entropy,Examples of increase of entropy in natural processes. Entropy and disorder. Brief review of the terms and Laws of Thermodynamics, Carnot's Cycle (Using S-T

diagram), Entropy changes in Carnot's Cycle. entropy of a perfect gas. Equation of state of ideal gas from simple statistical consideration. Heat death of the universe.

Derivation of Maxwell's thermodynamical relations and applications, cooling produced by adiabatic stretching, adiabatic compression, change of internal energy with volume. Expression for (Cp-Cv), change of state and Clayperon Equation. Thermodynamical treatment of Joule-Thomson effect. Use of Joule-Thomson effect for liquification of helium.

Books Suggested:

Essential Readings:

1. "Statistical Physics and Thermodynamics", V.S. Bhatia, (Shoban Lal Nagin Chand, Jalandhar).
2. "A Treatise on Heat" Saha and Srivastava (Indian Press, Ahmedabad, 1972).

Further Readings:

1. Thermal Physics by C. Kittel & H. Kroemer, CBS Pub., 1987.
2. Thermal Physics, S.C. Garg, R.M. Bansal, and C.K. Ghosh, TMH, 2000.

PHYSICS PRACTICALS(PHY-DSC-6 Maj-402)

Max. Marks: 25 (20+5 internal assessment) **Total Credit: 1**

Teaching hours: 30 Hrs. / 2 Hrs. per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i) One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error). 09

(ii) One exercise based on experiment 04

(iii) Viva-Voce 04

(iv) Record (Practical file) 03

(v) Internal Assessment 05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

List of Practicals

1. To verify some of laws of probability distribution by throwing of similar coins like objects.
2. To determine the Stefan's constant using black body radiations.
3. To study adiabatic expansion of a gas.
4. To study the thermal expansion of a crystal using interreference fringes.
5. To measure thermal conductivity in poor conductor by Lee's method.
6. Thermo-emf calibration, comparison.
7. Total Radiation Law, temperature dependence of radiation.
8. To determine temperature coefficient of resistance by platinum resistance thermometer (PRT).
9. To determine the coefficient of thermal conductivity of copper by Searle's apparatus.
10. To determine the heating efficiency of an electric kettle with varying voltage.

Texts and Reference Books :

1. "B.Sc. Practical Physics by C.L. Arora, S.Chand& Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal.
3. "Programming with C, Schaum series" by Byron Gottfried&JitenderChhabra.

B.Sc. Semester 4
(PHY-DSC-7 Maj-403)
(Condensed Matter Physics I)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Teaching hours: 45 Hrs. / 3 Hrs. per week
Total Credits: 3 (Theory)+1 (Practical)

Learning Outcomes

- **Understand Crystal Structures**: Describe symmetry operations, Bravais lattices, Miller indices, and basic crystal structures like diamond and NaCl.
- **Apply Crystal Diffraction Concepts**: Use Bragg's Law, Laue equations, and reciprocal lattices to analyze crystal structures and Brillouin zones.
- **Explain Band Theory**: Understand energy band formation, Bloch's theorem, and distinguish between conductors, semiconductors, and insulators.
- **Analyze Electrical Properties**: Apply free electron theory, effective mass, carrier mobility, and study temperature effects, Fermi levels, and Hall effect.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Crystal structure: Symmetry operations for a two-dimensional crystal. Two dimensional Bravais lattices, Three dimensional Bravais lattices, Basic primitive cells, Crystal planes and Miller indices, Diamond and NaCl structure. Crystal diffraction: Bragg's Law, Determination of crystal structure, Laue equations, Reciprocal lattices of SC, BCC and FCC, Bragg's law in reciprocal lattice, Brillouin zones and its derivation in two dimensions, structure factor and atomic form factor.

UNIT-II

Band Theory of solids, periodic potential and Bloch theorem, Kronig-Penney model, band gaps, band structures in conductors, direct and indirect semiconductors and insulators. Free electron theory of metals, effective mass, drift current, mobility and conductivity (carrier concentration and mobility of carriers) and their variation with temperature in semi-conductors, Fermi level positions in intrinsic and extrinsic semiconductors, Wiedemann-Franz law, Hall effect in metals and semiconductors.

Recommended Books:

Essential Readings:

` 1. Introduction to Solid State Physics, C. Kittel, Wiley Eastern
2. Elements of Modern Physics,S.H. Patil, Tata McGraw Hill, 1985.
3. Solid State Physics, 6th Edition, S.O. Pillai, New Age International Publishers.
4. Physics for Degree Students, C.L. Arora and P.S. Hemne, S. Chand & Co., 2014.

Further Readings:

1. Elements of Solid State Physics, 2nd Edition, J.P. Srivastava, Prentice Hall.
2. Elementary Solid State Physics, M. Ali Omar, Pearson.
3. Crystallography for Solid State Physics, A.R. Verma, O.N. Srivastava, Wiley Eastern.

PHYSICS PRACTICALS(PHY-DSC-7 Maj-403)

Max. Marks: 25 (20+5 internal assessment) **Total Credit: 1**
Teaching hours: 30 Hrs. / 2 Hrs. per week

General Guidelines for Physics Practical Examinations:

The distribution of marks is as follows:

(i) One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error). 09
(ii) One exercise based on experiment 04
(iii) Viva-Voce 04
(iv) Record (Practical file) 03
(v) Internal Assessment 05

Note for Examiners:

1. The marks scored under each head must be clearly written on the answer sheet.
2. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
3. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
4. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
5. Number of candidates in a group for practical examination should not exceed 12.
6. In a single group, no experiment to be allotted to more than three examinees in the group.

List of Practicals

1. Measurement of reverse saturation current in p-n junction diode at various temperatures and to find the approximate value of energy gap.
2. To measure the magnetic susceptibility of FeCl2 solution by Quincke's method.
3. To trace the B-H curves for different materials using CRO and find the magnetic parameters from these.
4. To find the conductivity of a given semi-conductor crystal using four probe method.
5. To determine the Hall coefficient for a given semiconductor.
6. To study the characteristics of a thermistor and find its parameters.
7. To draw the characteristics of a Light emitting diode.
8. To study the characteristics of photo voltaic cell
9. To determine the resistivity of wires of different materials.
10. To determine the mass susceptibility of paramagnetic solution using Quincke's method.

Texts and Reference Books:

1. "B.Sc. Practical Physics by C.L. Arora, S.Chand & Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal.
3. "Programming with C, Schaum series" by Byron Gottfried&JitenderChhabra.

B.Sc. Semester 4
(PHY-DSC-8 Maj-404)
(Electronics and Solid-State Devices – I)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Teaching hours: 45 Hrs. / 3 Hrs. per week
Total Credits: 3 (Theory)+1 (Practical)

Learning Outcomes

- **Apply Network Theorems and CRO Usage**: Understand and apply Thevenin's/Norton's theorems, source conversions, and use CRO for waveform analysis, frequency, and phase measurements.
- **Explain Semiconductor Device Behavior**: Interpret energy band diagrams, Fermi level positions, and diode characteristics including Zener, LED, photodiode, and solar cell.
- **Analyze Diode-Based Circuits**: Design and evaluate rectifiers, filter circuits, clipping circuits, and voltage regulators using Zener diodes.
- **Understand and Use BJT Configurations**: Explain BJT operation in CB, CE, CC modes, perform biasing, load-line analysis, and evaluate CE amplifier performance.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Concepts of current and voltage sources, Thevenin's theorem, Norton's theorem, Source conversion. CRO, Block diagram, construction and principle of working, Use of CRO for frequency, time period, special features of dual trace, phase measurements. Energy band diagrams in semiconductors, Direct and indirect semiconductors, Formula to calculate Position of Fermi level in p and n semiconductors, Barrier formation, energy band diagram of p-n junction, Formula for Depletion width, Qualitative ides of current flow mechanism in forward and reverse biased diode, v-i characteristics, static and dynamic resistance, Depletion and diffusion capacitance, zener diode, LED, photodiode and solar cell.

UNIT-II

Diode circuits, Clipping circuits, Rectification: half wave, full wave and bridge rectifiers, filter circuits (C, LC and π filters), rectification efficiency and ripple factor in LC filter, voltage regulation circuit using zener diode, voltage multiplier circuits. Bipolar Junction transistors: Structure and working, different currents in transistor, switching action. Characteristics of CB, CE and CC configurations, Active, cutoff and saturation regions. Load line analysis of transistors, Q-point, Transistor biasing and stabilization of operating point, fixed bias, collector to base bias, bias circuit

with emitter resistor, voltage divider biasing circuit. Working analysis of CE amplifier using h-parameters, current, voltage and power gain, input and output impedance.

Recommended Books:

Essential Readings:

1. Electronic Devices and Circuit Theory, 7th Ed., R. Boylestad, L. Nashelsky, Prentice Hall Inc.
2. Electronic Principles, A.P. Malvino, and D.J. Bates, 7th ed. McGraw Hill
3. Solid State Electronic Devices, 6th Ed., Ben G. Streetman and S. Banerjee, Eastern Economy Edition.

Further Readings:

1. Basic Electronics, 5 th Edition, B.L. Thareja, S. Chand.
2. Basic Electronics and Linear Circuits, N.N. Bhargave, D.C. Kulshreshtha, and S.C. Gupta,Tata Mc Graw Hill.
3. Foundations of Electronics, D. Chatopadhyay, P.C. Rakshit, B. Saha, and N.N. Purkit, New Age International

PHYSICS PRACTICALS (PHY-DSC-8 Maj-404)

Max. Marks: 25 (20+5 internal assessment) **Total Credit: 1**

Teaching hours: 30 Hrs. / 2 Hrs. per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i) One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error). 09
(ii) One exercise based on experiment 04
(iii) Viva-Voce 04
(iv) Record (Practical file) 03
(v) Internal Assessment 05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

List of Practicals

1. To draw the characteristics of a P-N Junction diode.
2. To draw the characteristics of a Zener diode.
3. To study the stabilization of output voltage of a power supply with Zener diode.
4. To measure and plot Common Emitter Characteristics of a transistor (pnp or npn).
5. To plot Common Base Characteristics and determine h-parameters of a given transistor.
6. To measure the efficiency and ripple factors for (a) Half-wave, (b) Full wave, and (c) Bridge rectifier circuits.
7. To study the reduction in the ripples in the rectified output with RC, LC and π-filters.
8. To study the function of diode as a clipping element.
9. To obtain the waveform of a given oscillator using a CRO/DSO.

Texts and Reference Books :

1. "B.Sc. Practical Physics by C.L. Arora, S. Chand & Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal.

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Statistics

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR(Table 1*) and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY (Table 2*) B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH

Table 1*						**Table 2***					
Sem	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**
III	L-200	(STAT-DSC-3 Maj/Min-301)	Descriptive Statistics-II	3+1=4 (x2)	24	III	L-200	(STAT-DSC-3 Maj/Min-301)	Descriptive Statistics-II	3+1=4 (x3)	24
		(STAT-DSC-4 Maj/Min-302)	Probability Distributions	3+1=4 (x2)				(STAT-DSC-4 Min-302)	Probability Distributions	3+1=4	
		(STAT-MDC-3-301)	Applied Statistics	3				(STAT-MDC-3-301)	Applied Statistics	3	
		(STAT-SEC-3-301)	Statistics for Business Analytics with Microsoft Excel	3				(STAT-SEC-3-301)	Statistics for Business Analytics with Microsoft Excel	3	
		(AEC-3-301)	Eng-3/MIL-1	2				(AEC-3-301)	Eng-3/MIL-1	2	
IV	L-200	(STAT-DSC-5 Maj/Min-401)	Statistical Inference	3+1=4 (x2)	24	IV	L-200	(STAT-DSC-4 Maj-402)	Probability Distributions	3+1=4 (x2)	24
		(STAT-DSC-6 Maj-402)	Sampling Distributions	4				(STAT-DSC-5 Maj/Min-401)	Statistical Inference	3+1=4(X3)	
		(STAT-DSC-7 Maj-403)	Applied Statistics-I	4							
		(STAT-DSC-8 Maj-404)	Practical	4							
		(VAC-3-401)*	---	2				(VAC-3-401)*	----	2	
		(AEC-4-401)	Eng-4/MIL-2	2				(AEC-4-401)	Eng-4/MIL-2	2	
The student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secures 4 credits from summer internship											
***Value Added Courses are not being offered by the Deaprtment of Statistics, Panjab University, Chandigarh**											

SEMESTER-III
Course Code: (STAT-DSC-3-Maj/Min-301)
Course Name: Descriptive Statistics-II

Credit: 4
L-3, P-1

Notes:
1. **Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).**
2. **Practical examination will be held in the respective semester.**

Paper Title with code	Lecture(L) /Credit	Practical(P)/ Credit	Marks Distribution	
Descriptive Statistics-II (STAT-DSC-3-Maj/Min-301)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Theory Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours	Practical Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

DESCRIPTIVE STATISTICS-II

Objectives: The main objective of this paper isto provide students with a comprehensive understanding of correlation analysis, curve fitting techniques, linear regression, and multivariate data analysis. Additionally, students will learn about multivariate data analysis techniques such as multiple and partial correlation, and gain the ability to interpret and analyze relationships among multiple variables.

Learning Outcomes: On completion of this course students will be able to:
1. Understand the concepts and types of correlation, regression, multivariate data and their utilization in real life.
2. Utilize the principle of least squares for curve fitting, including fitting straight lines, second degree parabolas, power curves, and exponential curves.
3. Analyze multivariate data by understanding the plane of regression, properties of residuals, and variance of the residual.
4. Apply multiple and partial correlation techniques for three variables, including understanding the coefficient of multiple correlation, coefficient of partial correlation, and coefficient of determination.

Note: There will be nine questions in all. Question No.1 will be compulsory covering whole of the syllabus and comprising 5 to 8 short answer type questions. Rest of the eight questions will be set from the four units uniformly i.e. two from each unit. The candidate will be required to attempt five questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks except the compulsory question, the distribution of marks for which will be as follows:-

UNIT-I

Correlation: Concept and types of correlation, methods of finding correlation - scatter diagram, Karl Pearson's Coefficient of correlation (r), its properties, coefficient of correlation for a bivariate frequency distribution. Rank correlation with its derivation, its merits and demerits, limits of rank correlation coefficient, tied or repeated ranks.

UNIT-II

Curve Fitting: Principle of least squares, fitting of straight line, second degree parabola, power curves of the type $Y=aX^b$, exponential curves of the types $Y=ab^X$ and $Y=ae^{bX}$, regression curves.

UNIT-III

Linear Regression: Two lines of regression, regression coefficients, properties of regression coefficients, angle between two regression lines, standard error of estimate obtained from regression line, correlation coefficient between observed and estimated values, distinction between correlation and regression.

UNIT-IV

Multivariate Data: Plane of regression, properties of residuals, variance of the residual. Multiple and partial correlation for three variables: coefficient of multiple correlation and its properties, coefficient of partial correlation and its properties, multiple correlation in terms of total and partial correlations and coefficient of determination.

Books recommended

S. No.	Title of Book	Name of author	Publisher
1.	Introduction to Theory of Statistics	Mood A.M., Graybill F.A. &Boes D.C.	McGraw Hill
2.	Applied General Statistics	Croxton F.E., Cowden D.J. &Kelin S.	Prentice Hall
3.	Introduction to Mathematical Statistics	Kapoor&Saxena.	S.Chand
4.	Statistical Methods	Snedecor G.W. & Cochran W.G.	Iowa State Uni. Press
5.	Fundamentals of Mathematical Statistics	Gupta S.C.& Kapoor V.K.	Sultan Chand & Sons

(STAT-DSC-3-Maj/Min-301): **DESCRIPTIVE STATISTICS-II (PRACTICAL)**

Notes:

i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

ii) The practical of paper (STAT-DSC-3-Maj/Min-301) will be based on the following topics:

1. Problems based on Karl Pearson's Coefficient of correlation.
2. Problems based on Spearman's Rank Correlation coefficient without ties.
3. Problems based on Spearman's Rank Correlation coefficient with ties.
4. Problems based on fitting of straight line.
5. Problems based on second degree parabola curve.
6. Problems based on exponential curve.
7. Problem based on two lines of regression.
8. Problem based on regression coefficients.
9. Problem based on multiple correlation coefficients.
10. Problem based on partial correlation coefficients.

Course Code: (STAT-DSC-4 Maj/Min-302)
Course Name: Probability Distributions

Credit: 4
L-3, P-1

Notes:

1. **Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).**
2. **Practical examination will be held in the respective semester.**

Paper Title with code	Lecture(L)/ Credit	Practical(P)/ Credit	Marks Distribution	
Probability Distributions (STAT-DSC-4 Maj/Min-302)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Theory Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours	Practical Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

PROBABILITY DISTRIBUTIONS

Objectives: The main objective of the paper is to provide students with a comprehensive understanding of various probability distributions including Bernoulli, Binomial, Poisson, Negative Binomial, Geometric, Continuous Uniform, Gamma, Exponential, and Normal distributions. This includes learning their key characteristics, moments, moment generating functions, mean deviation, mode, characteristic functions, and their applications in real-world scenarios such as modeling and inference.

Learning Outcomes: On completion of this course students will be able to:

1. Understand about various discrete and continuous distributions and their properties.
2. Compute moments, moment generating functions, and characteristic functions for different probability distributions.
3. Analyze various types of numerical problems with the help of probability distributions.
4. Apply properties of discrete as well as continuous probability distributions to solve the real life problems.

Note: There will be nine questions in all. Question No.1 will be compulsory covering whole of the syllabus and comprising 5 to 7 short answer type questions. Rest of the eight questions will be set from the four units uniformly i.e. two from each unit. The candidate will be required to attempt five questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

UNIT-I

Bernoulli distribution and its moments, Binomial distribution: Moments, recurrence relation for the moments, mean deviation about mean, mode, moment generating function (m.g.f), characteristic function, additive property and recurrence relation for the probabilities of Binomial distribution.

UNIT-II

Poisson distribution: Poisson distribution as a limiting case of Binomial distribution, moments, mode, recurrence relation for moments, m.g.f., additive property of independent Poisson variates. Negative Binomial distribution: m.g.f., deduction of moments of negative binomial distribution from those of binomial distribution. Geometric distribution: moments, m.g.f,and lack of memory.

UNIT-III

Continuous uniform distribution: Moments, m.g.f.,characteristic function and mean deviation. Gamma distribution: m.g.f., and additive property. Exponential distribution: m.g.f., moments and lack of memory.

UNIT-IV

Normal distribution as a limiting form of binomial distribution, chief characteristics of Normal distribution; mode, median, m.g.f.,and moments of Normal Distribution, A linear combination of independent normal variates, points of inflexion, mean deviation about mean, area property of Normal distribution, importance and fitting of normal distribution.

Books recommended

S. No.	Title of Book	Name of author	Publisher
1.	Statistics:A Beginner's Text Vol. II	Bhat B.R., Srivenkatramana T. & RaoMadhava K.S.	New Age International
2.	Fundamentals of Mathematical Statistics	Gupta S.C. & Kapoor V.K.	Sultan chand & Sons
3.	Introduction to Mathematical Statistics	Kapoor&Saxena.	S.Chand
4.	Statistics	Johnson R.	Wiley Publishers
5.	Mathematical Statistics With Applications	Freund's J.E.	Prentice Hall

(STAT-DSC-4 Maj/Min-302): **PROBABILITY DISTRIBUTIONS (PRACTICAL)**

Notes:

i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

ii) The practical of paper (STAT-DSC-4 Maj/Min-302) will be based on the following topics:

1. Practical problem based on binomial distribution.
2. To fit binomial distribution of the given data.
3. Practical problem based on Poisson distribution.
4. To fit Poisson distribution of the given data.
5. Practical problem based on negative binomial distribution.
6. Practical problem based on geometric distribution.
7. Practical problem based on continuous uniform distribution.
8. Practical problem based on exponential distribution.
9. Practical problem based on area property of normal distribution.
10. To fit normal distribution of the given data.

Semester-IV

Course Code: (STAT-DSC-5 Maj/Min-401)
Course Name: Statistical Inference

Credit: 4
L-3, P-1

Notes:

1. **Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).**
2. **Practical examination will be held in the respective semester.**

Paper Title with code	Lecture(L) /Credit	Practical(P) / Credit	Marks Distribution	
Statistical Inference (STAT-DSC-5 Maj/Min-401)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Theory Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours	Practical Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

STATISTICAL INFERENCE

Objectives: The learning objectives of this course are as follows:

1. To understand the concept of estimation theory and testing of hypothesis.
2. To draw inferences about the unknown population parameters based on random samples.
3. To validate the estimation/ inference about the population parameters.

Learning Outcomes: After successful completion of this course, students should be able to:

1. Understanding of estimation theory, Point and interval estimations.
2. Get knowledge about various characteristics of a good estimator and different methods of estimation.
3. Develop the best/most powerful statistical tests to test the hypotheses regarding unknown population parameters by using Neyman-Pearson theory.
4. Apply the likelihood ratio test and Sequential probability ratio test (SPRT) to test the different problems.

Note: There will be nine questions in all. Question No.1 will be compulsory covering whole of the syllabus and comprising 5 to 8 short answer type questions. Rest of the eight questions will be set from the four units uniformly i.e. two from each unit. The candidate will be required to attempt five questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks except the compulsory question, the distribution of marks for which will be as follows:-

UNIT I

Estimation Theory: Parameter space, sample space, point estimation, requirement of a good estimator, consistency, unbiasedness, efficiency, sufficiency, Minimum variance unbiased estimators, Factorization theorem, Fisher- Neyman Criterion: statement and applications, Cramer-Rao inequality: statement and application, MVB estimators and their applications, Statement of Rao-Blackwell theorem.

UNIT II

Method of Estimation: Maximum likelihood, method of moments, least squares and minimum variance, Properties of maximum likelihood estimators (illustration), Interval Estimation: confidence interval and confidence limits for the parameters of normal distribution, confidence intervals for large samples.

UNIT III

Test of Significance: Principles of test of significance: Null and alternative hypotheses, simple and composite, Type-Iand Type-II errors, critical region, level of significance, power of the test, best critical region, most powerful test, uniformly most powerful test, uniformly most powerful unbiased (UMPU)critical region, Neyman- Pearson Lemma: statement and its applications to construct most powerful test.

UNIT-IV

Likelihood Ratio Test: Introduction of likelihood ratio test, its properties, test for the mean of a normal population, test for the equality of means of two normal populations, test for the variance of a normal population, test for equality of variances of two normal populations.

Sequential Analysis: Sequential probability ratio test (SPRT), operating characteristic (OC) Function of SPRT, average sample number (ASN).

Books recommended

S. No.	Title of Book	Name of author	Publisher
1.	A First Course on Parametric Inference	Kale B.K.	Narosa
2.	Introduction to Theory of Statistics	Mood A.M., Graybill F.A. &Boes D.C.	McGraw Hill
3.	Mathematical Statistics With Applications	Freund's J.E.	Prentice Hall
4.	Fundamentals of Mathematical Statistics	Gupta S.C. & Kapoor V.K.	Sultan chand & Sons

(STAT-DSC-5 Maj/Min-401): **STATISTICAL INFERENCE (PRACTICAL)**

Notes:

i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

ii) The practical of paper (STAT-DSC-5 Maj/Min-401)will be based on the following topics:

1. Problem based on unbiased estimators.
2. Problem based on consistent estimators.
3. Problem based on efficient estimators.
4. Problem based on sufficiency.
5. Problem based on maximum likelihood estimators of standard distributions.
6. Problem based on method of moments.
7. Problem based on least square method of estimation.
8. Problem based on Best critical region or most powerful test.
9. Problem based on uniformly most powerful (UMP) test.
10. Problem based on likelihood ratio test.
11. Testing problem based on SPRT and find OC and ASN functions also.

Semester-IV

Course Code: (STAT-DSC-6 Maj-402)
Course Name: Sampling Distributions

Credit: 4
L-4

Note: Students are to be engaged to 4 hours of teaching per week.

Paper Title with code	Lecture(L)/Credit	Marks Distribution
Sampling Distributions (STAT-DSC-6 Maj-402)	Lecture : 4 Credit : 4	Theory Max. Marks: 100 Theory: 80 Internal Assessment : 20 Time: 3 Hours

SAMPLING DISTRIBUTIONS

Objectives: The learning objectives of this course are as follows:

1. To introduce the modes of convergence and their relation to limit laws, with a focus on the central limit theorem.
2. To introduce the concept of sampling distributions and their applications in statistical inference.
3. To describe the statistical ideas behind the procedure of hypothesis testing.
4. To explain the assumptions and conditions under which to apply different tests of hypothesis about population parameters and draw appropriate conclusions from the analysis.

Learning Outcomes: After successful completion of this course, students should be able to:

1. Understand the basics of convergence theory and its importance in limit laws.
2. Apply the concept of the central limit theorem and the relevance of the theorem in
3. inferential statistics.
4. Analyze data by using suitable hypothesis testing procedures in real-life applications related to large and small samples.
5. Apply the knowledge of the idea of sampling distributions and appreciate their importance in the field of statistics.
6. Integrate the knowledge of various sampling distributions like chi-square, t, and F distributions in hypothesis testing problems.

Note: There will be nine questions in all. Question No.1 will be compulsory covering whole of the syllabus and comprising 5 to 8 short answer type questions. Rest of the eight questions will be set from the four units uniformly i.e. two from each unit. The candidate will be required to attempt five questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks except the compulsory question, the distribution of marks for which will be as follows:-

UNIT I

Generating functions and Central Limit Theorem: Moment generating function, its limitations, properties, cumulants and their properties, characteristic function and its properties, Chebyshev's inequality, Weak Law of Large Numbers (WLLN),along with examples and applications.
Basic idea and relevance of Central Limit Theorem (CLT), De-Moivre Laplace theorem,Lindeberg Levy theorem, Liapunov Theorem (only statement), and applications of CLT.

UNIT II

Sampling Distributions and Test of Hypotheses: Concepts of parameter, statistic, sampling distribution of a statistic, standard error. Sampling distribution of sample mean, standard errors of the sample mean, sample variance, and sample proportion.
Null and alternative hypotheses, level of significance, Type I and Type II errors, their probabilities and critical region, determination of sample size, confidence intervals, and pvalue.
Tests of significance and confidence intervals for - single proportion, difference of two proportions, single mean, difference of two means, and difference of two standard deviations.

UNIT III

Exact Sampling Distribution: Chi-Square distribution: Definition and derivation of the probability distribution of Chi-square distribution with n degrees of freedom, nature of the curve for different degrees of freedom, mean, mode, variance, moment generating function, cumulant generating function, additive property, and limiting form of the Chi-square distribution, Applications of Chi-Square distribution.

UNIT IV

Exact Sampling Distributions (continued): Student's t-statistic and Fishers t-statistic: definition and derivation of their sampling distributions, nature and characteristics of graph of t distribution, moments, limiting form and applications of the t distribution.
F-statistic: Definition and derivation of the sampling distribution, the graph of F distribution, moments, and applications of the F distribution. Relationship between t, F, and Chi-square distributions.

Books recommended

- Gupta, S. C. and Kapoor, V. K. (4010). Fundamentals of Mathematical Statistics, Twelfth Edition, S. Chand and Sons. Delhi.
- Goon, A.M., Gupta, M.K. and Dasgupta, B. (3016). An Outline of Statistical Theory,Volume I, The World Press, Kolkata.
- Mukhopadhyay, P. (3016). Mathematical Statistics, Books and Allied, India.
- Hogg, R.V., Tanis, E.A. and Rao, J.M. (2009). Probability and Statistical Inference,Seventh Edition, Pearson Education, New Delhi.
- Miller, I. and Miller, M. (2006). John E. Freund's Mathematical Statistics with Applications, Eight Edition, Pearson Education, Asia.
- Johnson, R.A. and Bhattacharya, G.K. (2001). Statistics-Principles and Methods, Fourth Edition, John Wiley and Sons.

SUGGESTED READINGS

- Bhat, B.R. (3016). Modern Probability Theory- An Introductory Textbook, Fourth Edition, New Age International Publishers.
- Rohatgi, V. K and Saleh M. E. (3015). An Introduction to Probability and Statistics, Third Edition, John Wiley and Sons, Inc., New Jersey.
- Mood, A.M. Graybill, F.A. and Boes, D.C. (2007). Introduction to the Theory of Statistics, Third Edition, (Reprint), Tata McGraw-Hill Pub. Co. Ltd.

Semester-IV
Course Code: (STAT-DSC-7 Maj-403)
Course Name: Applied Statistics-I

Credit: 4
L-4

Note: Students are to be engaged to 4 hours of teaching per week.

Paper Title with code	Lecture(L)/Credit	Marks Distribution
Applied Statistics-I (STAT-DSC-7 Maj-403)	Lecture : 4 Credit : 4	Theory Max. Marks: 100 Theory: 80 Internal Assessment : 20 Time: 3 Hours

APPLIED STATISTICS-I

Objectives: The Learning Objectives of this course are as follows:

1. Introduce the concept of time series, its components, and their estimation.
2. Introduce the application of time series.
3. Introduce the concept, formulation, and application of index numbers.

Learning outcomes: After completion of this course, the students will be able to:

1. Understand the concepts of time series and index numbers.
2. Formulate, solve, and analyse the use of time series and index numbers for real-world problems.
3. Apply index number and time series concepts for making future planing and policy related to business.

Note: There will be nine questions in all. Question No.1 will be compulsory covering whole of the syllabus and comprising 5 to 8 short answer type questions. Rest of the eight questions will be set from the four units uniformly i.e. two from each unit. The candidate will be required to attempt five questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks except the compulsory question, the distribution of marks for which will be as follows:-

UNIT-I

Index Numbers: Definition, problems involved in the construction of index numbers, types of index numbers, methods of construction of price index numbers–un weighted and weighted aggregates method, simple average of price relatives, weighted average of price relatives, quantity and value indexes, errors in index numbers, Uses and limitations of index numbers.

UNIT-II

Index Numbers (Continued):Tests of consistency of index numbers-time reversal test, factor reversal test and circular test, chain base index numbers, uses and limitations of chin base index numbers, base shifting, splicing and deflating of index numbers, cost of living or consumer price index numbers, its construction and uses.

UNIT-III

Time Series Analysis: Definition, components of time series-trend, seasonal variations, cyclic variations, irregular component, illustrations, additive and multiplicative models, determination of trend: graphic method, semi-averages method, method of curve fitting by principle of least squares-fitting of straight line, second degree parabola and exponential curves, growth curves and their fittings, moving average method and uses of time series.

UNIT-IV

Time Series Analysis (Continued):Analysis of seasonal fluctuations, construction of seasonal indices using method of simple averages, ratio to trend method and ratio to moving average method, link relative method, Deseasonalisation of data, Measurement of cyclic fluctuations, measurement of irregular or random variations, Serial correlation and correlogram and forcasting in time series.

Books recommended

S. No.	Title of Book	Name of author	Publisher
1.	Applied General Statistics	Croxton F.E., Cowden D.J. &Kelin S.	Prentice Hall
2	Fundamental of Statistics, Goon Vol II (3011)	A.M., Gupta, M.K. and Dasgupta, B.	World Press, Kolkata.
2.	Fundamentals of Applied Statistics	Gupta S.C. & Kapoor V.K.	Sultan Chand & Sons

Semester-IV

Course Code: (STAT-DSC-8 Maj-404)
PRACTICAL

Credit: 4
L-4

Time: 3 Hours **Max. Marks: 80+20***

*** Internal Assessment**
Viva voce: 10
Record of the practical: 10
Written practical Exam: 60

Notes:

(i) **Students are to be engaged to 8 hours of teaching per week.**

(ii) **The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 20 marks, in three hours duration.**

(iii) **Practical work to be conducted using calculators/Microsoft Excel.**

1. The practical will be based on the following topics:
2. Problem based on moment generating function and characteristic function.
3. Problem based on Chebyshev's inequality and Weak Law of Large Numbers.
4. To apply large sample test of significance for single mean and to obtained confidence interval.
5. To apply large sample test of significance for difference between two
6. means and standard deviations.
7. To apply large sample test of significance for single proportion and to
8. Obtained confidence interval.
9. To apply large sample test of significance for difference between two
10. Proportions and to obtained confidence interval.
11. To apply Chi-square test for testing the single variance.
12. To apply Chi-square test for goodness of fit.
13. To apply Chi-square test for independence of attributes.
14. Testing based on a 2 x 2 contingency table without and with Yates' corrections.
15. To apply t-test for testing single mean and difference between means and to obtain their confidence intervals.
16. To apply paired t-test for difference between two means.
17. To apply test of significance of sample correlation coefficient.
18. To apply F-test for testing difference of two variances.
19. Construction of price and quantity index numbers by simple aggregate method.
20. Construction of price and quantity index numbers by Laspeyres, Paasche's, Drobish-Bowley, Marshall-Edgeworth, Walsch and Fisher's Formula.
21. Construction of price and quantity index numbers by simple and weighted average of price relatives.
22. Construction of consumer price index number by family budget method and Aggregate expenditure method
23. Time Reversal Test and Factor Reversal Test Measurement of trend by using the least square method with the help of semi-average method.
24. Problem based on chain base index numbers.
25. Problem based on cost of living index numbers.
26. Problem based on base shifting, splicing and deflation of index numbers.

27. Fit the trend with the help of a straight line by using semi-average method.
28. Fit the trend with the help of a straight line by using least square method.
29. Fit the trend with the help second degree parabola curve by using least square method.
30. Fit the trend with the help of an exponential curve by using least square method.
31. Measure the trend by using moving average method.
32. Measurement of seasonal variations using simple Average method.
33. Measurement of seasonal variations using simple ratio to trend method.
34. Measurement of seasonal variations using ratio to moving average method.
35. Measurement of seasonal variations using link relative method.

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Zoology

STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR (Table1*) AND STUDENTS OPTING FOR MULTI-DISCIPLINARY/INTER-DISCIPLINARY (Table 2*)

B.Sc.,B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH IN ZOOLOGY

Table1*						Table2*					
Sem	**Level**	**Course Code**	**Course Name**	**Credits Th(L) + Pr(P)**	**Total Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits Th(L) + Pr(P)**	**Total Credits**
III	L-200	(ZOO-DSC-3 Maj/Min-301)	Animal Physiology	3+1=4 (x2)	24	III	L-200	(ZOO-DSC-3 Maj/Min-301)	Animal Physiology	3+1=4(x3)	24
		(ZOO-DSC-4 Maj/Min-302)	Developmental Biology	3+1=4 (x2)				(ZOO-DSC-4 Min-302)	Developmental Biology	3+1=4	
		(ZOO-MDC-3-301)	First Aid and Emergency Management	2+1=3				(ZOO-MDC-3-301)	First Aid and Emergency Management	2+1=3	
		(ZOO-SEC-3-301)	Poultry Farming	2+1=3				(ZOO-SEC-3-301)	Poultry Farming	2+1=3	
		(AEC-3-301)	Eng-3/MIL-1	2				(AEC-3-301)	Eng-3/MIL-1	2	
IV	L-200	(ZOO-DSC-5 Maj/Min-401)	Biochemistry	3+1=4 (x2)	24	IV	L-200	(ZOO-DSC-5 Maj/Min-401)	Biochemistry	3+1=4(x3)	24
		(ZOO-DSC-6 Maj-402)	Biomolecules and Nutrition	3+1=4				(ZOO-DSC-4 Maj-402)	Developmental Biology	3+1=4(x2)	
		(ZOO-DSC-7 Maj-403)	Basic Biochemical Techniques	3+1=4							
		(ZOO-DSC-8 Maj-404)	Endocrinology	3+1=4							
		(ZOO-VAC-3-401)	Reproductive Health	1+1=2				(ZOO-VAC-3-401)	Reproductive Health	1+1=2	
		(AEC-4-401)	Eng-4/MIL-2	2				(AEC-4-401)	Eng-4/MIL-2	2	
The student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secures 4 credits from summer internship											

***Note:The candidate will opt for eitherTemplate1or 2 from Semester III onwards.**

Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain at least 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

Semester- III (L-200)

Course Code: ZOO-DSC-3-Maj/Min-301
Course Title: Animal Physiology

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) +08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) +03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:

- *Students will be able to explore basic physiological principles common to animals.*
- *Students will understand the fundamental concepts of digestion, respiration, blood circulation and excretion.*
- *Students will learn about the structure and functioning of nerves and muscles, andthe role of hormones in physiological processes.*

UNIT- 1
Digestion: Extra and Intra Cellular Digestion, Absorption of carbohydrates, proteins and fats.
Respiration: Exchange and transport of respiratory gases, Oxygen dissociation curve, Bohr Effect, Chloride shift and Haldane effect.

UNIT- 2
Blood: Composition and functions of blood and lymph,Blood Clotting, Blood groups
Heart: Origin and conduction of heart beat, cardiac cycle, cardiac output, electrocardiogram and blood pressure.

UNIT- 3
Excretion: Structure of nephron, urine formation, composition of urine, osmoregulation and counter current mechanism.
Muscles: Types of muscles and their structure. Skeletal Muscle Contraction.

UNIT- 4
Neural Integration: Structure of neuron, resting membrane potential, origin and propagation of nerve impulse.
Endocrine System: Hormones and functions of Endocrine glands: Hypothalamus, pituitary, thyroid, parathyroid, adrenal, pancreas and gonads.

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-3-Maj/Min-301: Animal Physiology

Practical

1. Study of blood smear of a mammal.
2. Demonstration of estimation of haemoglobin content in blood.
3. Recording of blood pressure of man
4. Determination of heart rate and pulse rate
5. Study of waveforms in electrocardiogram.
6. Study of stained sections of different mammalian tissues and organs
7. An idea of location of endocrine glands in mammals through charts / models/ video clippings

SUGGESTED READINGS

1. Taneja, S.K.: Biochemistry & Animal Physiology, Trueman Book Co.,1997.
2. Guyton, A.S.: Text Book of Medical Physiology, 14th edition, W.B. Saunders Company, 2020.
3. H.R Singh &Neeraj Kumar, Animal Physiology & Biochemistry Edition. 2005,Vishal Publication Co. Jalandhar.
4. VasantikaKashyap: Animal Physiology & Biochemistry, KedarNath Ram Nath Publications, Meerut.
5. Berry, A.K. Text Book of Animal Physiology 2005 Edition, Emkay Publications Delhi.

Course Code: ZOO-DSC-4 Min-302
Course Title: Developmental Biology

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:

- *Students will get detailed knowledge about the formation and structure of gametes.*
- *Students will understand development patterns in different animals ranging from proto chordates to mammals.*
- *Students will be able to apply their knowledge and pursue research in the field of Developmental biology.*

UNIT- 1
Gametogenesis: Spermatogenesis, spermiogenesis, Oogenesis &Vitellogenesis
Types of eggs, egg membranes and polarity of egg.

UNIT- 2
Fertilization & parthenogenesis.
Cleavage: Types of cleavage patterns depending upon amount and distribution of yolk and position of spindle.
Blastulation and types of blastula.

UNIT- 3
Gastrulation: Definition, Morphogenetic movements & Fate of three germ layers
Development up to three germ layers in *Herdmania*, frog, chick and rabbit.

UNIT- 4
Foetal membranes, their formation and role.
Mammalian placenta – its formation, types and functions.
Metamorphosis in *Herdmania* and *Rana* (frog)

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-4 Min-302

Practical

1. Study of the following permanent slides for gametogenesis- T.S. Testis and T.S. ovary of mammals
2. Study of the development of frog from permanent slides.
3. Study of the development of chick embryo from permanent slides up to 96 hours.
4. Study of metamorphosis of *Herdmania* and Frog through charts/video/ models

SUGGESTED READINGS

1. Ballinsky, B.I. An Introduction to Embryology, 5th Edition, Saunders Co., 1981.
2. Gilbert, S.F. Developmental Biology, 10th Edition, Sinauer Ass., Inc. Publ., 2014.
3. Urspaung, H. Major Problems in Developmental Biology, Academic Press, New York, 1966.
4. Wolport, Lewis Principles of development, Oxford University press, Oxford, 2007
5. Veer BalaRastogi, Chordate Embryology (Developmental Biology) KedarNath Ram Nath, Meerut/ Delhi 2011

Semester- IV (L-200)

Course Code: ZOO-DSC-5 Maj/Min-401
Course Title: Biochemistry

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) +08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) +03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:

- *Students will gain knowledge about biomolecules, the basic building blocks of all living organisms.*
- *Students will get to know about the nature and mode of action of enzymes, the biological catalysts.*
- *Students will be able to understand the metabolic pathways of carbohydrates, lipids and proteins.*

UNIT- 1
Biomolecules: Carbohydrates, proteins, lipids and nucleic acids: their classifications and functions.

UNIT- 2
Enzymes: Nature, nomenclature, mode of action, their classification, coenzymes and cofactors.
Carbohydrate metabolism: EmbdenMeyerhof Parnas pathway (glycolysis), Tricarboxylic Acid Cycle, Glycogenesis.

UNIT- 3
Lipid metabolism: Beta-oxidation of fatty acids, gluconeogenesis, lipogenesis and ketosis.

UNIT- 4
Protein metabolism: Metabolism of amino acids: Oxidative deamination, Transamination Decarboxylation and Ornithine cycle.

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-5 Maj/Min-401: Biochemistry

Practical

1. Preparation of Molar, Normal Solutions and Buffers
2. Determination of pH
3. Study the action of salivary amylase under optimum conditions.
4. Study the effect of pH and temperature on the action of salivary amylase.
5. Qualitative analysis of carbohydrates, proteins and lipids.
6. Analysis of urine for urea, uric acid, chloride and glucose.

SUGGESTED READINGS

1. Taneja, S.K.: Biochemistry & Animal Physiology, Trueman Book Co.,1997.
2. Robert, K., Murray, R.K.: Harper's Biochemistry, 22nd edition, Daryl, M., Granner, K., Prentice – Hall International, Inc.,1990 Victor, W. and Woodwell.
3. Nelson, D.L. and Cox, M.M.: Lehninger Principles of Biochemistry, 5th edition, W.H., Freeman and Company, New York, 2008.
4. Murray, R.K., Bender, D.A., Botham, K.M., Kennelly, P.J., Rodwell, V.W. and Well, P.A. (2009). Harper's Illustrated Biochemistry. XXVIII Edition, International Edition, The McGraw- Hill Companies Inc. Suggestive readings
5. Stryer, L., Berg, J., Tymoczko, J., Gatto, G. (2019). Biochemistry (9th Ed.). New York, WH: Freeman. 4. Voet, D., Voet. J. G. (2013). Biochemistry (4th ed.). New Jersey, John Wiley & Sons, Asia Pvt. Ltd.
6. H.R Singh &Neeraj Kumar, Animal Physiology & Biochemistry, Vishal Publication Co. Jalandhar Ed. 2005
7. VasantikaKashyap: Animal Physiology & Biochemistry, KedarNath Ram Nath Publications, Meerut

Course Code: ZOO-DSC-4 Maj-402
Course Title: Developmental Biology

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:

- *Students will get detailed knowledge about the formation and structure of gametes.*
- *Students will understand development patterns in different animals ranging from proto chordates to mammals.*
- *Students will be able to apply their knowledge and pursue research in the field of Developmental biology.*

UNIT- 1
Gametogenesis: Spermatogenesis, spermiogenesis, Oogenesis &Vitellogenesis
Types of eggs, egg membranes and polarity of egg.

UNIT- 2
Fertilization & parthenogenesis.
Cleavage: Types of cleavage patterns depending upon amount and distribution of yolk and position of spindle.
Blastulation and types of blastula.

UNIT- 3
Gastrulation: Definition, Morphogenetic movements & Fate of three germ layers
Development up to three germ layers in *Herdmania*, frog, chick and rabbit.

UNIT- 4
Foetal membranes, their formation and role.
Mammalian placenta – its formation, types and functions.
Metamorphosis in *Herdmania* and *Rana* (frog)

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-4 Maj-402: Developmental Biology

Practical

1. Study of the following permanent slides for gametogenesis- T.S. Testis and T.S. ovary of mammals.
2. Study of the development of frog from permanent slides.
3. Study of the development of chick embryo from permanent slides up to 96 hours.
4. Study of metamorphosis of *Herdmania* and Frog through charts/video/ models

SUGGESTED READINGS

1. Ballinsky, B.I. An Introduction to Embryology, 5th Edition, Saunders Co., 1981.
2. Gilbert, S.F. Developmental Biology, 10th Edition, Sinauer Ass., Inc. Publ., 2014.
3. Urspaung, H. Major Problems in Developmental Biology, Academic Press, New York, 1966.
4. Wolport, Lewis Principles of development, Oxford University press, Oxford, 2007
5. Veer BalaRastogi, Chordate Embryology (Developmental Biology) KedarNath Ram Nath, Meerut/ Delhi 2011

Course Code: ZOO-DSC-6 Maj-402
Course Title: Biomolecules and Nutrition

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory= 75 + Practical= 25)
Theory Marks= 75 {67 (Theory examination) + 8 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 3 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning outcomes:
Students will be able to:

- *Acquire scientific knowledge about the basic biomolecules and their types*
- *Understand the importance of Nutrition and balanced diet for a healthy body*

UNIT- 1
Carbohydrates: Types (Monosaccharides, Disaccharides & Polysaccharides) andtheir sources
Lipids: **Types (triglycerides, phospholipids, and sterols) andtheir**sources

UNIT- 2
Proteins: Types (Simple, Conjugated, Fibrous, Globular) andtheir sources
Nucleic acids: Types (DNA & RNA) and their sources

UNIT- 3
Vitamins: Sources and functions of Fat-solubleVitamins(A,D,E and K) and Water-soluble Vitamins (BI, B2, B3, B6, B12 and C).
Minerals:Sources and functions of Macrominerals(sodium, potassium, calcium, magnesium, chloride) and Micro/Trace minerals (iodine, iron, zinc).

UNIT- 4
Balanced diet, Recommended Dietary Allowance (RDA), Undernutrition (Kwashiorkor, Marasmus), Overnutrition (Obesity, diabetes, hypertension), Micronutrient deficiency (Anemia, Goitre), Key objectives of the National Nutrition Policy of India (1993).

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-6 Maj-402: Biomolecules and Nutrition

Practical

- Qualitative analysis of carbohydrates (Molisch's Test, Benedict's test, Iodine test)
- Qualitative analysis of lipids (Solubility test, Spotting test)
- Qualitative analysis of proteins (Biuret test, Xanthoproteic test)
- Preparation of Balanced Diet Chart for different age groups- Children (1-9 years), Adolescents (10-18 years), Adults (19-59 years) and elderly (60+ years)
- Project on calculation of Body Mass Index (BMI) of individuals and assessment of their nutritional status (Underweight, Normal, Overweight and Obese) based on BMI

SUGGESTED READINGS

- Swaminathan, M. (2004). Handbook of food and nutrition. The Bangalore Printing and Publishing Co. Ltd.
- Joshi, S. (2023). Nutrition and dietetics (2nded.). Tata McGraw-Hill Publishing Company Limited. ISBN: 978-93-907278-27.
- Joshi, V. D. (1999). Handbook of nutrition and dietetics. Vora Medical Publications.
- Arora, M. P. (2004). Biomolecules. The Bangalore Printing and Publishing Co. Ltd. ISBN: 978-93-5024-579-8.
- Sihag, M. K. (2023). Biochemistry and human nutrition. Tata McGraw-Hill Publishing Company Limited. ISBN: 978-93-87973-54-1.
- Chahande, S. J. (2022). Textbook of biomolecules and nutritional biochemistry. ISBN: 978-81-976705-7-2.
- Gopalan, C., Rama Sastri, B. V., &Balasubramanian, S. C. (2003). Nutritive value of Indian foods (5th ed.). National Institute of Nutrition. ISBN: 978-81-85038-27-5.

Course Code: ZOO-DSC-7 Maj-403
Course Title: Basic Biochemical Techniques

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory= 75 + Practical= 25)
Theory Marks= 75 {67 (Theory examination) + 8 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) +3 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:
The students will be able to:
- *Familiarize themselves with the fundamentals of key biochemical techniques.*
- *Understand the principles and practical applications of chromatography, spectroscopy, centrifugation, and electrophoresis.*

Unit- 1
Chromatographic Techniques
General methodology of chromatography, Retention Factor (Rf value), Retention Time (Rt)
Principle and applications of Paper Chromatography, Thin Layer Chromatography, Column Chromatography (Adsorption, Partition, Gel/Size exclusion, Ion exchange)

Unit- 2
Spectroscopic Techniques
Beer-Lambert Law,Principle and applications of Colorimetry, UV-Visible Spectroscopy, Fluorescence Spectroscopy, Infrared (IR) Spectroscopy and Nuclear Magnetic Resonance (NMR) Spectroscopy

Unit- 3
Centrifugation Techniques
Principle of sedimentation, Types of centrifuges(Benchtop, High speed, Ultracentrifuge), Types of rotors (Fixed-angle, Swinging-bucket, Vertical)
Principle and applications of Differential Centrifugation, Density Gradient Centrifugationand Ultracentrifugation

Unit- 4
Electrophoretic Techniques
Principle and applications of Agarose Gel Electrophoresis, Polyacrylamide Gel Electrophoresis (SDS-PAGE, Native PAGE), Isoelectric focusing, Capillary Electrophoresis

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-7 Maj-403: Basic Biochemical Techniques

Practical

1. Separation of amino acids/plant pigments by paper chromatography.
2. Separation of sugars/lipids/leaf pigments/dyes by Thin Layer Chromatography.
3. Estimation of proteins by Lowry/Bradford method(Demonstration or Virtual simulation).
4. Operation and Calibration of Benchtop and High-Speed Centrifuges.
5. Separation of milk components using centrifuge.
6. Agarose gel electrophoresis for the separation of DNA fragments (Demonstration or Virtual simulation).
7. SDS-PAGE for protein separation (Demonstration or Virtual simulation).

SUGGESTED READINGS

- **Upadhyay A., Upadhyay K &Nath N. –*Biophysical Chemistry and Biochemical Techniques*,** 4th edition (2023), Himalaya Publishing House, Mumbai. ISBN: 978-93-5142-227-3.
- **Pattabiraman T.N. –*Laboratory Manual and Practical Biochemistry,*** 4th edition**, All India Publishers & Distributors**, New Delhi. **ISBN:**978-81-8004-002-3.
- **Gupta R.C. –*Practical Biochemistry*,** 5th edition (2018),CBS Publishers**& Distributors Pvt. Ltd.,** New Delhi. **ISBN:**978-81-239-2230-0.
- **Wilson K & Walker**J. **–*Principles and Techniques of Biochemistry and Molecular Biology, 8th*** edition (2018), Cambridge University Press. ISBN: 978-1-107-16227-3.
- **Plummer, D.T. –*An Introduction to Practical Biochemistry, 3rd***edition (2017), Tata McGraw-Hill Education, New Delhi**. ISBN:**978-0-07-099487-4.
- **Robyt J.F. & White B.J. –*Biochemical Techniques: Theory and Practice, 3rd edition, Pearson Publishers.* ISBN:**978-0-321-53657-2.

Course Code: ZOO-DSC-8 Maj-404
Course Title: Endocrinology

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory= 75 + Practical= 25)
Theory Marks= 75 {67 (Theory examination) + 8 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) +3 (Internal assessment)}
Duration for final examination: 3 hours

Theory

***Learning Outcome*:**
Students will be able to:

- *Understand the hormones and functions of endocrine glands*
- *Evaluate the mechanisms of hormone action*
- *Identify major endocrine disorders*

UNIT- 1
Introduction to Endocrinology: Classification of hormones, Mechanism of action of Hormones, Feedback mechanisms
Hormones of Gastrointestinal tract, and their functions

UNIT- 2
Hypothalamo- hypophysial portal system
Hormones of hypothalamus and pituitary gland, and their functions
Disorders of pituitary gland: Acromegaly, Cushing's Syndrome, Diabetes Insipidus
.

UNIT- 3
Hormones of Pancreas, Thyroid gland and Parathyroid glands, and their functions
Disorders: Diabetes mellitus, Goitre, Grave's disease

UNIT- 4
Hormones of Adrenal gland and their role in stress management
Disorders: Addison's disease, Cushing Syndrome
Hormones of Ovary and Testis, and their functions

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-8 Maj-404:Endocrinology

Practical

1. Demonstration of location of Endocrine glands in Rat/Human with the help of Charts/Models/Videos.
2. Study of the permanent slides of endocrine glands.
3. Demonstration of plasma level of any hormone using ELISA.
4. Demonstration of hormonal disorders mentioned in the theory with the help of Charts/Models/Videos.
5. Assignment on causes, symptoms and treatment of hormonal disorders.

SUGGESTED READINGS

- Turner, C. D. (1976). General Endocrinology, 6th edition. Pub- Saunders Toppan. ISBN-0721689337.
- Nussey, S. and Whitehead, S. (2001). Endocrinology: An Integrated Approach. Oxford: BIOS Scientific Publishers. ISBN- 9781859962527.
- Hadley, M.E. and Levine J.E. (2007). Endocrinology, 6th Edition. Pearson Prentice-Hall, Pearson Education Inc., New Jersey. ISBN- 9780131876064.
- Norris, D. O. and Carr. J. (2013). Vertebrate Endocrinology, 5th edition. Academic Press. ISBN- 9780123948151.
- Ramamurthy, V. and Durgadevi, K. (2020). Endocrinology, AkiNik Publications. **ISBN-**978-93-90420-39-1.